

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\E8001BEM.pvdb

Date of Analysis : Sep 9, 2026 10:21am

PV Elite 28, January 2026



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FileName : E8001_BEM

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Vessel Design Summary:

Step: 35 10:21am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1560.000 mm ID
Vessel Design Length, Tangent to Tangent 9557.00 mm
Specified Datum Line Distance 50.00 mm
Shell Side Design Temperature 180 °C
Channel Side Design Temperature 82 °C
Shell Side Design Pressure 2186.832 KPa.
Channel Side Design Pressure 1716.124 KPa.
Shop Shell Side Test Pressure 3198.695 KPa.
Field Shell Side Test Pressure 2230.960 KPa.
Shop Channel Side Test Pressure 2548.008 KPa.
Field Channel Side Test Pressure 2230.960 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F11	2	...	K11572	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Dish End	1730.858	0.00	1896.000	3.2000	No	No
LHS Ch shell	1730.858	0.00	3251.000	3.2000	N/A	No
LHS ch Flg	1730.827	0.00	2388.000	3.2000	N/A	No
Shell-1	2201.566	101.13	2564.000	3.2000	N/A	No
Shell-2	2201.566	101.13	2219.000	3.2000	N/A	No
Shell-3	2200.194	101.13	2565.000	3.2000	N/A	No
RHS Ch Flg	1730.827	0.00	1873.000	3.2000	N/A	No
RHS Ch Shell	1730.858	0.00	1872.000	3.2000	N/A	No
RHS Dish End	1716.124	103.42	1911.000	3.2000	No	No

Liquid Level: 1500.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000



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Vessel Design Summary:

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Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	50.0	16.0	14.0	13.0	...	1.00	1.00
Cylinder	1500.0	1500.0	22.0	22.0	13.1	...	1.00	1.00
Body Flg	1704.5	200.0	128.0	140.0	133.7	133.7	1.00	1.00
Cylinder	4034.5	2210.0	18.0	18.0	15.8	12.4	1.00	1.00
Cylinder	6244.5	2210.0	16.0	16.0	15.8	12.4	1.00	1.00
Cylinder	8423.5	2179.0	18.0	18.0	15.8	12.4	1.00	1.00
Body Flg	8757.0	209.0	164.0	163.0	158.6	158.6	1.00	1.00
Cylinder	9457.0	700.0	14.0	14.0	13.1	...	1.00	1.00
Ellipse	9507.0	50.0	16.0	14.0	12.9	7.3	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Tube outlet-	WRC-297	0.981	Shell	Passed
Tube Inlet	WRC-297	0.983	Shell	Passed
Shell Inlet	WRC-297	1.010	Shell	Failed
Shell outlet	WRC-297	0.993	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	325.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1050.000 mm
Wear Pad Width	500.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1360.000 mm
Baseplate Length	1300.000 mm
Baseplate Thickness	20.000 mm
Baseplate Width	350.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	370.000 mm
Number of Bolts in Baseplate	6

Baseplate Sketch



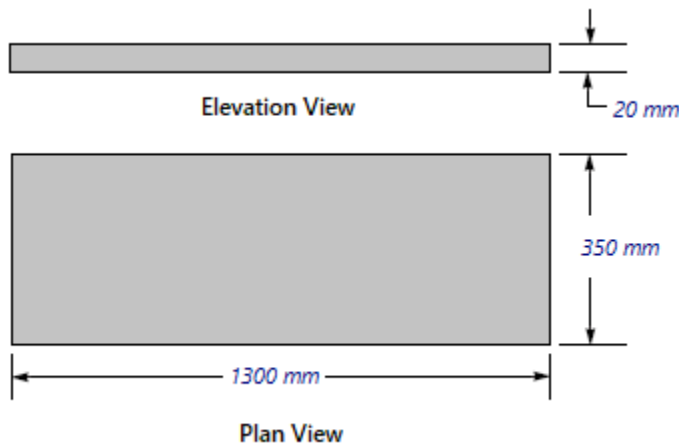
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Vessel Design Summary:

Step: 35 10:21am Sep 9, 2026

**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	35216.38	Kgf
Maximum Transverse Saddle Shear Load	2067.76	Kgf
Maximum Longitudinal Saddle Shear Load	3009.06	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	29771.29	Kgf
Maximum Transverse Saddle Shear Load	227.25	Kgf
Maximum Longitudinal Saddle Shear Load	156.74	Kgf

Weights:

Fabricated - Bare w/o removable internals	33001.9	kg.
Shop Test - Fabricated + water (full)	49874.6	kg.
Shipping - Fab. + rem. intls.+ shipping app.	33001.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	33501.9	kg.
Empty - Fab. + intls. + details + wghts.	35500.8	kg.
Operating - Empty + operating liquid (no ca)	51715.4	kg.
Field Test - Empty weight + test medium (full)	50617.6	kg.

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Nozzle Summary:

Step: 33 10:21am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Tube outlet-	2252	...	...	OK	11.33	OK	Passed
Tube Inlet	2238	...	...	OK	11.33	OK	Passed
Shell Inlet	2297	OK	...	OK	11.33	OK	Passed
Vent	2564	...	...	OK	6.42	OK	No Calc[*]
Drain	2564	...	...	OK	6.42	OK	No Calc[*]
Shell outlet	2346	OK	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2297 Nozzle : Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 2238 Nozzle : Tube Inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	Tube outlet-	750.000	0.000	1276.003
20	Tube Inlet	750.000	180.000	1276.003
40	Shell Inlet	2554.500	0.000	850.200
40	Vent	3674.500	0.000	89.925
40	Drain	3674.500	180.000	89.925
60	Shell outlet	7794.500	180.000	850.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
Vent	2.000 in	XXS	WNF	2.375	11.074	200.32	10.00	268.58	300
Drain	2.000 in	XXS	WNF	2.375	11.074	200.32	10.00	268.58	300
Shell Inlet	18.000 in	60	WNF	18.000	19.050	680.00	18.00	452.25	300
Shell outlet	18.000 in	60	WNF	18.000	19.050	697.20	18.00	452.25	300
Tube outlet-	25.984 in Actual	WNF	WNF	25.984	14.000	809.60	22.00	445.25	150
Tube Inlet	25.984 in Actual	WNF	WNF	25.984	14.000	809.60	22.00	445.25	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Vent	SA-106 B	14.000	20.000	10.000	10.000	...
Drain	SA-106 B	14.000	20.000	10.000	10.000	...
Shell Inlet	SA-106 B	14.000	10.000	12.000	18.000	...
Shell outle	SA-106 B	14.000	10.000	12.000	18.000	...
Tube outlet	SA-516 70	14.000	20.000	16.000	22.000	...
Tube Inlet	SA-516 70	14.000	20.000	16.000	22.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Vent	3624.500	0.0	250.00	0.00	Shell-1
Drain	3624.500	180.0	250.00	0.00	Shell-1
Shell Inlet	2504.500	0.0	400.00	0.00	Shell-1
Shell outlet	7744.500	180.0	400.00	0.00	Shell-3
Tube outlet-	700.000	0.0	350.00	0.00	LHS Ch shell
Tube Inlet	700.000	180.0	350.00	0.00	LHS Ch shell

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 1560.0mm ID X 50.0mm	SA-516 70
1	WELD NECK FLANGE: 140.0mm THK X 1840.0mm OD	SA-266 2
1	WELD NECK FLANGE: 164.0mm THK X 1840.0mm OD	SA-266 2
2	INSULATION: 425mm X 100mm THK	Hot Insulation
1	INSULATION: 1500mm X 100mm THK	Hot Insulation
1	INSULATION: 180mm X 100mm THK	Hot Insulation
13	BAFFLES: 1340.0mm DIA X 7.0mm THK	SA-516 70
10	TIE RODS: 25mm DIA X 6000mm	SA-193 B7
2	TUBESHEET: 1840.0mm X 120.0mm THK	SA-266 2
1780	TUBES: 6706.0mm X 25.4mm DIA X 2.3mm THK	SA-179
2	SADDLE: 325.0mm X 120 DEG	SA-516 70
2	INSULATION: 2210mm X 100mm THK	Hot Insulation
1	INSULATION: 2170mm X 100mm THK	Hot Insulation
1	INSULATION: 200mm X 100mm THK	Hot Insulation
1	INSULATION: 700mm X 100mm THK	Hot Insulation
2	GASKET: 1624.0mm OD X 1592.0mm ID	...
136	BODY FLANGE BOLTS: 28.6mm DIA	SA-193 B7
136	NUTS FOR BODY FLANGE BOLTS: 28.6mm DIA	...
2	CYLINDRICAL SEGMENT 22.0mm THK X 1946.7mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 1007.4mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1404.6mm X 2210.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 3496.2mm X 2210.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 3496.2mm X 2210.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 1404.6mm X 2210.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1892.3mm X 2179.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1878.6mm X 2179.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1129.9mm X 2179.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 4485.7mm X 700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 415.1mm X 700.0mm	SA-516 70
1	NAMEPLATE	...



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Bill of Material:

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MDMT Summary:

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell-1	[8]	B	-11	-25	-29	0.739	18.000	1.00	No
Shell-2	[8]	B	-14	-22	-29	0.854	16.000	1.00	No
Shell-3	[8]	B	-11	-25	-29	0.739	18.000	1.00	No
RHS Dish End	[10]	B	-18	-18	-29	1.000	14.000	1.00	No
RHS Dish End	[7]	B	-14	-22	-29	0.854	16.000	1.00	No
Left Saddle	[24]	B	-29	-43	-29	0.739	10.000	1.00	No
Sdl 2 Fr40	[24]	B	-29	-43	-29	0.739	10.000	1.00	No
Shell Inlet	[1]	B	-11	-29	-29	0.847	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Vent	[1]	B	-29	-37	-29	0.847	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Drain	[1]	B	-29	-37	-29	0.852	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Shell outlet	[1]	B	-11	-29	-29	0.852	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Tubesheet: SS	[13]	B	4	-12		0.717	30.000	1.00	No
Warmest MDMT:			4	-12					
LHS ch Flg	[11]	B	-5	-31	-29	0.558	22.000	1.00	No
RHS Ch Flg	[11]	B	-18	-32	-29	0.746	14.000	1.00	No
LHS Dish End	[10]	B	-18	-21	-29	0.953	14.000	1.00	No
LHS Dish End	[7]	B	-14	-24	-29	0.814	16.000	1.00	No
LHS Ch shell	[8]	B	-5	-31	-29	0.554	22.000	1.00	No
RHS Ch Shell	[8]	B	-18	-20	-29	0.964	14.000	1.00	No
Tube outlet-	[1]	B	-5	-34	-29	0.522	22.000	1.00	No
Nozzle Flg	[5]	A	-18	-25					
Tube Inlet	[1]	B	-5	-34	-29	0.527	22.000	1.00	No
Nozzle Flg	[5]	B	-18	-24					
Tubesheet: CS	[14]	B	4	-12		0.717	30.000	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						
Warmest MDMT:			4	-12					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-12.0	-12.0	Pass
Channel/Tube	-12.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required



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MDMT Summary:

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- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 10:21am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	2186.8	KPa.
Channel Side Design Pressure	1716.1	KPa.
Shell Side Design Temperature	180.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

Step: 1 10:21am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	0	mm
Distance Upwind of Crest	0	mm
Distance from Crest to the Vessel	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration	0.147	[Ss]
Max. Eff. Ground Acceleration	0.065	[S]
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	0.000	z/h
Amplification Factor	1.000	Ap
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 10:21am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	LHS Dish End
Distance "FROM" to "TO"	50 mm
Inside Diameter	1560 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1716.1 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1500 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation



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Input Echo:

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Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	-375 mm
Height/Length of Insulation	425 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	2/9
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	LHS Ch shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	1560 mm
Element Thickness	22 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1716.1 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1500 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube outlet-
Dist. from "FROM" Node / Offset dist	700 mm
Nozzle Diameter	25.98 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	111.36 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube Inlet
Dist. from "FROM" Node / Offset dist	700 mm
Nozzle Diameter	25.98 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0



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Input Echo:

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Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	335 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element#	3/9
Element From Node	30
Element To Node	40
Element Type	Flange
Description	LHS ch Flg
Distance "FROM" to "TO"	200 mm
Flange Inside Diameter	1560 mm
Element Thickness	140 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	128 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1716.1 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1500 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	30
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	180 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	4/9
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	2210 mm
Inside Diameter	1560 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm



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Input Echo:

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Design Internal Pressure	2186.8	KPa.
Design Temperature Internal Pressure	180	°C
Design External Pressure	101.13	KPa.
Design Temperature External Pressure	180	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	227.4	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1360	mm
Width of Saddle	325	mm
Height of Saddle at Bottom	1050	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	500	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1500	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2210	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell Inlet	
Dist. from "FROM" Node / Offset dist	680	mm
Nozzle Diameter	18	in.
Nozzle Schedule	60	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	261.33	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



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Input Echo:

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Element From Node	40
Detail Type	Nozzle
Detail ID	Vent
Dist. from "FROM" Node / Offset dist	1800 mm
Nozzle Diameter	2 in.
Nozzle Schedule	XXS
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.0576 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Nozzle
Detail ID	Drain
Dist. from "FROM" Node / Offset dist	1800 mm
Nozzle Diameter	2 in.
Nozzle Schedule	XXS
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.0576 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Weight
Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	1360 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	1050 mm

Element#	5/9
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	2210 mm
Inside Diameter	1560 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2186.8 KPa.
Design Temperature Internal Pressure	180 °C
Design External Pressure	101.13 KPa.
Design Temperature External Pressure	180 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1500 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	50
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Input Echo:

Step: 1 10:21am Sep 9,2026

Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2210 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	2179 mm
Inside Diameter	1560 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2186.8 KPa.
Design Temperature Internal Pressure	180 °C
Design External Pressure	101.13 KPa.
Design Temperature External Pressure	180 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Saddle
Detail ID	Sdl 2 Fr40
Dist. from "FROM" Node / Offset dist	820 mm
Width of Saddle	325 mm
Height of Saddle at Bottom	1050 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	500 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.40000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1360 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2170 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	60
Detail Type	Nozzle



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Input Echo:

Step: 1 10:21am Sep 9,2026

Detail ID	Shell outlet
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	18 in.
Nozzle Schedule	60
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	261.33 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Weight
Detail ID	Fireproofing
Dist. from "FROM" Node / Offset dist	820 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	1050 mm

Element#	7/9
Element From Node	70
Element To Node	80
Element Type	Flange
Description	RHS Ch Flg
Distance "FROM" to "TO"	209 mm
Flange Inside Diameter	1560 mm
Element Thickness	163 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	164 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1716.1 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1500 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	70
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	201 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³



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Input Echo:

Step: 1 10:21am Sep 9,2026

Element#	8/9
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	RHS Ch Shell
Distance "FROM" to "TO"	700 mm
Inside Diameter	1560 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1716.1 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1500 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	80
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	700 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	9/9
Element From Node	90
Element To Node	110
Element Type	Elliptical
Description	RHS Dish End
Distance "FROM" to "TO"	50 mm
Inside Diameter	1560 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1716.1 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C



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Input Echo:

Step: 1 10:21am Sep 9,2026

Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 90
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	425 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

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XY Coordinate Calculations:

Step: 2 10:21am

Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Dish End		50	...	50	...
LHS Ch shell		1550	...	1500	...
LHS ch Flg		1754.5	...	200	...
Shell-1		4084.5	...	2210	...
Shell-2		6294.5	...	2210	...
Shell-3		8473.5	...	2179	...
RHS Ch Flg		8807	...	209	...
RHS Ch Shell		9507	...	700	...
RHS Dish End		9557	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: LHS ch Flg

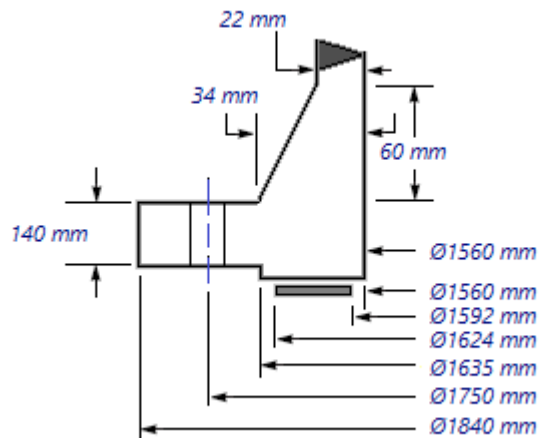
Flng: 5 10:21am Sep 9,2026

Flange Input Data Values

Description: LHS ch Flg :

LHS ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1730.83	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1560.000	mm
Flange Outside Diameter	A	1840.000	mm
Flange Thickness	t	140.0000	mm
Thickness of Hub at Small End	go	22.0000	mm
Thickness of Hub at Large End	gl	34.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1750.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		68	
Flange Face Outside Diameter	Fod	1635.000	mm
Flange Face Inside Diameter	Fid	1560.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1624.000	mm
Gasket Inside Diameter	Gi	1592.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: LHS ch Flg

Flng: 5 10:21am

Sep 9, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1731 \cdot (1560/2 + 3.2)) / (138 \cdot 1 - 0.6 \cdot 1731) + Ca$$

$$= 13.0983 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 18.8) / (783 + 0.6 \cdot 18.8)$$

$$= 3265.284 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 22) / (780 + 0.6 \cdot 22)$$

$$= 3827.242 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	136.800 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1566.400 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	30.800 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	18.800 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	61.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $bo = N/2$	8.000 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1609.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1610^2 \cdot 1731$$

$$= 359214.812 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1610 \cdot 3 \cdot 1731$$

$$= 38171.578 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1566^2 \cdot 1731$$

$$= 340130.344 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 359215 - 340130$$

$$= 19084.482 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(359215 + 38172 + 0, 0)$$

$$= 397386.406 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1610 + 0 \cdot 0$$

$$= 253422.000 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((397386 + 0 + 4 \cdot 0/1610)/172, 253422/172)$$

$$= 226.580 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 28.6 + 6 \cdot 137/(3 + 0.5)$$

$$= 291.664 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1750 \cdot \sin(\pi/68)$$

$$= 80.821 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{80.8/(2 \cdot 28.6 + 137)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	226.580	319.380	
Radial Distance between Hub and Bolts:	38.100	61.000	
Radial Distance between Bolts and Edge:	28.575	45.000	
Circ. Spacing between the Bolts:	63.500	80.821	291.664

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 319 \cdot 172 / (68.9 \cdot \pi(1624 + 1592))$$

$$= 7.885 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 319$$

$$= 560143.94 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 397386 - 359215$$

$$= 38171.57 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1750 - 1610) / 2$$

$$= 70.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 (C - B) / 2 + hg$$

$$= 0.5 (1750 - 1566) / 2 + 70.1$$

$$= 80.9637 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1750 - 1566 - 30.8) / 2$$

$$= 76.4000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	340130.	76.4000	1.0000	25986.
Face Pressure, Mt	19084.	80.9637	1.0000	1545.
Gasket Load, Mg	38172.	70.1274	1.0000	2677.
Gasket Seating, Matm	560144.	70.1274	1.0000	39282.

Total Moment for Operation, Mo 30209. Kg-m.
Total Moment for Gasket seating, Mg 39282. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 171.605 mm
Hub Ratio, h/h0 = HL / H0 0.350
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.638

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.867
Factor V per 4.16.5 0.328
Factor f per 4.16.5 1.226

Stress factor from Table 4.16.4 [K]:

= A / B
= 1840/1566
= 1.175

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.17² • log10(1.17)))/(1.17² - 1))/(1.17 - 1)
= 12.140

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.17²(1 + 8.55246 • log10(1.17)) - 1)/((1.0472 + 1.9448 • 1.17²) • (1.17 - 1))
= 1.849

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.17²(1 + 8.55246 • log10(1.17)) - 1)/(1.36136(1.17² - 1) • (1.17 - 1))
= 13.340

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.17² + 1)/(1.17² - 1)
= 6.265

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 13.3 • 18.8² • 172/0.33
= 2469306.750 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.87/172
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (137 • 0.0051 + 1)/1.85 + 137³/2469306
= 1.951

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.23•30209/1566)/(1.95•30.8²)
= 125.26 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$$

$$= (1.23 \cdot 39282 / 1566) / (1.95 \cdot 30.8^2)$$

$$= 162.88 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.92 \cdot 30209 / 1566) / (1.95 \cdot 137^2)$$

$$= 9.95 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.92 \cdot 39282 / 1566) / (1.95 \cdot 137^2)$$

$$= 12.94 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (12.1 \cdot 30209 / (137^2 \cdot 1566)) - 6.27 \cdot 9.95$$

$$= 60.34 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (12.1 \cdot 39282 / (137^2 \cdot 1566)) - 6.27 \cdot 12.9$$

$$= 78.46 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (125 + \max(9.95, 60.3)) / 2$$

$$= 92.80 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (163 + \max(12.9, 78.5)) / 2$$

$$= 120.67 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 397386 / 319$$

$$= 122.02 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (253422 / 319)$$

$$= 77.82 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	125.26	207.00	162.88	207.00
Radial Flange	9.95	138.00	12.94	138.00
Tangential Flange	60.34	138.00	78.46	138.00
Maximum Average	92.80	138.00	120.67	138.00
Bolting	122.02	172.00	77.82	172.00

Minimum Required Flange Thickness [Rigidity] 133.706 mm

Estimated M.A.W.P. (Operating) 2402 KPa.

Estimated Finished Weight of Flange at given Thk. 876.3 kg.

Estimated Unfinished Weight of Forging at given Thk 1159.0 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	240.54	374.99	0.641	240.54	374.99	0.641
Tangential Flange	Sto	160.92	250.00	0.644	160.92	250.00	0.644
Combined	Sh/f + St	647.85	749.99	0.864	647.85	749.99	0.864
Combined	Sr + St	280.21	749.99	0.374	280.21	749.99	0.374
Combined	Sh + Sto	660.91	749.99	0.881	660.91	749.99	0.881
Radial Flange	Sr	39.67	499.99	0.079	39.67	499.99	0.079
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.63/1.33)

= 1.2275

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.51/1.28) if $h/h_0 < 0.35$, else max (0.25; 0.3/0.76)

= 0.3951

In this case, the hub ratio h/h_0 , 0.3496 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	527.3 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	4.6968 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.573 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1624^2 - 1592^2)$$

$$= 808.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 208 \cdot 808 / (68 \cdot 4.7)$$

$$= 526.4 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(526; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 527)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 508 \geq 140 (808 / (68 \cdot 4.7))$$

$$\therefore 507.5 \geq 354.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 508 \geq (97 \cdot 808 + \pi/4 \cdot 1731 \cdot 1592^2) / (0.7 \cdot 4.7 \cdot 68)$$

$$\therefore 507.5 \geq 504.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 508 \leq 275 (808 / (68 \cdot 4.7))$$

$$\therefore 507.5 \leq 696.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 508 \leq 527 (1/0.57)$$



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:: 507.5 ≤ 920.6 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 508 \cdot 0.26 \cdot 4.7 \cdot 28.6$$

$$= 180.6 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 39282 / 1 \cdot 9807 \cdot 0.33 / (1.8 \cdot 201334 \cdot 18.8^2 \cdot 172 \cdot 0.3)$$

$$= 0.999 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 30209 / 1 \cdot 9807 \cdot 0.33 / (1.8 \cdot 197847 \cdot 18.8^2 \cdot 172 \cdot 0.3)$$

$$= 0.782 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 39282 / 1 \cdot 9807 \cdot 0.33 / (1.95 \cdot 201334 \cdot 18.8^2 \cdot 172 \cdot 0.3)$$

$$= 0.920 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 30209 / 1 \cdot 9807 \cdot 0.33 / (1.95 \cdot 197847 \cdot 18.8^2 \cdot 172 \cdot 0.3)$$

$$= 0.720 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.56, Temperature Reduction per Fig. UCS 66.1 = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	113.04	207.00	144.19	207.00
Radial Flange	10.74	138.00	13.71	138.00
Tangential Flange	51.32	138.00	65.46	138.00
Maximum Average	82.18	138.00	104.82	138.00
Bolting	122.02	172.00	77.82	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 116.078 mm

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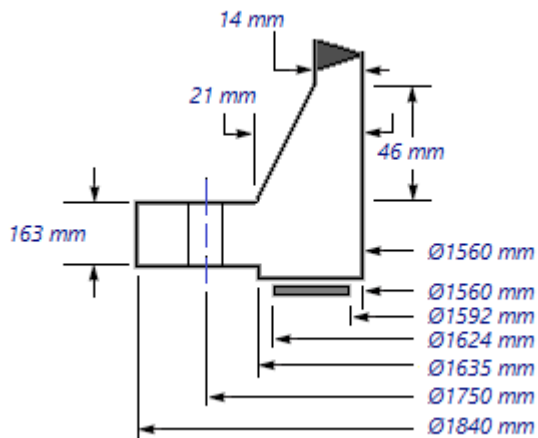
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Flange Input Data Values

Description: RHS Ch Flg :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1730.83	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1560.000	mm
Flange Outside Diameter	A	1840.000	mm
Flange Thickness	t	163.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	46.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1750.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		68	
Flange Face Outside Diameter	Fod	1635.000	mm
Flange Face Inside Diameter	Fid	1560.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1624.000	mm
Gasket Inside Diameter	Gi	1592.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1731 \cdot (1560/2 + 3.2)) / (138 \cdot 1 - 0.6 \cdot 1731) + Ca$$

$$= 13.0983 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 10.8) / (783 + 0.6 \cdot 10.8)$$

$$= 1887.203 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (780 + 0.6 \cdot 14)$$

$$= 2450.346 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	159.800 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1566.400 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	17.800 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	10.800 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	74.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $bo = N/2$	8.000 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1609.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1610^2 \cdot 1731$$

$$= 359214.812 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1610 \cdot 3 \cdot 1731$$

$$= 38171.578 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1566^2 \cdot 1731$$

$$= 340130.344 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 359215 - 340130$$

$$= 19084.482 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(359215 + 38172 + 0, 0)$$

$$= 397386.406 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1610 + 0 \cdot 0$$

$$= 253422.000 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((397386 + 0 + 4 \cdot 0/1610)/172, 253422/172)$$

$$= 226.580 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 28.6 + 6 \cdot 160/(3 + 0.5)$$

$$= 331.093 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1750 \cdot \sin(\pi/68)$$

$$= 80.821 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{80.8/(2 \cdot 28.6 + 160)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	226.580	319.380	
Radial Distance between Hub and Bolts:	38.100	74.000	
Radial Distance between Bolts and Edge:	28.575	45.000	
Circ. Spacing between the Bolts:	63.500	80.821	331.093

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 319 \cdot 172 / (68.9 \cdot \pi(1624 + 1592))$$

$$= 7.885 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 319$$

$$= 560143.94 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 397386 - 359215$$

$$= 38171.57 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1750 - 1610) / 2$$

$$= 70.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1750 - 1566) / 2 + 70.1)$$

$$= 80.9637 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1750 - 1566 - 17.8) / 2$$

$$= 82.9000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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Flg Calc [Int P]: RHS Ch Flg Flng: 6 10:21am Sep 9, 2026

End Pressure, Md	340130.	82.9000	1.0000	28197.
Face Pressure, Mt	19084.	80.9637	1.0000	1545.
Gasket Load, Mg	38172.	70.1274	1.0000	2677.
Gasket Seating, Matm	560144.	70.1274	1.0000	39282.

Total Moment for Operation, Mo 32419. Kg-m.
Total Moment for Gasket seating, Mg 39282. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 130.066 mm
Hub Ratio, h/h0 = HL / H0 0.354
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.648

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.866
Factor V per 4.16.5 0.325
Factor f per 4.16.5 1.228

Stress factor from Table 4.16.4 [K]:

= A / B
= 1840/1566
= 1.175

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.17² • log10(1.17))) / (1.17² - 1))/(1.17 - 1)
= 12.140

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.17²(1 + 8.55246 • log10(1.17)) - 1) / ((1.0472 + 1.9448 • 1.17²) • (1.17 - 1))
= 1.849

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.17²(1 + 8.55246 • log10(1.17)) - 1) / (1.36136(1.17² - 1) • (1.17 - 1))
= 13.340

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.17² + 1) / (1.17² - 1)
= 6.265

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 13.3 • 10.8² • 130 / 0.32
= 623490.562 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.87/130
= 0.007 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (160 • 0.0067 + 1) / 1.85 + 160³/623491
= 7.661

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.23•32419/1566)/(7.66•17.8²)
= 102.65 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.23 \cdot 39282 / 1566) / (7.66 \cdot 17.8^2)$$

$$= 124.37 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.42 \cdot 32419 / 1566) / (7.66 \cdot 160^2)$$

$$= 2.51 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.42 \cdot 39282 / 1566) / (7.66 \cdot 160^2)$$

$$= 3.04 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (12.1 \cdot 32419 / (160^2 \cdot 1566)) - 6.27 \cdot 2.51$$

$$= 80.77 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (12.1 \cdot 39282 / (160^2 \cdot 1566)) - 6.27 \cdot 3.04$$

$$= 97.87 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (103 + \max(2.51, 80.8)) / 2$$

$$= 91.71 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (124 + \max(3.04, 97.9)) / 2$$

$$= 111.12 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 397386 / 319$$

$$= 122.02 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (253422 / 319)$$

$$= 77.82 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	102.65	207.00	124.37	207.00
Radial Flange	2.51	138.00	3.04	138.00
Tangential Flange	80.77	138.00	97.87	138.00
Maximum Average	91.71	138.00	111.12	138.00
Bolting	122.02	172.00	77.82	172.00

Minimum Required Flange Thickness [Rigidity] 158.623 mm

Estimated M.A.W.P. (Operating) 1887 KPa.

Estimated Finished Weight of Flange at given Thk. 975.5 kg.

Estimated Unfinished Weight of Forging at given Thk 1211.2 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	469.12	499.99	0.938	469.12	499.99	0.938
Tangential Flange	St	368.50	374.99	0.983	368.50	374.99	0.983
Tangential Flange	Sto	154.04	250.00	0.616	154.04	250.00	0.616
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	379.94	749.99	0.507	379.94	749.99	0.507
Combined	Sh + Sto	623.15	749.99	0.831	623.15	749.99	0.831
Radial Flange	Sr	11.45	499.99	0.023	11.45	499.99	0.023
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.64/1.33)

= 1.2297

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)

= max(0.25; 0.52/1.29) if h/ho < 0.35, else max (0.25; 0.31/0.76)

= 0.4038

In this case, the hub ratio h/ho, 0.3537 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	647.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	4.6968 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.549 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1624^2 - 1592^2)$$

$$= 808.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 808 / (68 \cdot 4.7)$$

$$= 523.9 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(524; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 648)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 508 \geq 140 (808 / (68 \cdot 4.7))$$

$$\therefore 507.5 \geq 354.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 508 \geq (97 \cdot 808 + \pi/4 \cdot 1731 \cdot 1592^2) / (0.7 \cdot 4.7 \cdot 68)$$

$$\therefore 507.5 \geq 504.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 508 \leq 276 (808 / (68 \cdot 4.7))$$

$$\therefore 507.5 \leq 698.5 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 508 \leq 648 (1/0.55)$$



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Flg Calc [Int P]: RHS Ch Flg

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:: 507.5 ≤ 1178.6 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi_b$$

$$= 508 \cdot 0.2 \cdot 4.7 \cdot 28.6$$

$$= 138.9 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 39282 / 1 \cdot 9807 \cdot 0.32 / (7.12 \cdot 201334 \cdot 10.8^2 \cdot 130 \cdot 0.3)$$

$$= 0.999 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 32419 / 1 \cdot 9807 \cdot 0.32 / (7.12 \cdot 197847 \cdot 10.8^2 \cdot 130 \cdot 0.3)$$

$$= 0.839 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 39282 / 1 \cdot 9807 \cdot 0.32 / (7.66 \cdot 201334 \cdot 10.8^2 \cdot 130 \cdot 0.3)$$

$$= 0.928 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 32419 / 1 \cdot 9807 \cdot 0.32 / (7.66 \cdot 197847 \cdot 10.8^2 \cdot 130 \cdot 0.3)$$

$$= 0.780 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.75, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -32 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	101.38	207.00	120.72	207.00
Radial Flange	3.39	138.00	4.04	138.00
Tangential Flange	71.71	138.00	85.40	138.00
Maximum Average	86.55	138.00	103.06	138.00
Bolting	122.02	172.00	77.82	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 148.565 mm

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Internal Pressure Calculations: Step: 5 10:21am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Dish End		1730.9	16	3.2	1560	138	1
LHS Ch shell		1730.9	22	3.2	1560	138	2
LHS ch Flg		1730.8	128	3.2	1560	138	3
Shell-1		2201.6	18	3.2	1560	138	4
Shell-2		2201.6	16	3.2	1560	138	5
Shell-3		2200.2	18	3.2	1560	138	6
RHS Ch Flg		1730.8	164	3.2	1560	138	7
RHS Ch Shell		1730.9	14	3.2	1560	138	8
RHS Dish End		1716.1	16	3.2	1560	138	9

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Dish End		1716.12	1896	2472	14	12.9831	1
LHS Ch shell		1716.12	3251	3827	22	13.0985	2
LHS ch Flg		1716.12	2388	2440	140	133.706	3
Shell-1		2186.83	2564	3141	18	15.8164	4
Shell-2		2186.83	2219	2796	16	15.8164	5
Shell-3		2186.83	2565	3141	18	15.8085	6
RHS Ch Flg		1716.12	1873	2440	163	158.623	7
RHS Ch Shell		1716.12	1872	2450	14	13.0985	8
RHS Dish End		1716.12	1911	2472	14	12.8997	9

Warning: Channel Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1911 KPa.
 Shell Side MAPnc = 2472 KPa.
 Channel Side MAWP = 1820 KPa.
 Channel Side MAPnc = 1960 KPa.

Elements Suitable for Design Internal Pressure.

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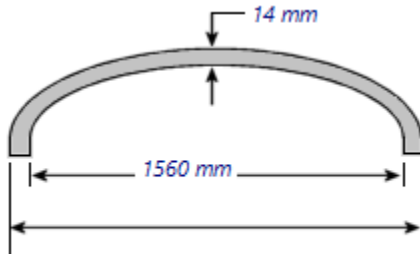
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Element: LHS Dish End

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Dish End

Elliptical Head Design Information:					
Design Pressure	P	1730.858 KPa.	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.2000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1560.000 mm	Actual Stress		125.0 N./mm ²
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		29973.9 cm ²	SG of Contents		1.000
Element Volume		592623.1 cm ³	Weight of Contents		589.7 Kg
Empty Weight		399.3 Kg	Operating Weight		989.0 Kg
			Comp. All. Ope. Stress		119.9 N./mm ²
			Aspect Ratio		2.00
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 19.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 9.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1731 \cdot 1566 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1731)$$

$$= 9.7831 + 3.2000 = 12.9831 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 14.734 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 10.8) / (0.99 \cdot 1566 + 0.2 \cdot 10.8)$$

$$= 1911 - 14.7 = 1896 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 14) / (1 \cdot 1560 + 0.2 \cdot 14)$$

$$= 2472 \text{ KPa.}$$



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Element: LHS Dish End

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1731 \cdot (0.99 \cdot 1566 + 0.2 \cdot 10.8)) / (2 \cdot 1 \cdot 10.8)$$

$$= 125.020 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (1731 \cdot 783) / (138 \cdot 1 - 0.6 \cdot 1731) + 3.2$$

$$= 13.098 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 14.734 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12.8) / (783 + 0.6 \cdot 12.8)$$

$$= 2233 - 14.7 = 2219 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1566 / (2 \cdot 393)))^2 / 6$$

$$= 0.994586$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.392 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 14, tr = 10.3, c = 3.2 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.95$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -21 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 10.4, c = 3.2 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.81$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1820.434 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.814	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-24 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 19.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 9.750 mm

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FileName : E8001_BEM

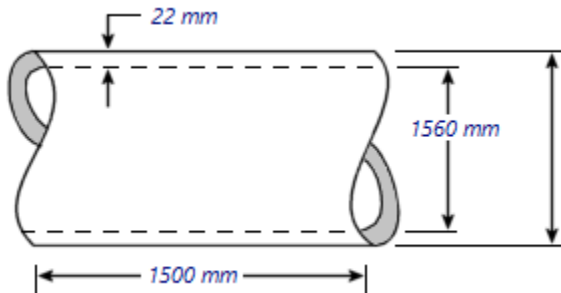
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Element: LHS Ch shell

Step: 7 10:21am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS Ch shell

Cylindrical Shell Design Information:			
Design Pressure P	1730.858 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1560.000 mm	Actual Stress	73.1 N./mm ²
Spec. Min. Thk t,e, or sa	22.0000 mm	Spec. Nominal Thk.	22.000 mm
Surface Area	75586.7 cm ²	SG of Contents	1.000
Element Volume	2867531.2 cm ³	Weight of Contents	2829.5 Kg
Empty Weight	1271.2 Kg	Operating Weight	4100.7 Kg
		Comp. All. Ope. Stress	133.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1731 \cdot 783) / (138 \cdot 1 - 0.6 \cdot 1731) \\
 &= 9.8985 + 3.2000 = 13.0985 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 14.734 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18.8) / (783 + 0.6 \cdot 18.8) \\
 &= 3265 - 14.7 = 3251 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 22) / (780 + 0.6 \cdot 22) \\
 &= 3827 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1731 \cdot (783 + 0.6 \cdot 18.8)) / (1 \cdot 18.8) \\
 &= 73.150 \text{ N./mm}^2
 \end{aligned}$$



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Element: LHS Ch shell

Step: 7 10:21am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.391 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 22$, $t_r = 10.4$, $c = 3.2$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.55$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1820.434 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.554	Temperature Difference	26 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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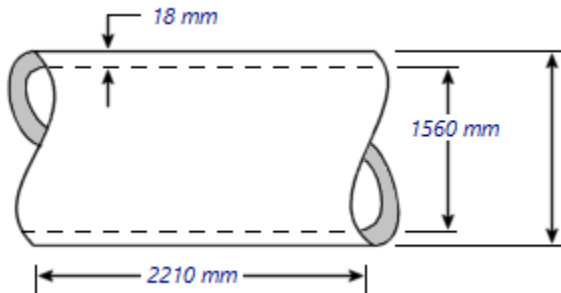
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Element: Shell-1

Step: 8 10:21am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:			
Design Pressure P	2201.566 KPa.	Int. Design Temperature	180 °C
Material	SA-516 70	Ext. Design Temperature	180 °C
Length	2210.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1560.000 mm	Actual Stress	117.8 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	110809.0 cm ²	SG of Contents	1.000
Element Volume	2231186.5 cm ³	Weight of Contents	2229.8 Kg
Empty Weight	1528.5 Kg	Operating Weight	3758.3 Kg
		Comp. All. Ope. Stress	119.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2202 \cdot 783) / (138 \cdot 1 - 0.6 \cdot 2202) \\
 &= 12.6164 + 3.2000 = 15.8164 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 14.734 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14.8) / (783 + 0.6 \cdot 14.8) \\
 &= 2578 - 14.7 = 2564 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18) / (780 + 0.6 \cdot 18) \\
 &= 3141 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2202 \cdot (783 + 0.6 \cdot 14.8)) / (1 \cdot 14.8) \\
 &= 117.832 \text{ N./mm}^2
 \end{aligned}$$



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Element: Shell-1

Step: 8 10:21am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.141 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18$, $t_r = 10.9$, $c = 3.2$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.74$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -25 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1910.527 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.739	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	-25 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 180 °C

Elastic Modulus from Chart: CS-2 at 180 °C: 196084384 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.800	1596.00	6599.00	107.84	4.1347	0.0002658	26.07

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 26.1) / (3 \cdot 108) = 322 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.225	1596.00	6599.00	173.01	4.1347	0.0001338	13.12

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 13.1) / (3 \cdot 173) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.800	1596.00	141428.31	107.84	50.0000	0.0000961	9.42

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.42) / (3 \cdot 108) = 116 \text{ KPa.}$$



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FileName : E8001_BEM

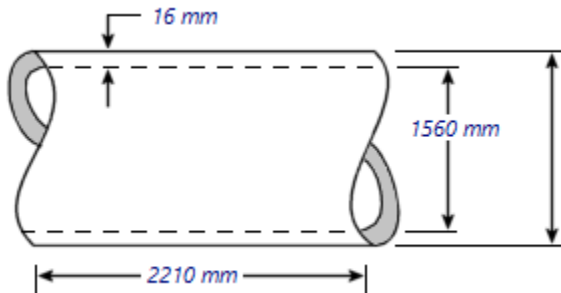
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Element: Shell-2

Step: 9 10:21am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:			
Design Pressure P	2201.566 KPa.	Int. Design Temperature	180 °C
Material	SA-516 70	Ext. Design Temperature	180 °C
Length	2210.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1560.000 mm	Actual Stress	136.0 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	110531.3 cm ²	SG of Contents	1.000
Element Volume	2231186.5 cm ³	Weight of Contents	2229.8 Kg
Empty Weight	1356.9 Kg	Operating Weight	3586.7 Kg
		Comp. All. Ope. Stress	116.3 N./mm ²
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2202 \cdot 783) / (138 \cdot 1 - 0.6 \cdot 2202) \\
 &= 12.6164 + 3.2000 = 15.8164 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 14.734 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 12.8) / (783 + 0.6 \cdot 12.8) \\
 &= 2233 - 14.7 = 2219 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 16) / (780 + 0.6 \cdot 16) \\
 &= 2796 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2202 \cdot (783 + 0.6 \cdot 12.8)) / (1 \cdot 12.8) \\
 &= 136.037 \text{ N./mm}^2
 \end{aligned}$$



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Element: Shell-2

Step: 9 10:21am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.015 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 16$, $t_r = 10.9$, $c = 3.2$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp. at Required thickness (UCS 66.1) -22 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1910.527 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.854	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-22 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 180 °C

Elastic Modulus from Chart: CS-2 at 180 °C: 196084384 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.800	1592.00	6599.00	124.38	4.1451	0.0002134	20.92

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 20.9) / (3 \cdot 124) = 224 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.212	1592.00	6599.00	172.81	4.1451	0.0001337	13.11

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 13.1) / (3 \cdot 173) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.800	1592.00	13642.46	124.38	8.5694	0.0000963	9.44

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.44) / (3 \cdot 124) = 101 \text{ KPa.}$$



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FileName : E8001_BEM

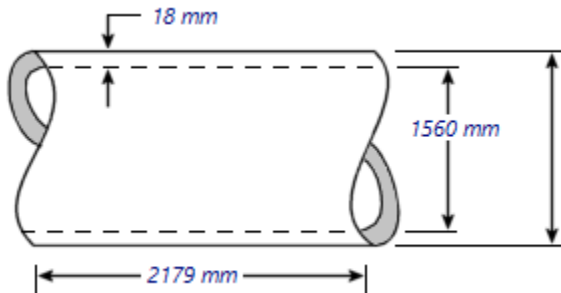
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Element: Shell-3

Step: 10 10:21am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:			
Design Pressure P	2200.194 KPa.	Int. Design Temperature	180 °C
Material	SA-516 70	Ext. Design Temperature	180 °C
Length	2179.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1560.000 mm	Actual Stress	117.8 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	109254.7 cm ²	SG of Contents	1.000
Element Volume	2199889.2 cm ³	Weight of Contents	2198.5 Kg
Empty Weight	1507.0 Kg	Operating Weight	3705.6 Kg
		Comp. All. Ope. Stress	119.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2200 \cdot 783) / (138 \cdot 1 - 0.6 \cdot 2200) \\
 &= 12.6085 + 3.2000 = 15.8085 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 13.362 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14.8) / (783 + 0.6 \cdot 14.8) \\
 &= 2578 - 13.4 = 2565 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18) / (780 + 0.6 \cdot 18) \\
 &= 3141 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2200 \cdot (783 + 0.6 \cdot 14.8)) / (1 \cdot 14.8) \\
 &= 117.759 \text{ N./mm}^2
 \end{aligned}$$



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FileName : E8001_BEM

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Element: Shell-3

Step: 10 10:21am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.141 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18$, $t_r = 10.9$, $c = 3.2$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.74$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -25 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1910.527 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.739	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	-25 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 180 °C

Elastic Modulus from Chart: CS-2 at 180 °C: 196084384 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.800	1596.00	6599.00	107.84	4.1347	0.0002658	26.07

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 26.1) / (3 \cdot 108) = 322 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.225	1596.00	6599.00	173.01	4.1347	0.0001338	13.12

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 13.1) / (3 \cdot 173) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.800	1596.00	141428.31	107.84	50.0000	0.0000961	9.42

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.42) / (3 \cdot 108) = 116 \text{ KPa.}$$



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FileName : E8001_BEM

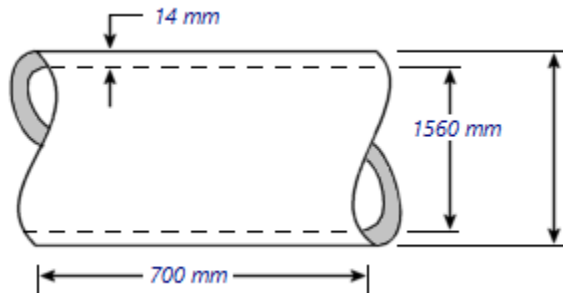
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Element: RHS Ch Shell

Step: 11 10:21am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1730.858 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1560.000 mm	Actual Stress	126.6 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	34921.9 cm ²	SG of Contents	1.000
Element Volume	1338181.2 cm ³	Weight of Contents	1320.4 Kg
Empty Weight	375.6 Kg	Operating Weight	1696.0 Kg
		Comp. All. Ope. Stress	119.9 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1731 \cdot 783) / (138 \cdot 1 - 0.6 \cdot 1731) \\
 &= 9.8985 + 3.2000 = 13.0985 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 14.734 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 10.8) / (783 + 0.6 \cdot 10.8) \\
 &= 1887 - 14.7 = 1872 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14) / (780 + 0.6 \cdot 14) \\
 &= 2450 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1731 \cdot (783 + 0.6 \cdot 10.8)) / (1 \cdot 10.8) \\
 &= 126.565 \text{ N./mm}^2
 \end{aligned}$$



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FileName : E8001_BEM

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Element: RHS Ch Shell

Step: 11 10:21am

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.889 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 14$, $t_r = 10.4$, $c = 3.2$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.96$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -20 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1820.434 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.964	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-20 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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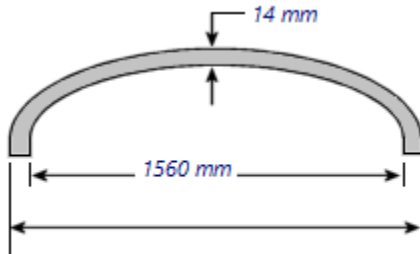
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Element: RHS Dish End

Step: 12 10:21am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:			
Design Pressure P	1716.124 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1560.000 mm	Actual Stress	124.0 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	29973.9 cm ²	SG of Contents	0.000
Element Volume	592623.1 cm ³	Weight of Contents	0.0 Kg
Empty Weight	399.3 Kg	Operating Weight	399.3 Kg
		Comp. All. Ope. Stress	119.9 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 19.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 9.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1716 \cdot 1566 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1716)$$

$$= 9.6997 + 3.2000 = 12.8997 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 10.8) / (0.99 \cdot 1566 + 0.2 \cdot 10.8)$$

$$= 1911 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 14) / (1 \cdot 1560 + 0.2 \cdot 14)$$

$$= 2472 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:



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Element: RHS Dish End

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$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1716 \cdot (0.99 \cdot 1566 + 0.2 \cdot 10.8)) / (2 \cdot 1 \cdot 10.8)$$

$$= 123.956 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (1716 \cdot 783) / (138 \cdot 1 - 0.6 \cdot 1716) + 3.2$$

$$= 13.014 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12.8) / (783 + 0.6 \cdot 12.8)$$

$$= 2233 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1566 / (2 \cdot 393)))^2 / 6$$

$$= 0.994586$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.392 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 14, tr = 10.8, c = 3.2 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 10.9, c = 3.2 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -22 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1910.527 KPa.
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.854	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-22 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 19.500 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 9.750 mm

EXTERNAL PRESSURE RESULTS for: RHS Dish End

Elliptical Head From 90 to 110 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure

MAEP



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Element: RHS Dish End

Step: 12 10:21am

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Tca	Outer Dia	Do/t	Factor A	Factor B
10.800	1588.00	147.04	0.0009446	83.53

$$MAEP = B / (K0 * Do / t) = 83.5 / (0.9 * 147) = 631 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
4.112	1588.00	386.22	0.0003596	35.95

$$MAEP = B / (K0 * Do / t) = 36 / (0.9 * 386) = 103 \text{ KPa.}$$

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Shop Pressure Test Results:

Step: 13 10:21am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2483.685 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2230.960 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3198.695 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2101.580 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2454.057 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2454.057 KPa.
App 27-4	= M.A.W.P.	1910.527 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2366.564 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2230.960 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2548.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2002.478 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2454.057 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2454.057 KPa.
App 27-4	= M.A.W.P.	1820.434 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	185.1	234.0	0.791	2563.30
LHS Ch shell	108.3	234.0	0.463	2563.30
Shell-1	172.0	234.0	0.735	3213.99
Shell-2	198.6	234.0	0.849	3213.99
Shell-3	172.0	234.0	0.735	3213.99
RHS Ch Shell	187.4	234.0	0.801	2563.30
RHS Dish End	232.1	234.0	0.992	3213.99

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube outlet-	Nozzle	138.00	138.00	1.000
Tube outlet-	Pad	138.00	138.00	1.000
Tube Inlet	Nozzle	138.00	138.00	1.000
Tube Inlet	Pad	138.00	138.00	1.000
Shell Inlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Pad	138.00	138.00	1.000
Vent	Nozzle	118.00	118.00	1.000
Vent	Pad	138.00	138.00	1.000
Drain	Nozzle	118.00	118.00	1.000
Drain	Pad	138.00	138.00	1.000
Shell outlet	Nozzle	118.00	118.00	1.000
Shell outlet	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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Shop Pressure Test Results:

Step: 13 10:21am Sep 9,2026

LHS Dish End	138.00	138.00	1.000
LHS Ch shell	138.00	138.00	1.000
LHS ch Flg	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Ch Shell	138.00	138.00	1.000
RHS Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube outlet-	75.88	234.00	0.324
Tube Inlet	75.88	234.00	0.324
Shell Inlet	44.49	216.00	0.206
Vent	10.72	216.00	0.050
Drain	10.72	216.00	0.050
Shell outlet	44.49	216.00	0.206

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 14 10:21am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2483.685 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2230.960 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3198.695 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2101.580 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2454.057 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2454.057 KPa.
App 27-4	= M.A.W.P.	1910.527 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2366.564 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2230.960 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2548.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2002.478 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2454.057 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2454.057 KPa.
App 27-4	= M.A.W.P.	1820.434 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1716 • 1

= 2230.960 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1716 • 1

= 2230.960 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	162.2	234.0	0.693	2246.25
LHS Ch shell	94.9	234.0	0.406	2246.25
Shell-1	120.2	234.0	0.514	2246.25
Shell-2	138.8	234.0	0.593	2246.25
Shell-3	120.2	234.0	0.514	2246.25
RHS Ch Shell	164.3	234.0	0.702	2246.25
RHS Dish End	162.2	234.0	0.693	2246.25

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube outlet-	Nozzle	138.00	138.00	1.000
Tube outlet-	Pad	138.00	138.00	1.000
Tube Inlet	Nozzle	138.00	138.00	1.000
Tube Inlet	Pad	138.00	138.00	1.000
Shell Inlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Pad	138.00	138.00	1.000
Vent	Nozzle	118.00	118.00	1.000
Vent	Pad	138.00	138.00	1.000
Drain	Nozzle	118.00	118.00	1.000



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Field Pressure Test Results: Step: 14 10:21am Sep 9,2026

Drain	Pad	138.00	138.00	1.000
Shell outlet	Nozzle	118.00	118.00	1.000
Shell outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Dish End	138.00	138.00	1.000
LHS Ch shell	138.00	138.00	1.000
LHS ch Flg	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Ch Shell	138.00	138.00	1.000
RHS Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube outlet-	66.49	234.00	0.284
Tube Inlet	66.49	234.00	0.284
Shell Inlet	31.10	216.00	0.144
Vent	7.49	216.00	0.035
Drain	7.49	216.00	0.035
Shell outlet	31.10	216.00	0.144

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 10:21a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell-1

Shell Test Pressure		2230.96 KPa.
Shell Thickness	ts	18.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1560.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch shell

Channel Type:		Cylinder
Channel Test Pressure		2230.96 KPa.
Channel Thickness	tc	22.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1560.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1780
Tube Wall Thickness	et	2.3200 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6706.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6466.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.3200 mm
Groove Weld Leg	ag	2.3200 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	726.462 mm
Largest Center-to-Center Tube Distance	U1	41.2750 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	15161.1 cm ²



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ASME TS Calc: Pressure Test

Case: 1 10:21a Sep 9,2026

Perimeter of Tube Layout Cp 4643.7 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 120.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 1840.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
Is the Tubesheet Flat on Tubeside? No
Step 1 Diameter on the Tubeside dt1 1632.00 mm
Step 1 Depth on the Tubeside ht1 8.00 mm
Step 2 Diameter on the Tubeside dt2 1557.00 mm
Step 2 Depth on the Tubeside ht2 8.00 mm
Is the Tubesheet Flat on Shellside? No
Calculated Tubesheet Diameter per Div 2, 4.18.4 1840.00 mm

Note: Tubesheet diameter is now: 1840.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 215.9 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 647.700 mm
Tube End condition corresponding to Span (l) k 0.80
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Shell Side Min. Design Pressure Psd,min -101.1300 KPa.
Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
Junction Stress Reduction option Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
Flange Outside Diameter A 1840.000 mm
Flange Inside Diameter B 1560.000 mm
Flange Face Outside Diameter Fod 1635.000 mm
Flange Face Inside Diameter Fid 1560.000 mm
Gasket Outside Diameter Go 1624.000 mm
Gasket Inside Diameter Gi 1592.000 mm
Small end Hub thk. g0 22.0000 mm
Large end Hub thk. g1 34.0000 mm
Gasket Factor, m 3.00
Gasket Design Seating Stress y 68.95 N./mm²
Flange Facing Sketch Code Sketch 1a
Gasket Thickness tg 4.5000 mm
Full face Gasket Flange Option Program Selects

Bolting Information:

Diameter of Bolt Circle C 1750.000 mm
Nominal Bolt Diameter dB 28.5750 mm



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ASME TS Calc: Pressure Test

Case: 1 10:21a Sep 9,2026

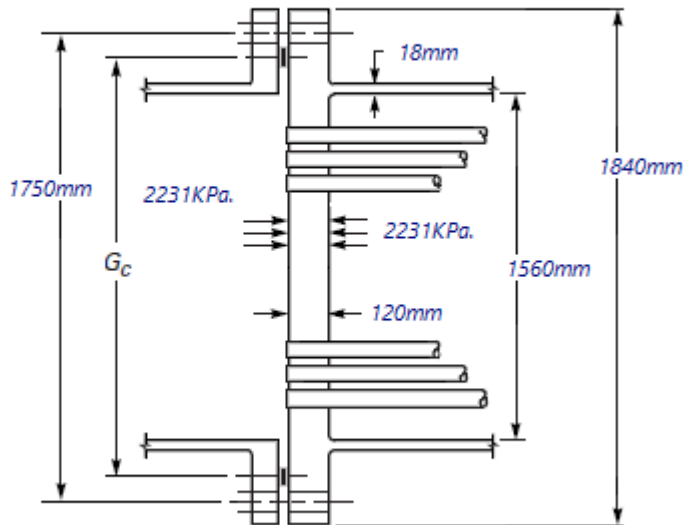
Thread Series	TEMA Thread Series
Number of Bolts	n 68
Bolt Material	SA-193 B7
Bolt Allowable Stress At Temperature	Sb 172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa 172.00 N./mm ²
Weld between Flange and Shell/Channel	0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 99.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1



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ASME TS Calc: Pressure Test

Case: 1 10:21a Sep 9, 2026

Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the

type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= (P*(D/2-CAE)) / (S*E+0.4*P) per Appendix 1-1 (a)(1)

= (2332*(25.4/2-0))/(92.4*1+0.4*2332)

= 0.3174 + 0.0000 = 0.3174 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.3200	25.40	6466.00	10.95	50.0000	0.0091770	95.84
EMAWP = (4*B)/(3*(D/T)) = (4 *95.8)/(3 *10.9) = 11671 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6315	25.40	6466.00	40.22	50.0000	0.0006799	67.31
EMAWP = (4*B)/(3*(D/T)) = (4 *67.3)/(3 *40.2) = 2231 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6315 mm
Allowable Internal Pressure at Corroded thickness	18208.61 KPa.
Required Internal Design Pressure	2332.09 KPa.
Allowable External Pressure at Corroded thickness	11671.24 KPa.
Required External Design Pressure	2230.96 KPa.
Required Thickness due to Shell Side pressure	0.6315 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Req'd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	79.516	43.754	174	356	11	190	Ps+Pt-Th	D1 Ok
D2uc	30.674	...	37	356	12	190	Ps+Pt-Th	D2 Ok
D3uc	85.918	...	185	356	2	190	Ps+Pt-Th	D3 Ok
Max:	85.9183	43.754 mm		0.521		0.064 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:



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ASME TS Calc: Pressure Test

Case: 1 10:21a Sep 9, 2026

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:
With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	28	370	...	...	:	...	...	...	...	Ok
D2uc	22	370	...	...	:	...	...	...	...	Ok
D3uc	51	370	...	...	:	...	...	...	...	Ok
Max RATIO	0.137				:	...				

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	23	170	-16	-170	396	2933	...	370	...	...	Ok
D2uc	9	170	-9	-170	151	2933	...	370	...	...	Ok
D3uc	14	170	-9	-170	237	2933	...	370	...	...	Ok
Max RATIO	0.135		0.092		0.135		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	85.918	120.000	Ok
Tubesheet thickness flanged extension :	43.754	99.000	Ok
Tube thickness :	0.632	2.320	Ok
Tube-tubesheet fillet weld leg :	1.449	2.320	Ok
Tube-tubesheet groove weld leg :	1.449	2.320	Ok

Min Shell length of thk, (18.000) adj. to tubesheet: 301.628 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
Due to Pressure Shell Compresses by : -0.097 mm
Due to Pressure + Thermal Shell Compresses by : -0.097 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load	Due to Pressure	Due to Thermal Exp.	Computed Delta Ms
------	-----------------	---------------------	-------------------



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ASME TS Calc: Pressure Test

Case: 1 10:21a Sep 9, 2026

Case#	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0414	...	...	...	...	...
2	-0.0975	...	...	...	...	...
3	-0.0605	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculations: Step: 15 10:21am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	1588	10.8	0.00094458	83.5279
20	30	1680	1604	18.8	0.0016376	99.0194
30	40	No Calc	...	136.8	No Calc	No Calc
40	50	6599	1596	14.8	0.00026585	26.0658
50	60	6599	1592	12.8	0.00021336	20.9199
60	70	6599	1596	14.8	0.00026585	26.0658
70	80	No Calc	...	159.8	No Calc	No Calc
80	90	880	1588	10.8	0.0014095	95.152
90	110	No Calc	1588	10.8	0.00094458	83.5279

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	14	No Calc	...	631
20	30	22	No Calc	...	1547
30	40	140	133.706	...	No Calc
40	50	18	12.425	101.13	322
50	60	16	12.4122	101.13	224
60	70	18	12.425	101.13	322
70	80	163	158.623	...	No Calc
80	90	14	No Calc	...	863
90	110	14	7.31161	103.419	631

Minimum

224

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	1680	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	6599	141428	No Calc	No Calc
50	60	6599	13642.5	No Calc	No Calc
60	70	6599	141428	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	880	No Calc	No Calc	No Calc
90	110	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 16 10:21am Sep 9,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	399.34	592623	319.472	599552	39.934
20	30	1271.16	2867531	1088.46	2891107	127.116
30	40	876.265	384538	851.906	385481	87.6265
40	50	1528.45	2231186	1259.28	2265922	152.845
50	60	1356.9	2231186	1087.73	2265922	135.69
60	70	1507.02	2199889	1241.62	2234138	150.702
70	80	981.308	404032	955.73	404755	98.1308
80	90	375.587	1338181	290.328	1349183	37.5588
90	110	399.34	592623	319.472	599552	39.934
Total		8695	12841791.00	7413	12995614.00	869



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	589.701	-130	30	...	Liquid: 10
10	Insl	139.092	-162.5	...	...	Hot Insulation
20	Liqd	2829.51	750	30	...	Liquid: 20
20	Insl	282.984	750	...	...	Hot Insulation
20	Nozl	122.497	700	1110	...	Tube outlet-
20	Nozl	368.5	700	-1110	...	Tube Inlet
30	Liqd	379.439	100	30	...	Liquid: 30
30	Insl	35.7251	90	...	...	Hot Insulation
40	Sadl	270.44	1360	901	...	Left Saddle
40	Liqd	2229.82	1105	30	...	Liquid: 40
40	Insl	414.972	1105	...	...	Hot Insulation
40	Nozl	287.464	680	1008.6	...	Shell Inlet
40	Nozl	9.96331	1800	799.088	...	Vent
40	Nozl	9.96331	1800	-799.088	...	Drain
40	Wght	250	1360	-1050	...	Fire proofing
50	Liqd	2229.82	1105	30	...	Liquid: 50
50	Insl	413.993	1105	...	...	Hot Insulation
60	Sadl	270.44	820	901	...	Sdl 2 Fr40
60	Liqd	2198.55	1089.5	100	...	Liquid: 40
60	Insl	407.461	1085	...	...	Hot Insulation
60	Nozl	287.464	1500	-1008.6	...	Shell outlet
60	Wght	250	820	-1050	...	Fireproofing
70	Liqd	398.675	104.5	30	...	Liquid: 70
70	Insl	34.7076	100.5	...	...	Hot Insulation
80	Liqd	1320.44	350	30	...	Liquid: 60
80	Insl	130.819	350	...	...	Hot Insulation
90	Insl	139.092	212.5	...	...	Ins: 90
30	FTsh	1797.64	264.5	...	...	Tubesheet
30	Tube	17119.1	3557.5	...	...	Tubes
30	RTsh	1797.64	6850.5	...	...	Tubesheet
40	Baf1	84.2079	647.7	...	...	Baffles
40	Baf1	84.2079	1125.43	...	...	Baffles
40	Baf1	84.2079	1603.16	...	...	Baffles
40	Baf1	84.2079	2080.89	...	...	Baffles
40	Baf1	84.2079	2558.62	...	...	Baffles



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Element and Detail Weights: Step: 16 10:21am Sep 9,2026

40 Baf1	84.2079	3036.35	...	...	Baffles
40 Baf1	84.2079	3514.08	...	...	Baffles
40 Baf1	84.2079	3991.81	...	...	Baffles
40 Baf1	84.2079	4469.54	...	...	Baffles
40 Baf1	84.2079	4947.27	...	...	Baffles
40 Baf1	84.2079	5425	...	...	Baffles
40 Baf1	84.2079	5902.73	...	...	Baffles
40 Baf1	85.4553	6380.46	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	540.9
Liquid	12176.0
Insulation	1998.8
Nozzles	1085.9
Weights	500.0
Exchanger Components	21810.3
Liquid in Tubes	4038.7

Sum of the Detail Weights 42150.5 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	9564.9	9564.9	9564.9	9564.9	9564.9	9564.9
Saddles	540.9	540.9	540.9	540.9	540.9	540.9
Nozzles	1085.9	1085.9	1085.9	1085.9	1085.9	1085.9
Exchanger	21810.3	21810.3	21810.3	21810.3	21810.3	21810.3
Insulation	...	...	...	...	1998.8	1998.8 *
Empty Weights	...	...	...	500.0	500.0	...
Ope Weights	...	...	...	...	...	500.0
Ope. Liquid	...	...	...	...	...	12176.0
Tube Ope Lqd	...	...	...	...	...	4038.7
Test Liquid	...	12834.0	...	...	...	...
Tube Tst Lqd	...	4038.7	...	...	...	...
Totals	33001.9	49874.6	33001.9	33501.9	35500.8	51715.4

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	33001.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	49874.6 kg.
Shipping Wt.	- Fab. weight + removable intl's.+ shipping app.	33001.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	33501.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	35500.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	51715.4 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	50305.9 kg.
Field Test Wt.	- Empty weight + test medium (full)	50617.6 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	4041159.0 cm ³
Weight of Ope Liq in tubes :	4038.7 kg.
Weight of Water in tubes :	4038.7 kg.



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Element and Detail Weights:

Step: 16 10:21am

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Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	29973.9	28830.3
20	30	75586.7	73513.3
30	40	18638.4	9801.77
40	50	110809	108310
50	60	110531	108310
60	70	109255	106790
70	80	19272.2	10291.9
80	90	34921.9	34306.2
90	110	29973.9	28830.3

Total 538962.1 508983.2 cm²Total 580.1 547.9 ft²

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Nozzle Flange MAWP:

Step: 17 10:21am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Tube outlet-	1820.4	1960.0	82	150	GR 1.1	...	...	...	...
Tube Inlet	1820.4	1960.0	82	150	GR 1.1	...	...	...	...
Shell Inlet	4432.0	5100.0	180	300	GR 1.1	1435.9	4432.0	4432.0	4432.0
Vent	4432.0	5100.0	180	300	GR 1.1	...	...	...	...
Drain	4432.0	5100.0	180	300	GR 1.1	...	...	...	...
Shell outlet	4432.0	5100.0	180	300	GR 1.1	1435.3	4432.0	4432.0	4432.0

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 4432.031 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 5100.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1820.434 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: None Selected

It appears that the ASME UG-44(b) method of pressure de-rating was not selected. If external loads (forces and bending moments) are also required to be evaluated on standard flanged joints with welding neck flanges as mentioned in UG-44(a)(2), (a)(9) and (a)(10) for vessels constructed per the ASME VIII-1 2019 Code Edition onward, they may be evaluated per the requirements specified in UG-44(b).

You can set the year of construction in Tools->Enter U1 Form Information.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 18 10:21am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$$

$$= \min(0.85, 0.91)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.580
Structure Height to Diameter ratio	5.818

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1050 + 300 = 1350 \text{ mm}$$

$$= 4.43 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (4.43 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Z_g)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.85$$

Type of Hill: No Hill



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Wind Load Calculation:

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Wind Directionality Factor [Kd]:
= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
= (1 + K1 • K2 • K3)²
= (1 + 0 • 0 • 0)²
= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)
= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)
= 16 psf [78.1] Kgs/m²

Force on the first element [F]:
= qz • G • Cf • WindArea
= 16 • 0.85 • 0.58 • 8.77
= 69.2 lbs. [31.4] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Dish End	1350.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch shell	1350.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS ch Flg	1350.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-1	1350.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-2	1350.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-3	1350.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch Flg	1350.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch Shell	1350.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Dish End	1350.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Fire proofing	0.00	300.00	78.12	0.00
Fireproofing	0.00	300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1350	2145.6	8150.43	78.12	34.5464
20	30	1350	2164.8	32472	78.12	137.636
30	40	1350	2088	4176	78.12	17.7004
40	50	1350	2155.2	47629.9	78.12	201.884
50	60	1350	2150.4	47523.8	78.12	201.435
60	70	1350	2154.21	46940.2	78.12	198.961
70	80	1350	2102.81	4394.88	78.12	18.6281
80	90	1350	2145.6	15019.2	78.12	63.6604
90	110	1350	2145.6	8150.43	78.12	34.5464

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.



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Wind Load Calculation:

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Earthquake Load Calculation: Step: 19 10:21am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(60.75)$
 $= 0.077$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.077, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 51175 \cdot 1.25$
 $= 3009.063 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 51175$
 $= 1604.8 \text{ Kgf}$

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	780	4652.23	273.551



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Earthquake Load Calculation:

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20	30	780	4652.23	273.551
30	40	780	4652.23	273.551
40	50	780	4652.23	273.551
40	50	780	4652.23	273.551
50	60	780	4652.23	273.551
60	0	780	4652.23	273.551
60	70	780	4652.23	273.551
70	80	780	4652.23	273.551
80	90	780	4652.23	273.551
90	110	780	4652.23	273.551

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 20 10:21am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5174.500 mm
Center of Gravity of the Liquid	4338.464 mm
Center of Gravity of the Insulation	4755.715 mm
Center of Gravity of the Nozzles	3146.317 mm
Center of Gravity of the Added Weights (Operating)	5174.500 mm
Center of Gravity of the Added Weights (Empty)	5174.500 mm
Center of Gravity of the Tubesheet(s)	5107.500 mm
Center of Gravity of the Tubes	5107.500 mm
Center of Gravity of the Baffles	5391.842 mm
Center of Gravity of Bare Shell New and Cold	4713.006 mm
Center of Gravity of Bare Shell Corroded	4703.240 mm
Vessel CG in the Operating Condition	4807.098 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4942.689 mm
Vessel CG in the Test Condition	4863.838 mm

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Saddle Calcs: Road Transport:

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Note:

Wear Pad Width (500.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

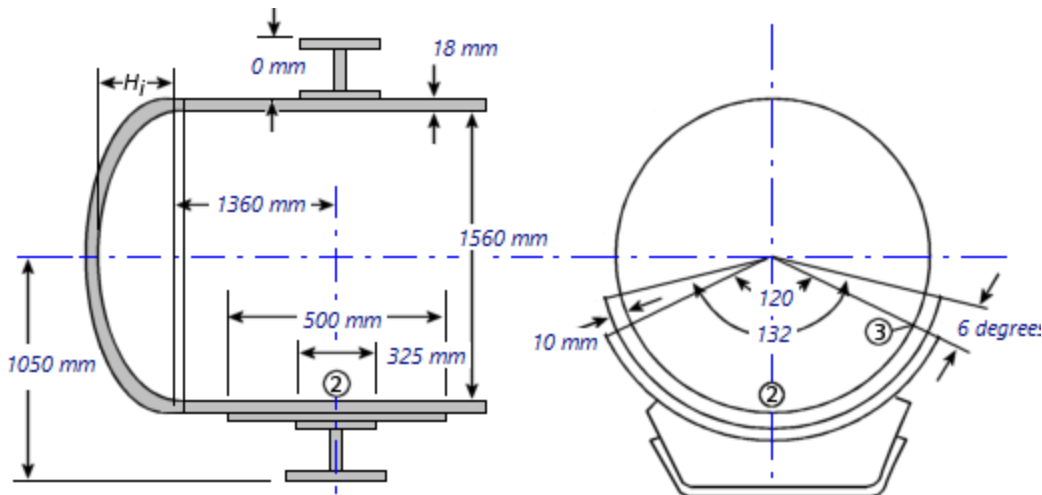
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{789 \cdot 18}, 2 \cdot 1360)$$

$$= 510.9085 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	789.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6649.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1360.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		21076.32	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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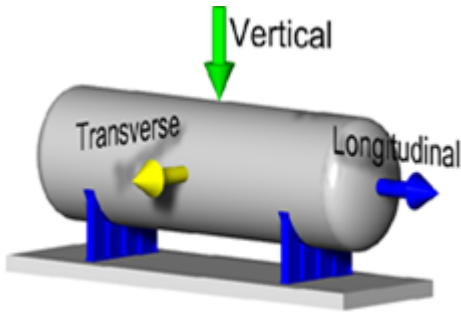
FileName : E8001_BEM

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Saddle Calcs: Road Transport:

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-2.31	-204.66
Long. Stress at Top of Midspan	2.31	138.00
Long. Stress at Bottom of Midspan	2.31	138.00
Long. Stress at Top of Saddles	12.74	138.00
Long. Stress at Bottom of Saddles	-7.06	-234.00
Long. Stress at Bottom of Saddles	7.06	138.00
Tangential Shear in Shell	10.07	110.40
Circ. Stress at Horn of Saddle	56.19	207.00
Circ. Compressive Stress in Shell	1.71	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 231 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 1960.8 \text{ lbf} \\
 &= 889.423 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 5.89^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 278.0 \text{ lbf} \\
 &= 126.1 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 32461 \cdot 1 + 889 \\
 &= 4135.5 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 32461 \cdot 1 + 126 \\
 &= 3372.2 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 4136/2 \cdot 1050/1300 \\
 &= 1670.1 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 3372 \cdot 1050/3880 \\
 &= 912.6 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 18482 \cdot 0.05
 \end{aligned}$$



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Saddle Calcs: Road Transport: Step: 21 10:21am Sep 9, 2026

= 924.105 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(FwL, Fwt)

= 18482 + 924 + max(913, 1670)

= 21076.3 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 21346.76 Kgf

Transverse Shear Load 2067.76 Kgf

Longitudinal Shear Load 3372.19 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 1300.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$

= $-21076 \cdot 1360 [1 - (1 - 1360/6649 + (789^2 - 0^2)/(2 \cdot 1360 \cdot 6649))/(1 + (4 \cdot 0)/(3 \cdot 6649))]$

= -4876.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$

= $21076 \cdot 6649/4 ((1 + 2(789^2 - 0^2)/(6649^2))/(1 + (4 \cdot 0)/(3 \cdot 6649)) - 4 \cdot 1360/6649)$

= 7357.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$

= $0 \cdot 789/(2 \cdot 16) - 7357/(\pi \cdot 789^2 \cdot 16)$

= -2.31 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$

= $0 \cdot 789/(2 \cdot 16) + 7357/(\pi \cdot 789^2 \cdot 16)$

= 2.31 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= $P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$

= $0 \cdot 789/(2 \cdot 18) - 4876/(0.11 \cdot \pi \cdot 789^2 \cdot 18)$

= 12.74 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= $P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$

= $0 \cdot 789/(2 \cdot 18) + 4876/(0.19 \cdot \pi \cdot 789^2 \cdot 18)$

= -7.06 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot hm/3))$

= $21076 (6649 - 2 \cdot 1360)/(6649 + (4 \cdot 0/3))$

= 12454.3 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= $K2 \cdot T / (Rm \cdot t)$

= $1.17 \cdot 12454/(789 \cdot 18)$



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$$= 10.07 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(789 \cdot 18)}$$

$$= 92.954 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 21076 \cdot 0.1 / (18(325 + 93 + 93))$$

$$= -1.71 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -21076 / (4 \cdot 18(325 + 93 + 93)) -$$

$$3 \cdot 0.053 \cdot 21076 / (2 \cdot 18^2)$$

$$= -56.19 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(789 \cdot 18)}, 2 \cdot 1360)$$

$$= 510.91 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1300.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(780 \cdot 18)} + 500)^3) / 12$$

$$= 33.283 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 33.3 + 123(88.7 - 9)^2$$

$$= 7862.062 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	684.8	18.0	9.0	123.3	110944.9	0.786E+04
Wearplate	500.0	10.0	23.0	50.0	115000.0	0.216E+04
Web	10.0	222.0	139.0	22.2	308580.2	0.147E+04
BasePlate	350.0	20.0	260.0	70.0	1820000.4	0.206E+05
Totals	...	...	...	265.5	2354525.5	0.321E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2354525 / 265$$

$$= 88.692 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$



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Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 21076$$

$$= 4289.4883 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.9582 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1050 - 780 \cdot \sin(120/2) / 1.05$$

$$= 404.945 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1737.0421 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.7120 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3 \cdot (21076 + 270) \cdot 350) / (4 \cdot 1300 \cdot 173)}$$

$$= 15.617 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot \text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}$$

$$= 2 \cdot \cos(90 - 120/2) \cdot (780 + 18 + 10)$$

$$= 1399.497 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1399 / (4 - 1)$$

$$= 466.499 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 466 \cdot 350 / 2$$

$$= 816.373 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 21347 / (1300 \cdot 350)$$

$$= 4.692 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 816 \cdot 4.69$$

$$= 3830.093 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 31.5 + 23.3$$

$$= 54.825 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 3830 / 54.8$$

$$= 6.851 \text{ N./mm}^2$$



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Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	325.0	...	32.5	...	0.286E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 325 - 10

= 315.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/54.8

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 325/2

= 162.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{2861/54.8}$

= 72.235 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1050 - cos(120/2)(780 + 18 + 10) - 20

= 626.000 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(120/2)(780 + 18 + 10)

= 1399.497 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 1300 - 50.8

= 1249.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (1399 - 1249)/2

= 75.149 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{626^2 + 75.1^2}$

= 630.495 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 630/72.2

= 8.728

Bending Moment [Rm]:



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$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 3372 / (2 \cdot 1300) \cdot 466 \cdot 630 / 2$$

$$= 190.743 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($8.73 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.73)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (8.73) / (8 \cdot 123) - (8.73^3) / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 6.85 / 153 + (191 \cdot 162 / 28606770) / 173$$

$$= 0.106$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	315.0	0.0	31.5	0.0	0.286E+04
Web	466.5	10.0	0.0	46.6	0.0	3.89
Totals	...	...	...	78.1	...	0.286E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 78.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 1050 - \sqrt{(808 + 10)^2 - (466/2)^2} - 20$$

$$= 256.399 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2864 / 78.1}$$

$$= 60.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 256 / 60.5$$

$$= 4.235$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 3372 / (2 \cdot 1300)$$

$$= 1.297 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.3 \cdot 466 \cdot 256$$

$$= 155.136 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 155 / 176276$$

$$= 8.631 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.24 < 123$) per AISC E2-1 [Sca]:



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$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (4.24)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (4.24) / (8 \cdot 123) - (4.24^3) / (8 \cdot 123^3))$$

$$= 154.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 9.62/155 + (155 \cdot 162/2864) / 173$$

$$= 0.112$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (500.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{789 \cdot 18}, 2 \cdot 1360)$$

$$= 510.9085 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	789.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6649.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1360.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		16873.01	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-1.85	-204.66	
Long. Stress at Top of Midspan	1.85	138.00	
Long. Stress at Bottom of Midspan	1.85	138.00	
Long. Stress at Top of Saddles	10.20	138.00	
Long. Stress at Bottom of Saddles	-5.65	-234.00	
Long. Stress at Bottom of Saddles	5.65	138.00	
Tangential Shear in Shell	8.06	110.40	
Circ. Stress at Horn of Saddle	44.99	207.00	
Circ. Compressive Stress in Shell	1.37	138.00	



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Saddle Calcs: Road Transport: Step: 21 10:21am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 231 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 1960.8 \text{ lbf} \\
 &= 889.423 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 5.89^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 278.0 \text{ lbf} \\
 &= 126.1 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 32461 \cdot 1 + 889 \\
 &= 4135.5 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [Fl]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 32461 \cdot 1 + 126 \\
 &= 3372.2 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 4136/2 \cdot 1050/1300 \\
 &= 1670.1 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= Fl \cdot B / Ls \\
 &= 3372 \cdot 1050/3880 \\
 &= 912.6 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot gy \\
 &= 14479 \cdot 0.05 \\
 &= 723.947 \text{ Kg}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + Frv + \max(Fwl, Fwt) \\
 &= 14479 + 724 + \max(913, 1670) \\
 &= 16873.0 \text{ Kg}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	17143.45	Kgf
Transverse Shear Load	2067.76	Kgf
Longitudinal Shear Load	3372.19	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \text{Baseplate Length} \\
 &= 1300.000 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -16873 \cdot 1360 [1 - (1 - 1360/6649 + (789^2 - 0^2)/
 \end{aligned}$$



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Saddle Calcs: Road Transport: Step: 21 10:21am Sep 9, 2026

$$\begin{aligned} & (2 \cdot 1360 \cdot 6649) / (1 + (4 \cdot 0) / (3 \cdot 6649)) \\ & = -3903.9 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} & \text{Moment per Equation 4.15.2 [M2]:} \\ & = Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\ & = 16873 \cdot 6649 / 4 \cdot ((1 + 2(789^2 - 0^2) / (6649^2)) / (1 + (4 \cdot 0) / (3 \cdot 6649)) - 4 \cdot 1360 / 6649) \\ & = 5889.9 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} & \text{Longitudinal Stress at Top of Shell (4.15.4) } [\sigma_1]: \\ & = P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\ & = 0 \cdot 789 / (2 \cdot 16) - 5890 / (\pi \cdot 789^2 \cdot 16) \\ & = -1.85 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} & \text{Longitudinal Stress at Bottom of Shell (4.15.5) } [\sigma_2]: \\ & = P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ & = 0 \cdot 789 / (2 \cdot 16) + 5890 / (\pi \cdot 789^2 \cdot 16) \\ & = 1.85 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} & \text{Longitudinal Stress at Top of Shell at Support (4.15.8) } [\sigma^*3]: \\ & = P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\ & = 0 \cdot 789 / (2 \cdot 18) - 3904 / (0.11 \cdot \pi \cdot 789^2 \cdot 18) \\ & = 10.20 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} & \text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) } [\sigma^*4]: \\ & = P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ & = 0 \cdot 789 / (2 \cdot 18) + 3904 / (0.19 \cdot \pi \cdot 789^2 \cdot 18) \\ & = -5.65 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} & \text{Maximum Shear Force in the Saddle (4.15.3) [T]:} \\ & = Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ & = 16873 (6649 - 2 \cdot 1360) / (6649 + (4 \cdot 0 / 3)) \\ & = 9970.5 \text{ Kg} \end{aligned}$$

$$\begin{aligned} & \text{Shear Stress in the shell no rings, not stiffened (4.15.12) } [\tau_2]: \\ & = K2 \cdot T / (Rm \cdot t) \\ & = 1.17 \cdot 9971 / (789 \cdot 18) \\ & = 8.06 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} & \text{Decay Length (4.15.20) } [x1, x2]: \\ & = 0.78 \cdot \sqrt{(Rm \cdot t)} \\ & = 0.78 \cdot \sqrt{(789 \cdot 18)} \\ & = 92.954 \text{ mm} \end{aligned}$$

$$\begin{aligned} & \text{Circumferential Stress in shell, no rings (4.15.21) } [\sigma_6]: \\ & = -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ & = -0.76 \cdot 16873 \cdot 0.1 / (18(325 + 93 + 93)) \\ & = -1.37 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} & \text{Circ. Comp. Stress at Horn of Saddle, } L \geq 8Rm \text{ (4.15.22) } [\sigma_7]: \\ & = -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\ & = -16873 / (4 \cdot 18(325 + 93 + 93)) - 3 \cdot 0.053 \cdot 16873 / (2 \cdot 18^2) \\ & = -44.99 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} & \text{Effective reinforcing plate width (4.15.24) } [B1]: \\ & = \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ & = \min(325 + 1.56 \cdot \sqrt{(789 \cdot 18)}, 2 \cdot 1360) \\ & = 510.91 \text{ mm} \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1300.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	350.0000	mm



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Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(780 \cdot 18)} + 500)^3) / 12$$

$$= 33.283 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 33.3 + 123 (88.7 - 9)^2$$

$$= 7862.062 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	684.8	18.0	9.0	123.3	110944.9	0.786E+04
Wearplate	500.0	10.0	23.0	50.0	115000.0	0.216E+04
Web	10.0	222.0	139.0	22.2	308580.2	0.147E+04
BasePlate	350.0	20.0	260.0	70.0	1820000.4	0.206E+05
Totals	...	...	...	265.5	2354525.5	0.321E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2354525 / 265$$

$$= 88.692 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 16873$$

$$= 3434.0227 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	2.3683	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	155.9972	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1050 - 780 \cdot \sin(120/2) / 1.05$$

$$= 404.945 \text{ mm}$$

Bending Moment, M = Fh * d	=	1390.6188	Kg-m.
----------------------------	---	-----------	-------

Bending Stress, Sb = (M * C1 / I)	=	3.7723	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	173.3302	N./mm ²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(16873 + 270) 350 / (4 \cdot 1300 \cdot 173)}$$



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Saddle Calcs: Road Transport: Step: 21 10:21am Sep 9,2026

= 13.995 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

= $2 \cdot \cos(90 - 120/2) (780 + 18 + 10)$

= 1399.497 mm

Distance between Ribs [e]:

= $\text{Web Length} / (\text{Nr ribs} - 1)$

= $1399 / (4 - 1)$

= 466.499 mm

Baseplate Pressure Area [Ap]:

= $e \cdot \text{Bpwid} / 2$

= $466 \cdot 350 / 2$

= 816.373 cm²

Bearing Pressure [Bp]:

= $(Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

= $17143 / (1300 \cdot 350)$

= 3.768 Kg/cm²

Axial Load [P]:

= $A_p \cdot B_p$

= $816 \cdot 3.77$

= 3075.924 Kg

Area of the Rib and Web [Ar]:

= $\text{Rib Area} + \text{Web Area}$

= $31.5 + 23.3$

= 54.825 cm²

Compressive Stress [Sc]:

= P / A_r

= $3076 / 54.8$

= 5.502 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	325.0	...	32.5	...	0.286E+04

Rib dimension [D]:

= $\text{Saddle Width} - \text{Web Thickness}$

= $325 - 10$

= 315.000 mm

Distance to Centroid from Datum [ytot]:

= A_Y / A

= $0 / 54.8$

= 0.000 mm

Distance to Centroid [C1]:

= $\text{Saddle Width} / 2$

= $325 / 2$

= 162.500 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(2861 / 54.8)}$



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Saddle Calcs: Road Transport: Step: 21 10:21am Sep 9,2026

= 72.235 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1050 - cos(120/2)(780 + 18 + 10) - 20

= 626.000 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(120/2)(780 + 18 + 10)

= 1399.497 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 1300 - 50.8

= 1249.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (1399 - 1249)/2

= 75.149 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$

= $\sqrt{(626^2 + 75.1^2)}$

= 630.495 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 630/72.2

= 8.728

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 3372 / (2 • 1300) • 466 • 630/2

= 190.743 Kg-m.

Compressive Allowable, KL/r < Cc (8.73 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (8.73)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (8.73) / (8 \cdot 123) - (8.73^3) / (8 \cdot 123^3))$

= 153.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba

= 5.5/153 + (191 • 162/28606770)/173

= 0.097

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	315.0	0.0	31.5	0.0	0.286E+04
Web	466.5	10.0	0.0	46.6	0.0	3.89
Totals	...	...	...	78.1	...	0.286E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/78.1



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Saddle Calcs: Road Transport: Step: 21 10:21am Sep 9, 2026

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 325/2

= 162.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1050 - $\sqrt{(808 + 10)^2 - (466/2)^2}$ - 20

= 256.399 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{2864/78.1}$

= 60.542 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 256/60.5

= 4.235

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 3372 / (2 • 1300)

= 1.297 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.3 • 466 • 256

= 155.136 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 155/176276

= 8.631 N./mm²

Compressive Allowable, KL/r < Cc (4.24 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (4.24)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (4.24) / (8 \cdot 123) - (4.24^3) / (8 \cdot 123^3))$

= 154.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 7.72/155 + (155 • 162/2864)/173

= 0.100

Review notes about nozzle loadings and supports in the Warnings report.

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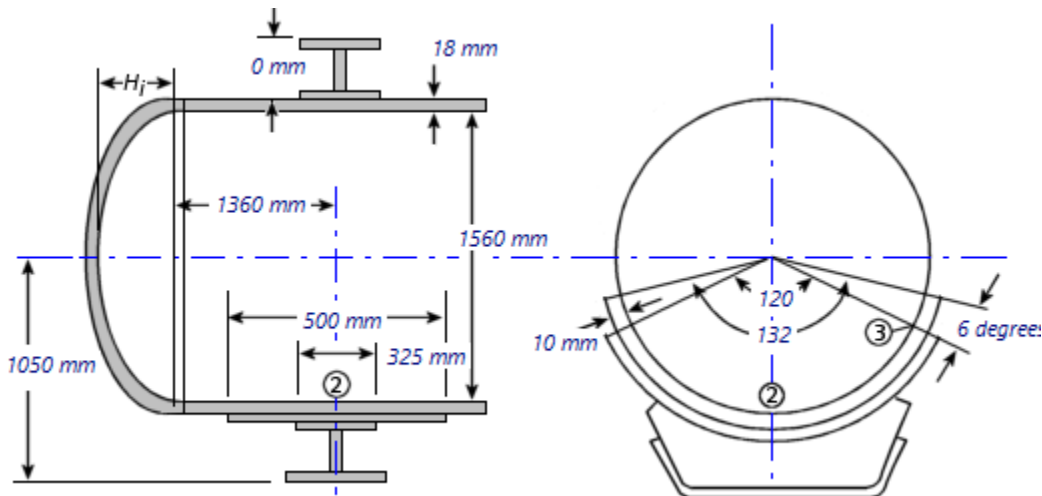
Saddle Calcs: Operating Case: Step: 22 10:21am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	790.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6649.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1360.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	500.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		34945.94	Kgf
Pressure used in Analysis	P	2193.889	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	63.00	138.00
Long. Stress at Bottom of Midspan	72.52	138.00
Long. Stress at Top of Saddles	84.17	138.00
Long. Stress at Bottom of Saddles	44.43	138.00
Tangential Shear in Shell	20.26	110.40
Circ. Stress at Horn of Saddle	51.17	172.50
Circ. Compressive Stress in Shell	2.13	138.00



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Saddle Calcs: Operating Case: Step: 22 10:21am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (909/2 + 0) \cdot 1050/1300$$

$$= 1101.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(313, 0) \cdot 1050/3880$$

$$= 84.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(3009, 0) \cdot 1050/3880$$

$$= 814.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (3009/2 + 0) \cdot 1050/1300$$

$$= 3645.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 31300 + \max(84.8, 1101, 814, 3646)$$

$$= 34945.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.8 \cdot 1.2)^2)$$

$$= 3074.221 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	35216.38	Kgf
Transverse Shear Load	1504.53	Kgf
Longitudinal Shear Load	3009.06	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1300.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -34946 \cdot 1360 [1 - (1 - 1360/6649 + (791^2 - 0^2)/(2 \cdot 1360 \cdot 6649))/(1 + (4 \cdot 0)/(3 \cdot 6649))]$$

$$= -8078.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 34946 \cdot 6649/4 ((1 + 2(791^2 - 0^2)/(6649^2))/(1 + (4 \cdot 0)/(3 \cdot 6649)) - 4 \cdot 1360/6649)$$



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Saddle Calcs: Operating Case: Step: 22 10:21am Sep 9, 2026

= 12205.2 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 2194 \cdot 791 / (2 \cdot 12.8) - 12205 / (\pi \cdot 791^2 \cdot 12.8)$$

$$= 63.00 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2194 \cdot 791 / (2 \cdot 12.8) + 12205 / (\pi \cdot 791^2 \cdot 12.8)$$

$$= 72.52 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 2194 \cdot 791 / (2 \cdot 14.8) - 8079 / (0.11 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 84.17 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2194 \cdot 791 / (2 \cdot 14.8) + 8079 / (0.19 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 44.43 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 34946 (6649 - 2 \cdot 1360) / (6649 + (4 \cdot 0 / 3))$$

$$= 20650.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 20650 / (791 \cdot 14.8)$$

$$= 20.26 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(791 \cdot 14.8)}$$

$$= 84.373 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(791 \cdot 14.8)}, 2 \cdot 1360)$$

$$= 493.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(138 / 138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 34946 \cdot 0.1 / (494(14.8 + 1 \cdot 10))$$

$$= -2.13 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -34946 / (4(14.8 + 1 \cdot 10)494) -$$

$$3 \cdot 0.053 \cdot 34946 / (2(14.8 + 1 \cdot 10)^2)$$

$$= -51.17 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3880.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12620\text{E-}04 \cdot 3880 (180 - 21.1)$$

$$= 7.780 \text{ mm}$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1300.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(783 \cdot 14.8)} + 500)^3) / 12$$

$$= 18.045 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 18 + 98.9 (94.2 - 7.4)^2$$

$$= 7469.539 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	668.0	14.8	7.4	98.9	73154.4	0.747E+04
Wearplate	500.0	10.0	19.8	50.0	99000.0	0.277E+04
Web	10.0	222.0	135.8	22.2	301476.2	0.130E+04
BasePlate	350.0	20.0	256.8	70.0	1797600.5	0.185E+05
Totals	...	...	...	241.1	2271231.0	0.301E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2271231 / 241$$

$$= 94.220 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 34946$$

$$= 7112.2563 \text{ Kgf}$$

$$\text{Tension Stress, St} = (Fh / A_s) = 4.9050 \text{ N./mm}^2$$

$$\text{Allowed Stress, Sa} = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1050 - 783 \cdot \sin(120 / 2) / 1.05$$

$$= 402.299 \text{ mm}$$

$$\text{Bending Moment, M} = Fh \cdot d = 2861.3088 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M * C_1 / I) = 8.7937 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{ (3(34946 + 270) 350 / (4 * 1300 * 175))}$
 $= 19.982 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (780 + 18 + 10)$
 $= 1399.497 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1399 / (4 - 1)$
 $= 466.499 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 466 * 350 / 2$
 $= 816.373 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 35216 / (1300 * 350)$
 $= 7.740 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 816 * 7.74$
 $= 6318.618 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 31.5 + 23.3$
 $= 54.825 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 6319 / 54.8$
 $= 11.302 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	325.0	...	32.5	...	0.286E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 325 - 10$
 $= 315.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= A_Y / A$
 $= 0 / 54.8$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$



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$$= 325/2$$

$$= 162.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2861/54.8}$$

$$= 72.235 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1050 - \cos(120/2) (780 + 18 + 10) - 20$$

$$= 626.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (780 + 18 + 10)$$

$$= 1399.497 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1300 - 50.8$$

$$= 1249.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (1399 - 1249) / 2$$

$$= 75.149 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{\text{Lv}^2 + \text{Dd}^2}$$

$$= \sqrt{626^2 + 75.1^2}$$

$$= 630.495 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199943008}{262}}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= \text{KL} / r$$

$$= 1 \cdot 630 / 72.2$$

$$= 8.728$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 3009 / (2 \cdot 1300) \cdot 466 \cdot 630 / 2$$

$$= 170.204 \text{ Kg-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($8.73 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (8.73)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (8.73) / (8 \cdot 123) - (8.73^3) / (8 \cdot 123^3))$$

$$= 154.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 11.3/154 + (170 \cdot 162/28606770) / 175$$

$$= 0.128$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	315.0	0.0	31.5	0.0	0.286E+04



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Web	466.5	10.0	0.0	46.6	0.0	3.89
Totals	...	...	...	78.1	...	0.286E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/78.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325/2$$

$$= 162.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1050 - \sqrt{(808 + 10)^2 - (466/2)^2} - 20$$

$$= 256.399 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2864/78.1)}$$

$$= 60.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 256/60.5$$

$$= 4.235$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3009 / (2 \cdot 1300)$$

$$= 1.157 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.16 \cdot 466 \cdot 256$$

$$= 138.431 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 138/176276$$

$$= 7.701 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.24 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.24)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (4.24) / (8 \cdot 123) - (4.24^3) / (8 \cdot 123^3))$$

$$= 155.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 15.9/156 + (138 \cdot 162/2864) / 175$$

$$= 0.146$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²



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Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load QO (Weight)	QO	31570.8 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	814.3 Kgf
Maximum Transverse Force	Ft	1504.5 Kgf
Maximum Longitudinal Force	Fl	3009.1 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 8699.8 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= max(Ff, Fl) / (Bolt Area • Number of Bolts)
= 8700 / (3.04 • 6)
= 46.7 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= Ft / (Bolt Area • Number of Bolts)
= 1505 / (3.04 • 6)
= 8.1 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= Fl / (BoltShearAllowable • Nbolts)
= 3009 / (82.8 • 6)
= 0.5940 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1050 • 1505 + 0
= 1579.79 Kg-m.

Eccentricity (e):
= Rmom / QO
= 1580 / 31571
= 50.04 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.59, 0]
= 0.5940 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10, c = 3.2 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.74, Temp. Reduction = 15 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -43 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius Rm 790.60 mm



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Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6649.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1360.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	500.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		25124.62	Kgf
Pressure used in Analysis	P	2192.517	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	64.29	138.00
Long. Stress at Bottom of Midspan	71.14	138.00
Long. Stress at Top of Saddles	76.95	138.00
Long. Stress at Bottom of Saddles	48.37	138.00
Tangential Shear in Shell	14.57	110.40
Circ. Stress at Horn of Saddle	36.79	172.50
Circ. Compressive Stress in Shell	1.53	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (909/2 + 0) \cdot 1050/1300$
 $= 1101.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(313, 0) \cdot 1050/3880$
 $= 84.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(3009, 0) \cdot 1050/3880$
 $= 814.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (3009/2 + 0) \cdot 1050/1300$
 $= 3645.6 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 21479 + \max(84.8, 1101, 814, 3646)$
 $= 25124.6 \text{ Kgf}$



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Longitudinal Wind Force [F]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.8 \cdot 1.2)^2)$$

$$= 3074.221 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25395.06	Kgf
Transverse Shear Load	1504.53	Kgf
Longitudinal Shear Load	3009.06	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1300.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -25125 \cdot 1360 [1 - (1 - 1360/6649 + (791^2 - 0^2)/(2 \cdot 1360 \cdot 6649)) / (1 + (4 \cdot 0)/(3 \cdot 6649))]$$

$$= -5808.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 25125 \cdot 6649/4 ((1 + 2(791^2 - 0^2)/(6649^2)) / (1 + (4 \cdot 0)/(3 \cdot 6649)) - 4 \cdot 1360/6649)$$

$$= 8775.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 2193 \cdot 791/(2 \cdot 12.8) - 8775/(\pi \cdot 791^2 \cdot 12.8)$$

$$= 64.29 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 2193 \cdot 791/(2 \cdot 12.8) + 8775/(\pi \cdot 791^2 \cdot 12.8)$$

$$= 71.14 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 2193 \cdot 791/(2 \cdot 14.8) - 5808/(0.11 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 76.95 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2193 \cdot 791/(2 \cdot 14.8) + 5808/(0.19 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 48.37 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4h_m/3))$$

$$= 25125 (6649 - 2 \cdot 1360) / (6649 + (4 \cdot 0/3))$$

$$= 14846.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$



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$$= 1.17 \cdot 14847 / (791 \cdot 14.8)$$

$$= 14.57 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(791 \cdot 14.8)}$$

$$= 84.373 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(791 \cdot 14.8)}, 2 \cdot 1360)$$

$$= 493.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 25125 \cdot 0.1 / (494(14.8 + 1 \cdot 10))$$

$$= -1.53 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ7,r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -25125 / (4(14.8 + 1 \cdot 10)494) -$$

$$3 \cdot 0.053 \cdot 25125 / (2(14.8 + 1 \cdot 10)^2)$$

$$= -36.79 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1300.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(783 \cdot 14.8)} + 500)14.8^3) / 12$$

$$= 18.045 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 18 + 98.9(94.2 - 7.4)^2$$

$$= 7469.539 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	668.0	14.8	7.4	98.9	73154.4	0.747E+04
Wearplate	500.0	10.0	19.8	50.0	99000.0	0.277E+04
Web	10.0	222.0	135.8	22.2	301476.2	0.130E+04
BasePlate	350.0	20.0	256.8	70.0	1797600.5	0.185E+05
Totals	...	...	...	241.1	2271231.0	0.301E+05

Distance to Centroid [C1]:



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Saddle Calcs: Operating Case: Step: 22 10:21am Sep 9, 2026

$$= AY / A$$

$$= 2271231/241$$

$$= 94.220 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 25125$$

$$= 5113.4053 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 3.5265 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1050 - 783 \cdot \sin(120/2) / 1.05$$

$$= 402.299 \text{ mm}$$

Bending Moment, M = Fh * d = 2057.1577 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 6.3223 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(25125 + 270)350 / (4 \cdot 1300 \cdot 175)}$$

$$= 16.968 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (780 + 18 + 10)$$

$$= 1399.497 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1399 / (4 - 1)$$

$$= 466.499 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 466 \cdot 350/2$$

$$= 816.373 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 25395 / (1300 \cdot 350)$$

$$= 5.581 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 816 \cdot 5.58$$

$$= 4556.451 \text{ Kgf}$$

Area of the Rib and Web [Ar]:



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Saddle Calcs: Operating Case: Step: 22 10:21am Sep 9,2026

= Rib Area + Web Area
= 31.5 + 23.3
= 54.825 cm²

Compressive Stress [Sc]:
= P/Ar
= 4556/54.8
= 8.150 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	325.0	...	32.5	...	0.286E+04

Rib dimension [D]:
= Saddle Width - Web Thickness
= 325 - 10
= 315.000 mm

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/54.8
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 325/2
= 162.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{2861/54.8}$
= 72.235 mm

Length of Outer Rib, step 1, [Lv]:
= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
= 1050 - cos(120/2)(780 + 18 + 10) - 20
= 626.000 mm

Length of Outer Rib, step 2, [Lw]:
= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
= 2 • sin(120/2)(780 + 18 + 10)
= 1399.497 mm

Length of Outer Rib, step 3, [Bd]:
= Baseplate length - clearance
= 1300 - 50.8
= 1249.200 mm

Length of Outer Rib, step 4, [Dd]:
= (Lw - Bd)/2
= (1399 - 1249)/2
= 75.149 mm

Length of Outer Rib [L]:
= $\sqrt{Lv^2 + Dd^2}$
= $\sqrt{626^2 + 75.1^2}$
= 630.495 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/262}$



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Saddle Calcs: Operating Case: Step: 22 10:21am Sep 9,2026

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 630/72.2$$

$$= 8.728$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 3009 / (2 \cdot 1300) \cdot 466 \cdot 630 / 2$$

$$= 170.204 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($8.73 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.73)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (8.73) / (8 \cdot 123) - (8.73^3) / (8 \cdot 123^3))$$

$$= 154.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.15 / 154 + (170 \cdot 162 / 28606770) / 175$$

$$= 0.107$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	315.0	0.0	31.5	0.0	0.286E+04
Web	466.5	10.0	0.0	46.6	0.0	3.89
Totals	...	...	...	78.1	...	0.286E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/78.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325/2$$

$$= 162.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1050 - \sqrt{ (808 + 10)^2 - (466/2)^2 } - 20$$

$$= 256.399 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 2864 / 78.1 }$$

$$= 60.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 256/60.5$$

$$= 4.235$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3009 / (2 \cdot 1300)$$

$$= 1.157 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.16 \cdot 466 \cdot 256$$



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Saddle Calcs: Operating Case: Step: 22 10:21am Sep 9,2026

= 138.431 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 138/176276

= 7.701 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.24 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c^2)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c^3)

= (1-(4.24)²/(2 • 123²))262/

(5/3+3•(4.24)/(8• 123)-(4.24³)/(8•123³)

= 155.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 11.4/156 + (138 • 162/2864)/175

= 0.117

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	21749.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	814.3 Kg
Maximum Transverse Force	Ft	1504.5 Kg
Maximum Longitudinal Force	F1	3009.1 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 3009/(3.04 • 6)

= 16.2 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 1505/(3.04 • 6)

= 8.1 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= 3009/(82.8 • 6)

= 0.5940 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= 1050 • 1505 + 0

= 1579.79 Kg-m.

Eccentricity (e):

= $Rmom / Q0$



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Saddle Calcs: Operating Case: Step: 22 10:21am Sep 9, 2026

= 1580/21749

= 72.63 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.59, 0]

= 0.5940 cm²

MDMT Calculations for: Sdl 2 Fr40, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10, c = 3.2 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.74, Temp. Reduction = 15 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -43 °C

[Review notes about nozzle loadings and supports in the Warnings report.](#)

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Saddle Calcs: Test Case:

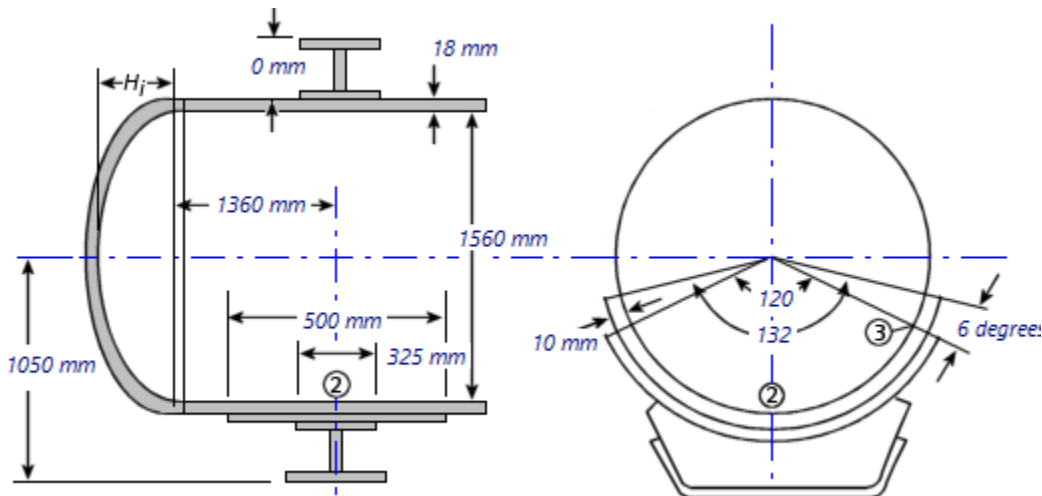
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	790.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6649.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1360.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	500.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		29500.85	Kgf
Pressure used in Analysis	P	2238.710	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	65.12	234.00
Long. Stress at Bottom of Midspan	73.16	234.00
Long. Stress at Top of Saddles	81.38	234.00
Long. Stress at Bottom of Saddles	47.83	234.00
Tangential Shear in Shell	17.10	138.58
Circ. Stress at Horn of Saddle	43.20	350.99
Circ. Compressive Stress in Shell	1.80	234.00



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Saddle Calcs: Test Case:

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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (454/2 + 0) \cdot 1050/1300$$

$$= 550.6 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(157, 0) \cdot 1050/3880$$

$$= 42.4 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 28950 + \max(42.4, 551, 0, 0)$$

$$= 29500.8 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(1.8 \cdot 1.2)^2)$$

$$= 3074.221 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	29771.29	Kgf
Transverse Shear Load	227.25	Kgf
Longitudinal Shear Load	156.74	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1300.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -29501 \cdot 1360 [1 - (1 - 1360/6649 + (791^2 - 0^2)/(2 \cdot 1360 \cdot 6649)) / (1 + (4 \cdot 0)/(3 \cdot 6649))]$$

$$= -6820.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 29501 \cdot 6649/4 ((1 + 2(791^2 - 0^2)/(6649^2)) / (1 + (4 \cdot 0)/(3 \cdot 6649)) - 4 \cdot 1360/6649)$$

$$= 10303.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 2239 \cdot 791/(2 \cdot 12.8) - 10303/(\pi \cdot 791^2 \cdot 12.8)$$

$$= 65.12 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 2239 \cdot 791/(2 \cdot 12.8) + 10303/(\pi \cdot 791^2 \cdot 12.8)$$



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Saddle Calcs: Test Case:

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$$= 73.16 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 2239 \cdot 791 / (2 \cdot 14.8) - 6820 / (0.11 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 81.38 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 2239 \cdot 791 / (2 \cdot 14.8) + 6820 / (0.19 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 47.83 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 29501 (6649 - 2 \cdot 1360) / (6649 + (4 \cdot 0/3))$$

$$= 17432.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 17433 / (791 \cdot 14.8)$$

$$= 17.10 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(791 \cdot 14.8)}$$

$$= 84.373 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(791 \cdot 14.8)}, 2 \cdot 1360)$$

$$= 493.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 29501 \cdot 0.1 / (494(14.8 + 1 \cdot 10))$$

$$= -1.80 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -29501 / (4(14.8 + 1 \cdot 10)494) -$$

$$3 \cdot 0.053 \cdot 29501 / (2(14.8 + 1 \cdot 10)^2)$$

$$= -43.20 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1300.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(783 \cdot 14.8)} + 500)^3) / 12$$

$$= 18.045 \text{ cm}^4$$



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Saddle Calcs: Test Case:

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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 18 + 98.9 (94.2 - 7.4)^2$$

$$= 7469.539 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	668.0	14.8	7.4	98.9	73154.4	0.747E+04
Wearplate	500.0	10.0	19.8	50.0	99000.0	0.277E+04
Web	10.0	222.0	135.8	22.2	301476.2	0.130E+04
BasePlate	350.0	20.0	256.8	70.0	1797600.5	0.185E+05
Totals	...	...	...	241.1	2271231.0	0.301E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2271231/241$$

$$= 94.220 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 29501$$

$$= 6004.0620 \text{ Kg}$$

$$\text{Tension Stress, } S_t = (F_h / A_s) = 4.1407 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1050 - 783 \cdot \sin(120/2) / 1.05$$

$$= 402.299 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 2415.4749 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C1 / I) = 7.4235 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(29501 + 270) \cdot 350) / (4 \cdot 1300 \cdot 173)}$$

$$= 18.442 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (780 + 18 + 10)$$

$$= 1399.497 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1399 / (4 - 1)$$

$$= 466.499 \text{ mm}$$

Baseplate Pressure Area [Ap]:



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$$= e \cdot Bpwid / 2$$

$$= 466 \cdot 350 / 2$$

$$= 816.373 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 29771 / (1300 \cdot 350)$$

$$= 6.543 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 816 \cdot 6.54$$

$$= 5341.645 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 31.5 + 23.3$$

$$= 54.825 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 5342 / 54.8$$

$$= 9.555 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	325.0	...	32.5	...	0.286E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 325 - 10$$

$$= 315.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 54.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (2861 / 54.8) }$$

$$= 72.235 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1050 - \cos(120 / 2) (780 + 18 + 10) - 20$$

$$= 626.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2) (radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(120 / 2) (780 + 18 + 10)$$

$$= 1399.497 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 1300 - 50.8$$



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Saddle Calcs: Test Case:

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$$= 1249.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (1399 - 1249) / 2$$

$$= 75.149 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(626^2 + 75.1^2)}$$

$$= 630.495 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 630 / 72.2$$

$$= 8.728$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 157 / (2 \cdot 1300) \cdot 466 \cdot 630 / 2$$

$$= 8.866 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($8.73 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.73)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (8.73) / (8 \cdot 123) - (8.73^3) / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 9.55 / 153 + (8.87 \cdot 162 / 28606770) / 173$$

$$= 0.065$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	315.0	0.0	31.5	0.0	0.286E+04
Web	466.5	10.0	0.0	46.6	0.0	3.89
Totals	...	...	...	78.1	...	0.286E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 78.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 1050 - \sqrt{(808 + 10)^2 - (466/2)^2} - 20$$

$$= 256.399 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2864 / 78.1)}$$



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Saddle Calcs: Test Case:

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= 60.542 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 256/60.5

= 4.235

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 157/(2 • 1300)

= 0.060 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.06 • 466 • 256

= 7.211 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 7.21/176276

= 0.401 N./mm²

Compressive Allowable, KL/r < Cc (4.24 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³)/(8•Cc³)

= (1-(4.24)²/(2 • 123²))260/

(5/3+3•(4.24)/(8• 123)-(4.24³)/(8•123³)

= 154.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 13.4/155 + (7.21 • 162/2864)/173

= 0.089

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	29220.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	42.4 Kg
Maximum Transverse Force	Ft	227.2 Kg
Maximum Longitudinal Force	Fl	156.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)

= 157/(3.04 • 6)

= 0.8 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)

= 227/(3.04 • 6)

= 1.2 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)



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Saddle Calcs: Test Case:

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Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
 $= 157 / (29 \cdot 6)$
 $= 0.0884 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1050 \cdot 227 + 0$
 $= 238.62 \text{ Kg-m.}$

Eccentricity (e):
 $= R_{\text{mom}} / QO$
 $= 239 / 29221$
 $= 8.17 \text{ mm} < B_{\text{plen}} / 6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 0.088, 0]$
 $= 0.0884 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	790.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6649.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1360.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	500.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		21434.13	Kgf
Pressure used in Analysis	P	2238.710	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		66.22	234.00
Long. Stress at Bottom of Midspan		72.06	234.00
Long. Stress at Top of Saddles		75.48	234.00
Long. Stress at Bottom of Saddles		51.11	234.00
Tangential Shear in Shell		12.43	138.58



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Saddle Calcs: Test Case:

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Circ. Stress at Horn of Saddle	31.39	350.99
Circ. Compressive Stress in Shell	1.31	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (454/2 + 0) \cdot 1050/1300$$

$$= 550.6 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(157, 0) \cdot 1050/3880$$

$$= 42.4 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 20883 + \max(42.4, 551, 0, 0)$$

$$= 21434.1 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.8 \cdot 1.2)^2)$$

$$= 3074.221 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	21704.57	Kgf
Transverse Shear Load	227.25	Kgf
Longitudinal Shear Load	156.74	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1300.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -21434 \cdot 1360 [1 - (1 - 1360/6649 + (791^2 - 0^2)/$$

$$(2 \cdot 1360 \cdot 6649)) / (1 + (4 \cdot 0)/(3 \cdot 6649))]$$

$$= -4955.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 21434 \cdot 6649/4 ((1 + 2(791^2 - 0^2)/(6649^2)) / (1 + (4 \cdot 0)/$$

$$(3 \cdot 6649)) - 4 \cdot 1360/6649)$$

$$= 7486.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 2239 \cdot 791/(2 \cdot 12.8) - 7486/(\pi \cdot 791^2 \cdot 12.8)$$

$$= 66.22 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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Saddle Calcs: Test Case:

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$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2239 \cdot 791 / (2 \cdot 12.8) + 7486 / (\pi \cdot 791^2 \cdot 12.8)$$

$$= 72.06 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2239 \cdot 791 / (2 \cdot 14.8) - 4955 / (0.11 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 75.48 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2239 \cdot 791 / (2 \cdot 14.8) + 4955 / (0.19 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 51.11 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 21434 (6649 - 2 \cdot 1360) / (6649 + (4 \cdot 0 / 3))$$

$$= 12665.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 12666 / (791 \cdot 14.8)$$

$$= 12.43 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{791 \cdot 14.8}$$

$$= 84.373 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{791 \cdot 14.8}, 2 \cdot 1360)$$

$$= 493.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 21434 \cdot 0.1 / (494(14.8 + 1 \cdot 10))$$

$$= -1.31 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -21434 / (4(14.8 + 1 \cdot 10)494) -$$

$$3 \cdot 0.053 \cdot 21434 / (2(14.8 + 1 \cdot 10)^2)$$

$$= -31.39 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1300.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$



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Saddle Calcs: Test Case:

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$$= ((1.56 \cdot \sqrt{(783 \cdot 14.8)} + 500) 14.8^3) / 12$$

$$= 18.045 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 18 + 98.9 (94.2 - 7.4)^2$$

$$= 7469.539 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	668.0	14.8	7.4	98.9	73154.4	0.747E+04
Wearplate	500.0	10.0	19.8	50.0	99000.0	0.277E+04
Web	10.0	222.0	135.8	22.2	301476.2	0.130E+04
BasePlate	350.0	20.0	256.8	70.0	1797600.5	0.185E+05
Totals	...	...	...	241.1	2271231.0	0.301E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2271231 / 241$$

$$= 94.220 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 21434$$

$$= 4362.3096 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.0085 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1050 - 783 \cdot \sin(120/2) / 1.05$$

$$= 402.299 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1754.9868 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.3936 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(21434 + 270) 350) / (4 \cdot 1300 \cdot 173)}$$

$$= 15.747 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (780 + 18 + 10)$$

$$= 1399.497 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1399 / (4 - 1)$$



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Saddle Calcs: Test Case:

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= 466.499 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 466 • 350/2

= 816.373 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)

= 21705/(1300 • 350)

= 4.770 Kg/cm²

Axial Load [P]:

= Ap • Bp

= 816 • 4.77

= 3894.292 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 31.5 + 23.3

= 54.825 cm²

Compressive Stress [Sc]:

= P/Ar

= 3894/54.8

= 6.966 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	325.0	...	32.5	...	0.286E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 325 - 10

= 315.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/54.8

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 325/2

= 162.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{2861/54.8}$

= 72.235 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1050 - cos(120/2)(780 + 18 + 10) - 20

= 626.000 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(120/2)(780 + 18 + 10)

= 1399.497 mm



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Saddle Calcs: Test Case:

Step: 23 10:21am Sep 9, 2026

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1300 - 50.8$$

$$= 1249.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1399 - 1249) / 2$$

$$= 75.149 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(626^2 + 75.1^2)}$$

$$= 630.495 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 630 / 72.2$$

$$= 8.728$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 157 / (2 \cdot 1300) \cdot 466 \cdot 630 / 2$$

$$= 8.866 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($8.73 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (8.73)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (8.73) / (8 \cdot 123) - (8.73^3) / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.97 / 153 + (8.87 \cdot 162 / 28606770) / 173$$

$$= 0.048$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	315.0	0.0	31.5	0.0	0.286E+04
Web	466.5	10.0	0.0	46.6	0.0	3.89
Totals	...	...	...	78.1	...	0.286E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 78.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1050 - \sqrt{(808 + 10)^2 - (466/2)^2} - 20$$

$$= 256.399 \text{ mm}$$



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Saddle Calcs: Test Case:

Step: 23 10:21am Sep 9, 2026

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{2864 / 78.1}$
 $= 60.542 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 256 / 60.5$
 $= 4.235$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 157 / (2 \cdot 1300)$
 $= 0.060 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.06 \cdot 466 \cdot 256$
 $= 7.211 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 7.21 / 176276$
 $= 0.401 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($4.24 < 123$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (4.24)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (4.24) / (8 \cdot 123) - (4.24^3) / (8 \cdot 123^3))$
 $= 154.7 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 9.78 / 155 + (7.21 \cdot 162 / 2864) / 173$
 $= 0.066$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	21153.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	42.4 Kg
Maximum Transverse Force	Ft	227.2 Kg
Maximum Longitudinal Force	F1	156.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 157 / (3.04 \cdot 6)$
 $= 0.8 \text{ N./mm}^2$. Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 227 / (3.04 \cdot 6)$
 $= 1.2 \text{ N./mm}^2$. Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss



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Saddle Calcs: Test Case:

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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
= $157 / (29 \cdot 6)$
= 0.0884 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1050 \cdot 227 + 0$
= 238.62 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $239 / 21154$
= 11.28 mm < $B_{\text{plen}}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 0.088, 0]$
= 0.0884 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Ext. Pressure:

Step: 24 10:21am

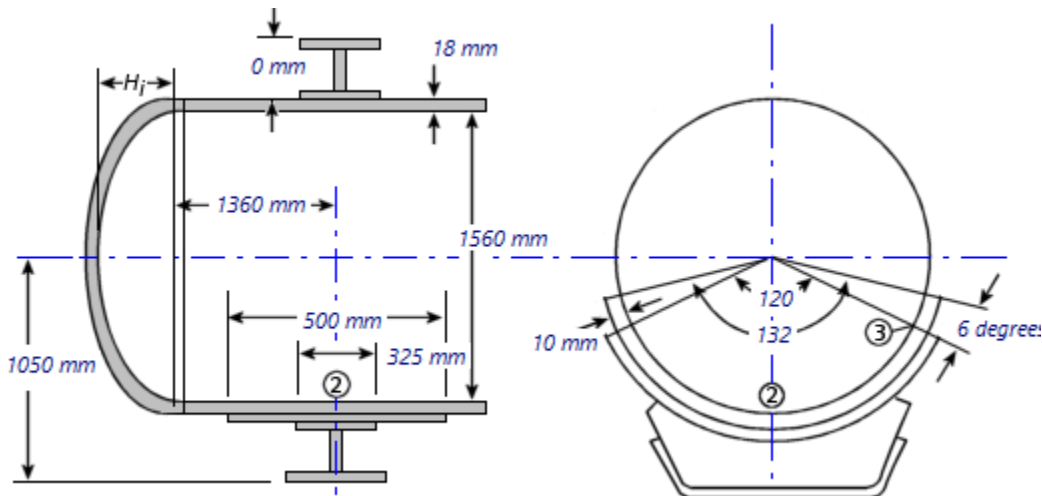
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	790.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6649.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1360.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	500.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		34945.94	Kgf
Pressure used in Analysis	P	-101.130	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-7.89	-204.66
Long. Stress at Top of Midspan		7.89	138.00
Long. Stress at Bottom of Midspan		1.64	138.00
Long. Stress at Top of Saddles		22.87	138.00
Long. Stress at Bottom of Saddles		-16.87	-204.66
Long. Stress at Bottom of Saddles		16.87	138.00
Tangential Shear in Shell		20.26	110.40



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9, 2026

Circ. Stress at Horn of Saddle	51.17	172.50
Circ. Compressive Stress in Shell	2.13	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (909/2 + 0) \cdot 1050/1300$$

$$= 1101.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(313, 0) \cdot 1050/3880$$

$$= 84.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl, eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(3009, 0) \cdot 1050/3880$$

$$= 814.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (3009/2 + 0) \cdot 1050/1300$$

$$= 3645.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 31300 + \max(84.8, 1101, 814, 3646)$$

$$= 34945.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.8 \cdot 1.2)^2)$$

$$= 3074.221 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	35216.38	Kgf
Transverse Shear Load	1504.53	Kgf
Longitudinal Shear Load	3009.06	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1300.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -34946 \cdot 1360 [1 - (1 - 1360/6649 + (791^2 - 0^2) / (2 \cdot 1360 \cdot 6649)) / (1 + (4 \cdot 0) / (3 \cdot 6649))]$$

$$= -8078.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$



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$$= 34946 \cdot 6649 / 4 \cdot ((1 + 2(791^2 - 0^2)) / (6649^2)) / (1 + (4 \cdot 0) / (3 \cdot 6649)) - 4 \cdot 1360 / 6649$$

$$= 12205.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 791 / (2 \cdot 12.8) - 12205 / (\pi \cdot 791^2 \cdot 12.8)$$

$$= -7.89 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 791 / (2 \cdot 12.8) + 12205 / (\pi \cdot 791^2 \cdot 12.8)$$

$$= 1.64 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 791 / (2 \cdot 14.8) - 8079 / (0.11 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= 22.87 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 791 / (2 \cdot 14.8) + 8079 / (0.19 \cdot \pi \cdot 791^2 \cdot 14.8)$$

$$= -16.87 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 34946 (6649 - 2 \cdot 1360) / (6649 + (4 \cdot 0 / 3))$$

$$= 20650.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 20650 / (791 \cdot 14.8)$$

$$= 20.26 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(791 \cdot 14.8)}$$

$$= 84.373 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(791 \cdot 14.8)}, 2 \cdot 1360)$$

$$= 493.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(138 / 138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.76 \cdot 34946 \cdot 0.1 / (494(14.8 + 1 \cdot 10))$$

$$= -2.13 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot t_r)^2)$$

$$= -34946 / (4(14.8 + 1 \cdot 10) 494) -$$

$$3 \cdot 0.053 \cdot 34946 / (2(14.8 + 1 \cdot 10)^2)$$

$$= -51.17 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1300.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9, 2026

Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw, c	370.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(783 \cdot 14.8) + 500}) 14.8^3) / 12$$

$$= 18.045 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 18 + 98.9 (94.2 - 7.4)^2$$

$$= 7469.539 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	668.0	14.8	7.4	98.9	73154.4	0.747E+04
Wearplate	500.0	10.0	19.8	50.0	99000.0	0.277E+04
Web	10.0	222.0	135.8	22.2	301476.2	0.130E+04
BasePlate	350.0	20.0	256.8	70.0	1797600.5	0.185E+05
Totals	...	...	...	241.1	2271231.0	0.301E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2271231 / 241$$

$$= 94.220 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 34946$$

$$= 7112.2563 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	4.9050	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	157.1972	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1050 - 783 \cdot \sin(120/2) / 1.05$$

$$= 402.299 \text{ mm}$$

Bending Moment, M = Fh * d	=	2861.3088	Kg-m.
----------------------------	---	-----------	-------

Bending Stress, Sb = (M * C1 / I)	=	8.7937	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	174.6635	N./mm ²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(34946 + 270) 350 / (4 \cdot 1300 \cdot 175)}$$



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9,2026

= 19.982 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

= $2 \cdot \cos(90 - 120/2) (780 + 18 + 10)$

= 1399.497 mm

Distance between Ribs [e]:

= $\text{Web Length} / (\text{Nr ribs} - 1)$

= $1399 / (4 - 1)$

= 466.499 mm

Baseplate Pressure Area [Ap]:

= $e \cdot \text{Bpwid} / 2$

= $466 \cdot 350 / 2$

= 816.373 cm²

Bearing Pressure [Bp]:

= $(Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

= $35216 / (1300 \cdot 350)$

= 7.740 Kg/cm²

Axial Load [P]:

= $A_p \cdot B_p$

= $816 \cdot 7.74$

= 6318.618 Kg

Area of the Rib and Web [Ar]:

= $\text{Rib Area} + \text{Web Area}$

= $31.5 + 23.3$

= 54.825 cm²

Compressive Stress [Sc]:

= P / A_r

= $6319 / 54.8$

= 11.302 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	325.0	...	32.5	...	0.286E+04

Rib dimension [D]:

= $\text{Saddle Width} - \text{Web Thickness}$

= $325 - 10$

= 315.000 mm

Distance to Centroid from Datum [ytot]:

= A_Y / A

= $0 / 54.8$

= 0.000 mm

Distance to Centroid [C1]:

= $\text{Saddle Width} / 2$

= $325 / 2$

= 162.500 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(2861 / 54.8)}$



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9,2026

= 72.235 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1050 - cos(120/2)(780 + 18 + 10) - 20

= 626.000 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(120/2)(780 + 18 + 10)

= 1399.497 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 1300 - 50.8

= 1249.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (1399 - 1249)/2

= 75.149 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$

= $\sqrt{(626^2 + 75.1^2)}$

= 630.495 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008 / 262)}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 630/72.2

= 8.728

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 3009 / (2 • 1300) • 466 • 630/2

= 170.204 Kg-m.

Compressive Allowable, KL/r < Cc (8.73 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (8.73)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (8.73) / (8 \cdot 123) - (8.73^3) / (8 \cdot 123^3))$

= 154.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba

= 11.3/154 + (170 • 162/28606770)/175

= 0.128

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	315.0	0.0	31.5	0.0	0.286E+04
Web	466.5	10.0	0.0	46.6	0.0	3.89
Totals	...	...	...	78.1	...	0.286E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/78.1



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9, 2026

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 325/2

= 162.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1050 - $\sqrt{(808 + 10)^2 - (466/2)^2}$ - 20

= 256.399 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{2864/78.1}$

= 60.542 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 256/60.5

= 4.235

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 3009 / (2 • 1300)

= 1.157 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.16 • 466 • 256

= 138.431 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 138/176276

= 7.701 N./mm²

Compressive Allowable, KL/r < Cc (4.24 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (4.24)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (4.24) / (8 \cdot 123) - (4.24^3) / (8 \cdot 123^3))$

= 155.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 15.9/156 + (138 • 162/2864)/175

= 0.146

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	790.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6649.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1360.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	500.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9, 2026

Shell Allowable Stress used in Calculation	138.00	N./mm ²
Head Allowable Stress used in Calculation	138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	
Distance from Saddle Base to Centerline	B	1050.00 mm
Coefficient of Friction	mu	0.40

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	25124.62	Kgf
Pressure used in Analysis	P	-101.130 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-6.55	-204.66
Long. Stress at Top of Midspan	6.55	138.00
Long. Stress at Bottom of Midspan	0.30	138.00
Long. Stress at Top of Saddles	15.68	138.00
Long. Stress at Bottom of Saddles	-12.89	-204.66
Long. Stress at Bottom of Saddles	12.89	138.00
Tangential Shear in Shell	14.57	110.40
Circ. Stress at Horn of Saddle	36.79	172.50
Circ. Compressive Stress in Shell	1.53	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (909/2 + 0) \cdot 1050/1300$
 $= 1101.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(313, 0) \cdot 1050/3880$
 $= 84.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake F_l [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(3009, 0) \cdot 1050/3880$
 $= 814.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake F_t [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (3009/2 + 0) \cdot 1050/1300$
 $= 3645.6 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 21479 + \max(84.8, 1101, 814, 3646)$
 $= 25124.6 \text{ Kgf}$

Longitudinal Wind Force [F_l]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (1.8 \cdot 1.2)^2)$
 $= 3074.221 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25395.06	Kgf
Transverse Shear Load	1504.53	Kgf
Longitudinal Shear Load	3009.06	Kgf



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9, 2026

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length

= 1300.000 mm

The Computed K values from Table 4.15.1:

$K1 = 0.1066$ $K2 = 1.1707$ $K3 = 0.8799$ $K4 = 0.4011$

$K5 = 0.7603$ $K6 = 0.0529$ $K7 = 0.0529$ $K8 = 0.3405$

$K9 = 0.2711$ $K10 = 0.0581$ $K1^* = 0.1923$ $K6p = 0.0434$

$K7p = 0.0434$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$

$= -25125 \cdot 1360 [1 - (1 - 1360/6649 + (791^2 - 0^2) /$

$(2 \cdot 1360 \cdot 6649)) / (1 + (4 \cdot 0) / (3 \cdot 6649))]$

$= -5808.3 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:

$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$

$= 25125 \cdot 6649 / 4 ((1 + 2(791^2 - 0^2)/(6649^2)) / (1 + (4 \cdot 0) /$

$(3 \cdot 6649)) - 4 \cdot 1360 / 6649)$

$= 8775.0 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$

$= -101 \cdot 791 / (2 \cdot 12.8) - 8775 / (\pi \cdot 791^2 \cdot 12.8)$

$= -6.55 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$

$= -101 \cdot 791 / (2 \cdot 12.8) + 8775 / (\pi \cdot 791^2 \cdot 12.8)$

$= 0.30 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$

$= -101 \cdot 791 / (2 \cdot 14.8) - 5808 / (0.11 \cdot \pi \cdot 791^2 \cdot 14.8)$

$= 15.68 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$

$= -101 \cdot 791 / (2 \cdot 14.8) + 5808 / (0.19 \cdot \pi \cdot 791^2 \cdot 14.8)$

$= -12.89 \text{ N./mm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$= Q(L - 2a) / (L + (4 \cdot hm/3))$

$= 25125 (6649 - 2 \cdot 1360) / (6649 + (4 \cdot 0/3))$

$= 14846.5 \text{ Kg}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$= K2 \cdot T / (Rm \cdot t)$

$= 1.17 \cdot 14847 / (791 \cdot 14.8)$

$= 14.57 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:

$= 0.78 \cdot \sqrt{(Rm \cdot t)}$

$= 0.78 \cdot \sqrt{(791 \cdot 14.8)}$

$= 84.373 \text{ mm}$

Effective reinforcing plate width (4.15.24) [B1]:

$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9,2026

$$= \min(325 + 1.56 \cdot \sqrt{791 \cdot 14.8}, 2 \cdot 1360)$$

$$= 493.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 25125 \cdot 0.1 / (494(14.8 + 1 \cdot 10))$$

$$= -1.53 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -25125 / (4(14.8 + 1 \cdot 10)494) -$$

$$3 \cdot 0.053 \cdot 25125 / (2(14.8 + 1 \cdot 10)^2)$$

$$= -36.79 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1300.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw, c	370.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{783 \cdot 14.8} + 500)14.8^3) / 12$$

$$= 18.045 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 18 + 98.9(94.2 - 7.4)^2$$

$$= 7469.539 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	668.0	14.8	7.4	98.9	73154.4	0.747E+04
Wearplate	500.0	10.0	19.8	50.0	99000.0	0.277E+04
Web	10.0	222.0	135.8	22.2	301476.2	0.130E+04
BasePlate	350.0	20.0	256.8	70.0	1797600.5	0.185E+05
Totals	...	...	...	241.1	2271231.0	0.301E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2271231/241$$

$$= 94.220 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:



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$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 25125$$

$$= 5113.4053 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.5265 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1050 - 783 \cdot \sin(120/2) / 1.05$$

$$= 402.299 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2057.1577 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 6.3223 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(25125 + 270) \cdot 350) / (4 \cdot 1300 \cdot 175)}$$

$$= 16.968 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (780 + 18 + 10)$$

$$= 1399.497 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1399 / (4 - 1)$$

$$= 466.499 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 466 \cdot 350 / 2$$

$$= 816.373 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 25395 / (1300 \cdot 350)$$

$$= 5.581 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 816 \cdot 5.58$$

$$= 4556.451 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 31.5 + 23.3$$

$$= 54.825 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 4556 / 54.8$$

$$= 8.150 \text{ N./mm}^2$$

Check of Outside Ribs:



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9, 2026

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	325.0	...	32.5	...	0.286E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 325 - 10

= 315.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/54.8

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 325/2

= 162.500 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(2861/54.8)}$

= 72.235 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - $\cos(\text{saddle bearing angle} / 2) \times (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$

= 1050 - $\cos(120/2) \times (780 + 18 + 10) - 20$

= 626.000 mm

Length of Outer Rib, step 2, [Lw]:

= $2 \times \sin(\text{saddle bearing angle} / 2) \times (\text{radius} + \text{shlthk} + \text{wpdthk})$

= $2 \times \sin(120/2) \times (780 + 18 + 10)$

= 1399.497 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 1300 - 50.8

= 1249.200 mm

Length of Outer Rib, step 4, [Dd]:

= $(Lw - Bd) / 2$

= $(1399 - 1249) / 2$

= 75.149 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{626^2 + 75.1^2}$

= 630.495 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \times \pi^2 \times \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \times \pi^2 \times 199943008/262)}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= $1 \times 630/72.2$

= 8.728

Bending Moment [Rm]:

= $Fl / (2 \times Bplen) \times e \times L / 2$

= $3009 / (2 \times 1300) \times 466 \times 630/2$



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9,2026

= 170.204 Kg-m.

Compressive Allowable, $KL/r < C_c$ (8.73 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (8.73)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (8.73) / (8 \cdot 123) - (8.73^3) / (8 \cdot 123^3))$

= 154.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$

= $8.15 / 154 + (170 \cdot 162 / 28606770) / 175$

= 0.107

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	315.0	0.0	31.5	0.0	0.286E+04
Web	466.5	10.0	0.0	46.6	0.0	3.89
Totals	...	...	...	78.1	...	0.286E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/78.1

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 325/2

= 162.500 mm

Length of Inner Rib [L]:

= $\text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$

= $1050 - \sqrt{(808 + 10)^2 - (466/2)^2} - 20$

= 256.399 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(2864/78.1)}$

= 60.542 mm

Slenderness ratio [KL/r]:

= KL/r

= $1 \cdot 256/60.5$

= 4.235

Unit Force [Force,u]:

= $FI / (2 \cdot \text{Baseplate Length})$

= $3009 / (2 \cdot 1300)$

= 1.157 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force $\cdot e \cdot L$

= $1.16 \cdot 466 \cdot 256$

= 138.431 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= $138/176276$

= 7.701 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.24 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (4.24)^2 / (2 \cdot 123^2)) 262 /$



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Saddle Calcs: Ext. Pressure: Step: 24 10:21am Sep 9, 2026

$$\begin{aligned} & (5/3+3\cdot(4.24)/(8\cdot 123)-(4.24^3)/(8\cdot 123^3) \\ & = 155.9 \text{ N./mm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} & = Sc/Sca + (Mbase,c \cdot C1/I)/Sba \\ & = 11.4/156 + (138 \cdot 162/2864)/175 \\ & = 0.117 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: Tube outlet-

Nozl: 13 10:21am Sep 9,2026

Input, Nozzle Desc: Tube outlet-

From: 20

Pressure for Reinforcement Calculations	P	1716.124	KPa.
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1560.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		750.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		25.9843	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	809.6000	mm
Thickness of Pad	te	22.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		74.7994	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Tube outlet-



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Nozzle Calcs.: Tube outlet-

Nozl: 13 10:21am Sep 9,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	25.984 in.
Actual Thickness Used in Calculation	0.551 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)= $1716 \cdot 783 / (138 \cdot 1 - 0.6 \cdot 1716)$

= 9.8136 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)= $1716 \cdot 330 / (138 \cdot 1 + 0.4 \cdot 1716)$

= 4.0838 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1276.0026 mm

Parallel to Vessel Wall, opening length d 638.0013 mm

Normal to Vessel Wall (Thickness Limit), pad side Tlwp 47.0000 mm

Weld Strength Reduction Factor [fr1]:

= min(1, S_n / S_v)

= min(1, 138/138)

= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)

= min(1, 138/138)

= 1.000

Weld Strength Reduction Factor [fr4]:

= min(1, S_p / S_v)

= min(1, 138/138)

= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1, 1)

= 1.000



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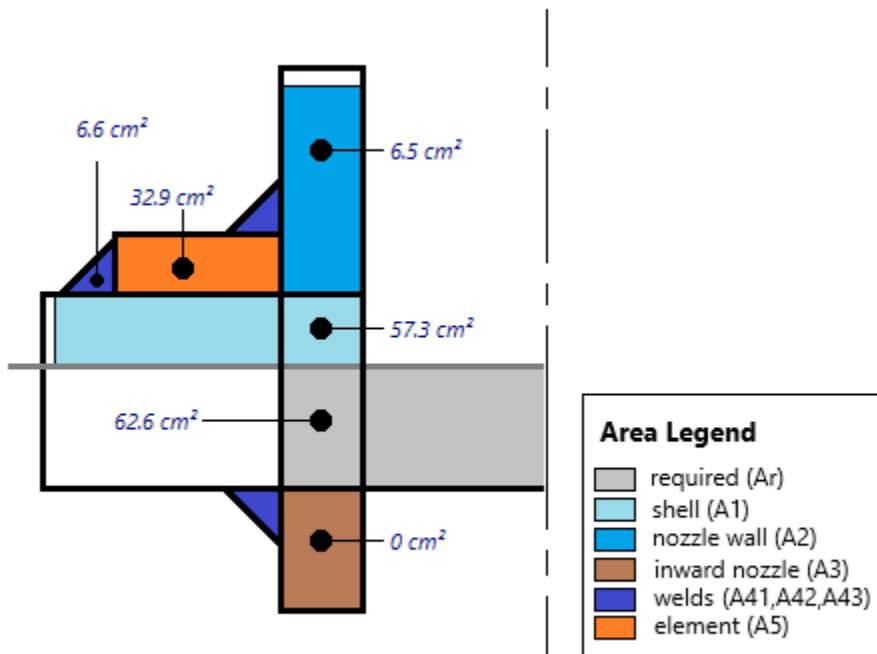


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	62.611	NA	NA
Area in Shell	A1	57.333	NA	NA
Area in Nozzle Wall	A2	6.501	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	6.560	NA	NA
Area in Element	A5	32.912	NA	NA
TOTAL AREA AVAILABLE	Atot	103.306	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (638 \cdot 9.81 \cdot 1 + 2 \cdot 11 \cdot 9.81 \cdot 1 \cdot (1 - 1)) \\
 &= 62.611 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 638(1 \cdot 18.8 - 1 \cdot 9.81) - 2 \cdot 11 \\
 &\quad (1 \cdot 18.8 - 1 \cdot 9.81) \cdot (1 - 1) \\
 &= 57.333 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 47) \cdot (11 - 4.08) \cdot 1
 \end{aligned}$$



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= 6.501 cm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $20^2 \cdot 1 + (0)^2 \cdot 1 + 16^2 \cdot 1$

= 6.560 cm²

Area Available in Element [A5]:

= $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4$

= $(810 - 660) \cdot 22 \cdot 1$

= 32.912 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 7.0838$ mm

Wall Thickness per UG16(b), $tr_{16b} = 5.7000$ mm

Wall Thickness, shell/head, internal pressure $tr_{b1} = 13.0136$ mm

Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 13.0136$ mm

Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 5.7000$ mm

Wall Thickness per table UG-45 $tb_3 = 11.3312$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[11.3, \max(13, 5.7)]$

= 11.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$

= $\max(7.08, 11.3)$

= 11.3312 mm

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 42.4, Allowable : 138.0 N./mm² Passed

Expansion : 0.0, Allowable : 302.6 N./mm² Passed

Occasional : 24.5, Allowable : 183.5 N./mm² Passed

Shear : 14.1, Allowable : 96.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 14$, $t_r = 4.08$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.37$, Temp. Reduction = 62 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 14$, $t_r = 4.08$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.37$, Temp. Reduction = 62 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 14$, $t_r = 4.08$, $c = 3$ mm, $E^* = 1$



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Thickness Ratio = $tr * E^* / (tg - c) = 0.37$, Temp. Reduction = 62 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22, tr = 9.81, c = 3.2 mm, $E^* = 1$

Thickness Ratio = $tr * E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-34 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 14, tr = 4.08, c = 3 mm, $E^* = 1$

Thickness Ratio = $tr * E^* / (tg - c) = 0.37$, Temp. Reduction = 62 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-34 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-34 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-25 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1716.12 / 1960.01 = 0.876$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	957.0018 mm
Parallel to Vessel Wall, opening length	d	478.5009 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	41.743	41.743
Area Available in Shell	A1	28.667	28.667
Area Available in Nozzle Wall	A2	3.804	6.501
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	4.000	6.560
Area Available in Pad	A5	0.000	32.912
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	36.471	74.640

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Tube outlet-

Intermediate Calc. for nozzle/shell Welds	Tmin	11.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	18.8000 mm



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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm
Pad Weld	9.4000 = 0.5 * TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (62.6 - 57.3 + 2 \cdot 11 \cdot 1 \cdot (1 \cdot 18.8 - 9.81)) \cdot 138)$$

$$= 10208.18 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (6.5 + 32.9 + 6.56 - 0 \cdot 1) \cdot 138$$

$$= 64691.21 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (6.5 + 0 + 4 + (4.14)) \cdot 138$$

$$= 20596.91 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (6.5 + 0 + 6.56 + 32.9 + (4.14)) \cdot 138$$

$$= 70511.21 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 660 \cdot 20 \cdot 0.49 \cdot 138$$

$$= 142966. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 810 \cdot 16 \cdot 0.49 \cdot 138$$

$$= 140297. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 325) \cdot (14 - 3) \cdot 0.7 \cdot 138$$

$$= 110458. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 660 \cdot 22 \cdot 0.74 \cdot 138$$

$$= 237499. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 660 \cdot (14 - 0) \cdot 0.74 \cdot 138$$

$$= 151136. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (140297 + 110458) = 250756 \text{ Kgf}$$



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Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (142966 + 237499 + 151136 + 0) = 531601 Kgf

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (140297 + 151136 + 0) = 291433 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 250755 Kgf, must exceed W = 10208 Kgf or W1 = 64691 Kgf

Path 2-2 = 531600 Kgf, must exceed W = 10208 Kgf or W2 = 20596 Kgf

Path 3-3 = 291432 Kgf, must exceed W = 10208 Kgf or W3 = 70511 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2252 KPa.

The Drop for this Nozzle is : 73.2471 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 445.2471 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 2.167 %

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 638.00 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Tube outlet- :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1560.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	660.001	mm
Nozzle Thickness	14.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	809.600	mm
Thickness of Reinforcing Pad	22.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	1716.12	KPa.
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Radial Load	P	-4586.98	Kgf
Circumferential Shear	Vc	3058.00	Kgf
Longitudinal Shear	Vl	3058.00	Kgf
Circumferential Moment	Mc	5138.00	Kg-m.
Longitudinal Moment	Ml	5138.00	Kg-m.
Torsional Moment	Mt	7340.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 660.001 + 2 \cdot 1.65 \cdot \sqrt{803.6(22.0 - 3.2)}$$

$$= 1065.615 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1607.200 mm
Nozzle Outside Diameter	(d) =	660.001 mm
Vessel Thickness used	(T) =	40.800 mm
Nozzle Thickness used	(t) =	11.000 mm
T / t	=	3.709
d / t	=	60.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.577$$

Nr/P	=	0.031
Mr/P	=	0.044
M0/P	=	0.047
N0/P	=	0.203
MrD/Mc	=	0.112
NrDL/Mc	=	0.071
M0d/Mc	=	0.152
N0DL/Mc	=	0.361
MrD/Ml	=	0.057
NrDL/Ml	=	0.040
M0D/Ml	=	0.067
N0DL/Ml	=	0.441

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-20	-20	20	20
Outplane Bending (Ml)	-18	18	18	-18
Normal Pressure Stress	32	33	32	33
Outplane Stress Summary	6	29	83	33

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7



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Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-15	15	15	-15
Inplane Pressure Stress	16	16	16	16

Inplane Stress Summary	6	23	41	-4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	9	31	83	37
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-16	-16	16	16
Outplane Bending (Mc)	-41	41	41	-41
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-29	39	87	-11
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-30	30	30	-30
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	33	32	33

Inplane Stress Summary	6	54	74	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2



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Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	35	55	88	12

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-20	-20	20	20
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	58	60	58	60
Hoop Stress Summary	43	46	84	86

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	64	-64	64	-64
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-139	139	139	-139
Axial Pressure Stress	29	29	29	29
Axial Stress Summary	-57	92	249	-158

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	12	12	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	104	95	249	245

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-16	-16	16	16
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0



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Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	58	60	58	60
Hoop Stress Summary	47	49	80	82

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	64	-64	64	-64
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-289	289	289	-289
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	29	29	29	29
Axial Stress Summary	-207	242	399	-308

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	255	242	399	392

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 660.001 + 2 \cdot 1.65 \cdot \sqrt{792.6(22.0 - 3.2)}$$

$$= 1062.829 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1585.200 mm
Nozzle Outside Diameter	(d) =	809.600 mm
Vessel Thickness used	(T) =	18.800 mm
Nozzle Thickness used	(t) =	85.799 mm
T / t	=	0.219
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.690$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.069
M0/P	=	0.023
N0/P	=	0.051
MrD/Mc	=	0.249
NrDL/Mc	=	0.086
M0d/Mc	=	0.075
N0DL/Mc	=	0.082



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MrD/M1 = 0.062
NrDL/M1 = 0.040
MØD/M1 = 0.019
NØDL/M1 = 0.048

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	70	72	70	72
Outplane Stress Summary	66	72	123	49

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	21	45	166	-70

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	66	73	166	119

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-78	78	78	-78
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	35	35	35	35



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Outplane Stress Summary	-33	88	152	-40
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-263	263	263	-263
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	70	72	70	72

Inplane Stress Summary	-148	272	407	-222
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	149	272	407	223
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	33	32	33	32	33	32	33
Circ. Pl (SUS)	-15	-15	25	25	-3	-3	3	3
Circ. Q (SUS)	-11	11	25	-25	-23	23	37	-37
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-1	-1	1	1	-11	-11	21	21
Long. Q (SUS)	-8	8	22	-22	-34	34	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	32.0	33.0	32.0	33.0	32.0	33.0	32.0	33.0
Pm+Pl (SUS)	17.0	18.0	57.0	58.0	29.0	30.0	37.0	37.0
Pm+Pl+Q (Total)	8.6	29.6	82.1	38.2	35.2	53.3	85.3	11.4

Vessel Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	58.00	207.00	Passed
Pm+Pl+Q (TOTAL)	85.30	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	70	72	70	72	70	72	70	72
Circ. Pl (SUS)	-2	-2	14	14	-10	-10	20	20
Circ. Q (SUS)	-3	3	37	-37	-210	210	316	-316
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	-2	-2	12	12	-8	-8	20	20
Long. Q (SUS)	-12	12	118	-118	-61	61	95	-95
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	70.0	72.0	70.0	72.0	70.0	72.0	70.0	72.0
Pm+Pl (SUS)	68.0	70.0	84.0	86.0	60.0	62.0	90.0	92.0
Pm+Pl+Q (Total)	65.4	73.6	165.1	120.1	150.0	272.0	406.1	224.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.00	138.00	Passed
Pm+Pl (SUS)	92.03	207.00	Passed
Pm+Pl+Q (TOTAL)	406.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	60	58	60	58	60	58	60
Circ. Pl (SUS)	-15	-15	25	25	-11	-11	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-13	-13	15	15	-13	-13	15	15
Long. Q (SUS)	-75	75	203	-203	-225	225	353	-353
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2



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Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	58.0	60.0	58.0	60.0	58.0	60.0	58.0	60.0
Pm+Pl (SUS)	43.1	45.1	83.1	85.1	47.1	49.1	79.1	81.1
Pm+Pl+Q (Total)	104.3	93.5	247.3	244.4	256.4	241.3	397.4	390.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	60.00	138.00	Passed
Pm+Pl (SUS)	85.10	207.00	Passed
Pm+Pl+Q (TOTAL)	397.38	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: Tube Inlet

From: 20

Pressure for Reinforcement Calculations	P	1730.827	KPa.
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1560.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		750.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		25.9843	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	809.6000	mm
Thickness of Pad	te	22.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		74.7994	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



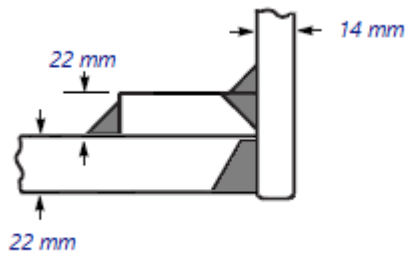
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Nozzle Calcs.: Tube Inlet

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Reinforcement CALCULATION, Description: Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	25.984 in.
Actual Thickness Used in Calculation	0.551 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1731 \cdot 783 / (138 \cdot 1 - 0.6 \cdot 1731)$
 $= 9.8983 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1731 \cdot 330 / (138 \cdot 1 + 0.4 \cdot 1731)$
 $= 4.1186 \text{ mm}$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1276.0026 mm
Parallel to Vessel Wall, opening length	d	638.0013 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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Nozzle Calcs.: Tube Inlet

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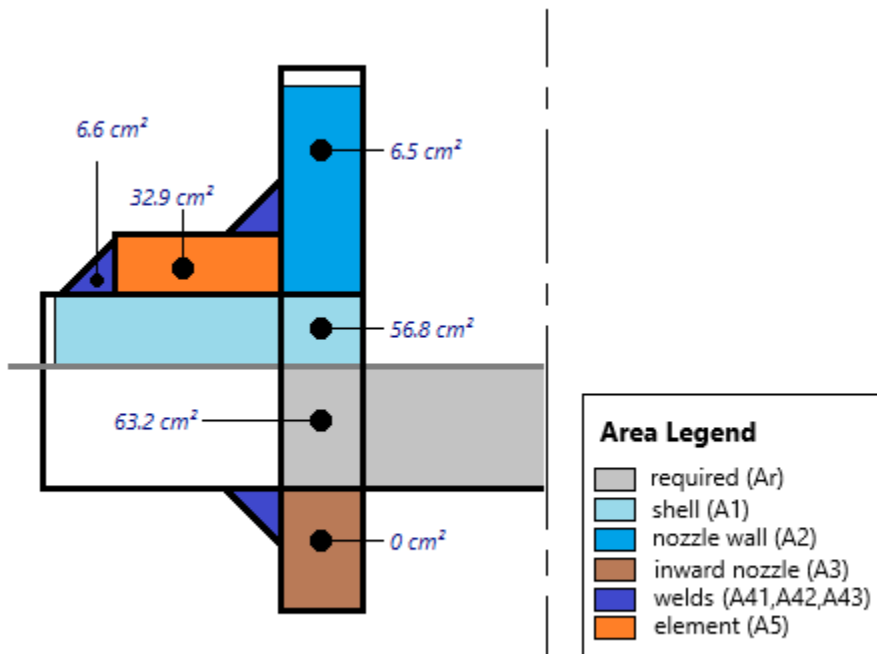


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	63.151	NA	NA
Area in Shell	A1	56.793	NA	NA
Area in Nozzle Wall	A2	6.469	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	6.560	NA	NA
Area in Element	A5	32.912	NA	NA
TOTAL AREA AVAILABLE	Atot	102.733	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (638 \cdot 9.9 \cdot 1 + 2 \cdot 11 \cdot 9.9 \cdot 1 \cdot (1 - 1)) \\
 &= 63.151 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 638(1 \cdot 18.8 - 1 \cdot 9.9) - 2 \cdot 11 \\
 &\quad (1 \cdot 18.8 - 1 \cdot 9.9) \cdot (1 - 1) \\
 &= 56.793 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 47) \cdot (11 - 4.12) \cdot 1
 \end{aligned}$$



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Nozzle Calcs.: Tube Inlet

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= 6.469 cm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $20^2 \cdot 1 + (0)^2 \cdot 1 + 16^2 \cdot 1$

= 6.560 cm²

Area Available in Element [A5]:

= $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$

= $(810 - 660) \cdot 22 \cdot 1$

= 32.912 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 7.1186$ mm

Wall Thickness per UG16(b), $tr_{16b} = 5.7000$ mm

Wall Thickness, shell/head, internal pressure $trb_1 = 13.0983$ mm

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 13.0983$ mm

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.7000$ mm

Wall Thickness per table UG-45 $tb_3 = 11.3312$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[11.3, \max(13.1, 5.7)]$

= 11.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$

= $\max(7.12, 11.3)$

= 11.3312 mm

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 42.6, Allowable : 138.0 N./mm² Passed

Expansion : 0.0, Allowable : 302.4 N./mm² Passed

Occasional : 24.7, Allowable : 183.5 N./mm² Passed

Shear : 14.1, Allowable : 96.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 14$, $t_r = 4.12$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.37$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 14$, $t_r = 4.12$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.37$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: Tube Inlet

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Govrn. thk, $t_g = 14$, $t_r = 4.12$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.37$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 22$, $t_r = 9.9$, $c = 3.2$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.53$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-34 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 14$, $t_r = 4.12$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.37$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-34 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-34 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1730.83/1960.01 = 0.883$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	957.0018	mm
Parallel to Vessel Wall, opening length	d	478.5009	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.0000	mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	42.103	42.103
Area Available in Shell	A1	28.396	28.396
Area Available in Nozzle Wall	A2	3.785	6.469
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	4.000	6.560
Area Available in Pad	A5	0.000	32.912
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	36.181	74.337

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Tube Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	11.0000	mm
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Nozzle Calcs.: Tube Inlet

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Intermediate Calc. for pad/shell Welds TminPad 18.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm
Pad Weld	9.4000 = 0.5 * TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (63.2 - 56.8 + 2 \cdot 11 \cdot 1 \cdot (1 \cdot 18.8 - 9.9)) \cdot 138)$$

$$= 11703.09 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (6.47 + 32.9 + 6.56 - 0 \cdot 1) \cdot 138$$

$$= 64645.16 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (6.47 + 0 + 4 + (4.14)) \cdot 138$$

$$= 20550.86 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (6.47 + 0 + 6.56 + 32.9 + (4.14)) \cdot 138$$

$$= 70465.16 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 660 \cdot 20 \cdot 0.49 \cdot 138$$

$$= 142966. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 810 \cdot 16 \cdot 0.49 \cdot 138$$

$$= 140297. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 325) \cdot (14 - 3) \cdot 0.7 \cdot 138$$

$$= 110458. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 660 \cdot 22 \cdot 0.74 \cdot 138$$

$$= 237499. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 660 \cdot (14 - 0) \cdot 0.74 \cdot 138$$

$$= 151136. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (140297 + 110458) = 250756 \text{ Kgf}$$



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Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (142966 + 237499 + 151136 + 0) = 531601 Kgf

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (140297 + 151136 + 0) = 291433 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 250755 Kgf, must exceed W = 11703 Kgf or W1 = 64645 Kgf

Path 2-2 = 531600 Kgf, must exceed W = 11703 Kgf or W2 = 20550 Kgf

Path 3-3 = 291432 Kgf, must exceed W = 11703 Kgf or W3 = 70465 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2238 KPa.

The Drop for this Nozzle is : 73.2471 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 445.2471 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 2.167 %

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 505.76 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 638.00 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1560.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	660.001	mm
Nozzle Thickness	14.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	809.600	mm
Thickness of Reinforcing Pad	22.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.*



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Design Internal Pressure	Dp	1730.83	KPa.
Radial Load	P	-4586.98	Kgf
Circumferential Shear	Vc	3058.00	Kgf
Longitudinal Shear	Vl	3058.00	Kgf
Circumferential Moment	Mc	5138.00	Kg-m.
Longitudinal Moment	Ml	5138.00	Kg-m.
Torsional Moment	Mt	7340.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 660.001 + 2 \cdot 1.65 \cdot \sqrt{803.6(22.0 - 3.2)}$$

$$= 1065.615 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1607.200 mm
Nozzle Outside Diameter	(d) =	660.001 mm
Vessel Thickness used	(T) =	40.800 mm
Nozzle Thickness used	(t) =	11.000 mm
T / t	=	3.709
d / t	=	60.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.577$$

Nr/P	=	0.031
Mr/P	=	0.044
M0/P	=	0.047
N0/P	=	0.203
MrD/Mc	=	0.112
NrDL/Mc	=	0.071
M0d/Mc	=	0.152
N0DL/Mc	=	0.361
MrD/Ml	=	0.057
NrDL/Ml	=	0.040
M0D/Ml	=	0.067
N0DL/Ml	=	0.441

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-20	-20	20	20
Outplane Bending (Ml)	-18	18	18	-18
Normal Pressure Stress	32	34	32	34
Outplane Stress Summary	6	30	84	33

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0



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Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-15	15	15	-15
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	6	23	41	-3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		2	2	2	2
Shear Stress Summary		3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		10	31	84	38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		5	5	5	5
Outplane Bending (P)		7	-7	7	-7
Outplane Membrane (Mc)		-16	-16	16	16
Outplane Bending (Mc)		-41	41	41	-41
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		16	16	16	16
Outplane Stress Summary		-29	39	87	-11

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		0	0	0	0
Inplane Bending (P)		7	-7	7	-7
Inplane Membrane (Mc)		-3	-3	3	3
Inplane Bending (Mc)		-30	30	30	-30
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		32	34	32	34
Inplane Stress Summary		6	55	74	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2



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Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	35	55	88	13

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-20	-20	20	20
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	59	61	59	61
Hoop Stress Summary	44	46	84	86

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	64	-64	64	-64
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-139	139	139	-139
Axial Pressure Stress	29	29	29	29
Axial Stress Summary	-57	92	249	-157

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	12	12	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	104	95	249	245

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-16	-16	16	16
Hoop Bending (Mc)	0	0	0	0



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Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	59	61	59	61

Hoop Stress Summary	48	50	81	83
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	64	-64	64	-64
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-289	289	289	-289
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	29	29	29	29

Axial Stress Summary	-207	242	399	-308
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9

Shear Stress Summary	6	6	12	12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	256	242	400	392
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 660.001 + 2 \cdot 1.65 \cdot \sqrt{792.6(22.0 - 3.2)}$$

$$= 1062.829 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1585.200 mm
Nozzle Outside Diameter	(d) =	809.600 mm
Vessel Thickness used	(T) =	18.800 mm
Nozzle Thickness used	(t) =	85.799 mm
T / t	=	0.219
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.690$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.069
M0/P	=	0.023
N0/P	=	0.051
MrD/Mc	=	0.249
NrDL/Mc	=	0.086
M0d/Mc	=	0.075



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NØDL/Mc = 0.082
MrD/Ml = 0.062
NrDL/Ml = 0.040
MØD/Ml = 0.019
NØDL/Ml = 0.048

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	71	72	71	72
Outplane Stress Summary	66	73	123	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	22	46	166	-69

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	67	74	166	119

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-78	78	78	-78
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



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Normal Pressure Stress	35	35	35	35
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Outplane Stress Summary	-33	88	152	-39
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-263	263	263	-263
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	72	71	72

Inplane Stress Summary	-148	273	408	-222
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	148	273	408	222
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	34	32	34	32	34	32	34
Circ. Pl (SUS)	-15	-15	25	25	-3	-3	3	3
Circ. Q (SUS)	-11	11	25	-25	-23	23	37	-37
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-1	-1	1	1	-11	-11	21	21
Long. Q (SUS)	-8	8	22	-22	-34	34	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	32.0	34.0	32.0	34.0	32.0	34.0	32.0	34.0
Pm+Pl (SUS)	17.0	19.0	57.0	59.0	29.0	31.0	37.0	37.0
Pm+Pl+Q (Total)	8.6	30.5	82.1	39.2	35.2	54.3	85.3	11.7

Vessel Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.00	138.00	Passed
Pm+Pl (SUS)	59.00	207.00	Passed
Pm+Pl+Q (TOTAL)	85.30	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	72	71	72	71	72	71	72
Circ. Pl (SUS)	-2	-2	14	14	-10	-10	20	20
Circ. Q (SUS)	-3	3	37	-37	-210	210	316	-316
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	-2	-2	12	12	-8	-8	20	20
Long. Q (SUS)	-12	12	118	-118	-61	61	95	-95
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	71.0	72.0	71.0	72.0	71.0	72.0	71.0	72.0
Pm+Pl (SUS)	69.0	70.0	85.0	86.0	61.0	62.0	91.0	92.0
Pm+Pl+Q (Total)	66.4	73.6	165.1	120.1	149.0	272.0	407.1	224.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.00	138.00	Passed
Pm+Pl (SUS)	92.03	207.00	Passed
Pm+Pl+Q (TOTAL)	407.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	59	61	59	61	59	61	59	61
Circ. Pl (SUS)	-15	-15	25	25	-11	-11	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-13	-13	15	15	-13	-13	15	15
Long. Q (SUS)	-75	75	203	-203	-225	225	353	-353
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	59.0	61.0	59.0	61.0	59.0	61.0	59.0	61.0
Pm+P1 (SUS)	44.1	46.1	84.1	86.1	48.1	50.1	80.1	82.1
Pm+P1+Q (Total)	105.3	93.5	247.3	245.4	257.4	241.3	397.4	391.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	138.00	Passed
Pm+P1 (SUS)	86.10	207.00	Passed
Pm+P1+Q (TOTAL)	397.38	413.99	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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Nozzle Calcs.: Shell Inlet

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Input, Nozzle Desc: Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	2186.832	KPa.
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	180	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1560.00	mm
Design Length of Section	L	6599.0000	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2554.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	680.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		111.4000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



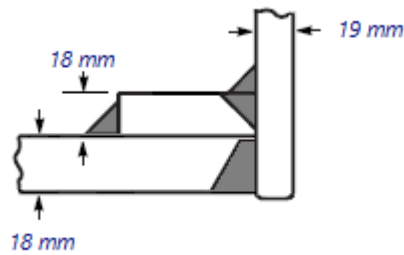
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Reinforcement CALCULATION, Description: Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	18.000 in.
Actual Thickness Used in Calculation	0.750 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2187 \cdot 783 / (138 \cdot 1 - 0.6 \cdot 2187)$
 $= 12.5312 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2187 \cdot 229 / (118 \cdot 1 + 0.4 \cdot 2187)$
 $= 4.2057 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3814 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	850.2000 mm
Parallel to Vessel Wall, opening length	d	425.1000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



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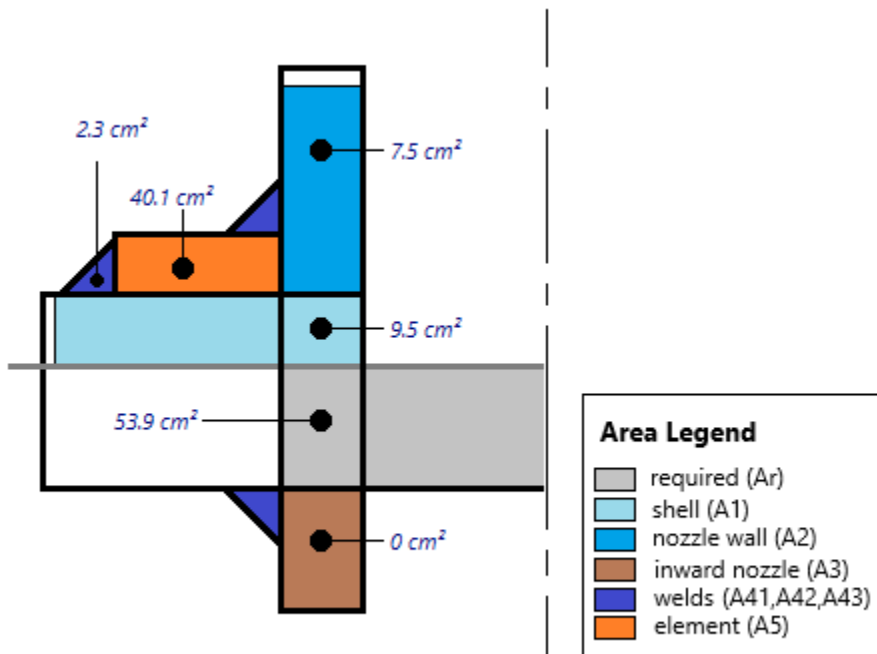


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	53.853	19.822	NA
Area in Shell	A1	9.539	23.440	NA
Area in Nozzle Wall	A2	7.495	9.282	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.295	2.295	NA
Area in Element	A5	40.104	40.104	NA
TOTAL AREA AVAILABLE	Atot	59.433	75.121	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 649.0003 18.0000 mm

Based on given Pad Diameter: 680.0000 15.4955 mm

Based on Shell or Nozzle Thickness: 649.0003 18.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (425 \cdot 12.5 \cdot 1 + 2 \cdot 16 \cdot 12.5 \cdot 1 \cdot (1 - 0.86))$$

$$= 53.853 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 425(1 \cdot 14.8 - 1 \cdot 12.5) - 2 \cdot 16$$

$$(1 \cdot 14.8 - 1 \cdot 12.5) \cdot (1 - 0.86)$$



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$$= 9.539 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot fr_2$$

$$= (2 \cdot 37) \cdot (16 - 4.21) \cdot 0.86$$

$$= 7.495 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$$

$$= 2.295 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (680 - 457) \cdot 18 \cdot 1$$

$$= 40.104 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 7.2057 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.7000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 15.7312 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 15.7312 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 3.7742 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.7000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(15.7, 5.7)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(7.21, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 19.050 = 16.669 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 26.0, Allowable : 118.0 N./mm² Passed

Expansion : 0.0, Allowable : 269.0 N./mm² Passed

Occasional : 14.0, Allowable : 156.9 N./mm² Passed

Shear : 9.7, Allowable : 82.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 16.7$, $t_r = 4.21$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16.7$, $t_r = 4.21$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -13 °C



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Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.7, tr = 4.21, c = 3 mm, E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18, tr = 12.5, c = 3.2 mm, E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.85$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-19 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.7, tr = 4.21, c = 3 mm, E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is:
Design Pressure/Ambient Rating = $2186.83 / 5100.00 = 0.429$

Weld Size Calculations, Description: Shell Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0500 mm
Intermediate Calc. for pad/shell Welds	TminPad	14.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$7.4000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (53.9 - 9.54 + 2 \cdot 16 \cdot 0.86 \cdot (1 \cdot 14.8 - 12.5)) \cdot 138)$
 $= 63232.61 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (7.49 + 40.1 + 2.3 - 0 \cdot 0.86) \cdot 138$



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= 70208.16 Kgf

Weld Load [W2]:

= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv

= (7.49 + 0 + 0.86 + (4.06)) • 138

= 17465.48 Kgf

Weld Load [W3]:

= (A2+A3+A4+A5+(2•tn•t•fr1))•S

= (7.49 + 0 + 2.3 + 40.1 + (4.06)) • 138

= 75924.42 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= (π/2) • Dlo • Wo • 0.49 • Snw

= (3.14/2.0) • 457 • 10 • 0.49 • 118

= 42342. Kgf

Shear, Pad Element Weld [Spew]:

= (π/2) • DP • WP • 0.49 • SEW

= (3.14/2.0) • 680 • 12 • 0.49 • 138

= 88379. Kgf

Shear, Nozzle Wall [Snw]:

= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn

= (3.14 • 221) • (19 - 3) • 0.7 • 118

= 93675. Kgf

Tension, Pad Groove Weld [Tpgw]:

= (π/2) • Dlo • Wgpn • 0.74 • Seg

= (3.14/2) • 457 • 18 • 0.74 • 138

= 134609. Kgf

Tension, Shell Groove Weld [Tngw]:

= (π/2) • Dlo • Wgnvi • 0.74 • Sng

= (3.14/2.0) • 457 • (14 - 0) • 0.74 • 138

= 104696. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SPEW + SNW) = (88379 + 93675) = 182054 Kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (42342 + 134609 + 104696 + 0) = 281646 Kgf

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (88379 + 104696 + 0) = 193075 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 182054 Kgf, must exceed W = 63232 Kgf or W1 = 70208 Kgf

Path 2-2 = 281645 Kgf, must exceed W = 63232 Kgf or W2 = 17465 Kgf

Path 3-3 = 193074 Kgf, must exceed W = 63232 Kgf or W3 = 75924 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2297 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 34.2507 mm



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The Cut Length for this Nozzle is, Drop + Ho + H + T : 452.2507 mm

Input Echo, WRC297 Item 1, Description: Shell Inlet :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.2000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1560.000	mm	
Shell Thickness	18.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	457.200	mm	
Nozzle Thickness	19.0500	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-106 B		
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	680.000	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2186.83	KPa.
Radial Load	P	-3058.00	Kgf
Circumferential Shear	Vc	2039.00	Kgf
Longitudinal Shear	Vl	2039.00	Kgf
Circumferential Moment	Mc	2284.00	Kg-m.
Longitudinal Moment	Ml	2284.00	Kg-m.
Torsional Moment	Mt	3262.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{799.6(18.0 - 3.2)}$$

$$= 816.189 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1599.200	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	32.800	mm
Nozzle Thickness used	(t) =	16.050	mm
T / t	=	2.044	
d / t	=	28.486	



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$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.996$$

Nr/P	=	0.086
Mr/P	=	0.080
M0/P	=	0.047
N0/P	=	0.151
MrD/Mc	=	0.182
NrDL/Mc	=	0.112
M0d/Mc	=	0.124
N0DL/Mc	=	0.180
MrD/Ml	=	0.104
NrDL/Ml	=	0.068
M0D/Ml	=	0.063
N0DL/Ml	=	0.273

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	51	53	51	53
Outplane Stress Summary	33	54	92	44

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	25	25	25	25
Inplane Stress Summary	9	40	72	-10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	34	55	93	55

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-33	33	33	-33
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25

Outplane Stress Summary	-4	47	79	-3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-49	49	49	-49
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	53	51	53

Inplane Stress Summary	12	87	121	-2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	16	87	122	7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36

Hoop Stress Summary	25	27	50	52
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-80	80	80	-80
Axial Pressure Stress	16	16	16	16

Axial Stress Summary	-46	63	133	-78
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary	7	7	4	4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	73	65	134	131

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36

Hoop Stress Summary	29	32	45	48
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-144	144	144	-144
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16

Axial Stress Summary	-110	127	197	-142
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6



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Shear Stress Summary	4	4	7	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	139	127	197	191

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{790.6(18.0 - 3.2)}$$

$$= 814.163 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1581.200 mm
Nozzle Outside Diameter	(d) =	680.000 mm
Vessel Thickness used	(T) =	14.800 mm
Nozzle Thickness used	(t) =	127.450 mm
T / t	=	0.116
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.445$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.048
Mr/P	=	0.075
M0/P	=	0.022
N0/P	=	0.035
MrD/Mc	=	0.248
NrDL/Mc	=	0.098
M0d/Mc	=	0.074
N0DL/Mc	=	0.063
MrD/Ml	=	0.057
NrDL/Ml	=	0.051
M0D/Ml	=	0.017
N0DL/Ml	=	0.036

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-5	-5	5	5
Outplane Bending (Ml)	-15	15	15	-15
Normal Pressure Stress	114	116	114	116
Outplane Stress Summary	116	113	158	93

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0



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Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	57	57	57	57

Inplane Stress Summary	66	45	184	-41
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	4	4	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	116	114	185	135
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-66	66	66	-66
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57

Outplane Stress Summary	4	101	156	-13
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-223	223	223	-223
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	114	116	114	116

Inplane Stress Summary	-55	270	421	-147
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	59	270	421	147

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	53	51	53	51	53	51	53
Circ. Pl (SUS)	-8	-8	16	16	-3	-3	7	7
Circ. Q (SUS)	-10	10	24	-24	-36	36	62	-62
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-1	-1	5	5	-4	-4	12	12
Long. Q (SUS)	-15	15	41	-41	-26	26	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	51.0	53.0	51.0	53.0	51.0	53.0	51.0	53.0
Pm+Pl (SUS)	43.0	45.0	67.0	69.0	48.0	50.0	58.0	60.0
Pm+Pl+Q (Total)	33.2	55.2	91.2	56.1	17.5	86.1	120.1	4.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed
Pm+Pl (SUS)	69.00	207.00	Passed
Pm+Pl+Q (TOTAL)	120.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	114	116	114	116	114	116	114	116
Circ. Pl (SUS)	-1	-1	9	9	-8	-8	20	20
Circ. Q (SUS)	3	-3	33	-33	-162	162	284	-284
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	-1	-1	13	13	-5	-5	13	13
Long. Q (SUS)	10	-10	112	-112	-48	48	84	-84
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2



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Nozzle Calcs.: Shell Inlet

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Pm (SUS)	114.0	116.0	114.0	116.0	114.0	116.0	114.0	116.0
Pm+Pl (SUS)	113.0	115.0	123.0	125.0	106.0	108.0	134.0	136.0
Pm+Pl+Q (Total)	116.2	112.1	182.0	134.0	60.0	270.0	418.0	148.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	116.00	138.00	Passed
Pm+Pl (SUS)	136.02	207.00	Passed
Pm+Pl+Q (TOTAL)	418.03	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	36	33	36	33	36	33	36
Circ. Pl (SUS)	-8	-8	16	16	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-55	55	105	-105	-119	119	169	-169
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	33.0	36.0	33.0	36.0	33.0	36.0	33.0	36.0
Pm+Pl (SUS)	25.1	28.0	49.0	52.0	29.0	32.0	45.1	48.0
Pm+Pl+Q (Total)	73.3	64.3	131.3	131.4	140.4	127.3	195.3	191.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	118.00	Passed
Pm+Pl (SUS)	52.04	177.00	Passed
Pm+Pl+Q (TOTAL)	195.33	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Nozzle Calcs.: Vent

Nozl: 16 10:21am Sep 9,2026

Input, Nozzle Desc: Vent

From: 40

Pressure for Reinforcement Calculations	P	2186.832	KPa.
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	180	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1560.00	mm
Design Length of Section	L	6599.0000	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3674.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	200.3250	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



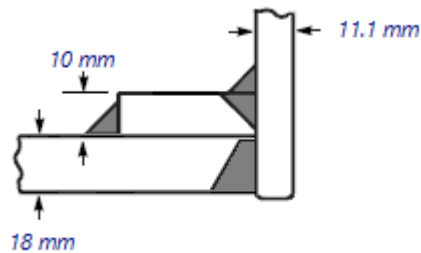
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Nozzle Calcs.: Vent

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Reinforcement CALCULATION, Description: Vent

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2187 \cdot 783 / (138 \cdot 1 - 0.6 \cdot 2187)$
 $= 12.5312 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2187 \cdot 22.1 / (118 \cdot 1 - 0.6 \cdot 2187)$
 $= 0.4140 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3490 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	89.9250 mm
Parallel to Vessel Wall	Rn+tn+t	44.9625 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.1860 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: Vent.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4140 mm
Wall Thickness per UG16(b),	tr16b = 5.7000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.7312 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.7312 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.7742 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.7000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(15.7, 5.7)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.41, 6.42)$
 $= 6.4200 \text{ mm}$



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Nozzle Calcs.: Vent

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Available Nozzle Neck Thickness = $0.875 * 11.074 = 9.690$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.69$, $t_r = 0.41$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.69$, $t_r = 0.41$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.69$, $t_r = 0.41$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 10$, $c = 3.2$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.85$, Temp. Reduction = 9 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 9.69$, $t_r = 0.41$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2186.83/5100.00 = 0.429$

Weld Size Calculations, Description: Vent

Intermediate Calc. for nozzle/shell Welds	T_{min}	8.0744	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	10.0000	mm

Results Per UW-16.1:



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Nozzle Calcs.: Vent

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	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min.}$	$14.1400 = 0.7 * W_o$ mm
Pad Weld	$5.0000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5834 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.5834 mm

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Nozzle Calcs.: Drain

Nozl: 17 10:21am Sep 9,2026

Input, Nozzle Desc: Drain

From: 40

Pressure for Reinforcement Calculations	P	2201.535	KPa.
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	180	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1560.00	mm
Design Length of Section	L	6599.0000	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3674.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	200.3250	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



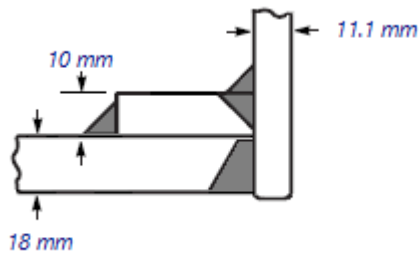
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Nozzle Calcs.: Drain

Nozl: 17 10:21am Sep 9, 2026



Reinforcement CALCULATION, Description: Drain

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.503 in.
Actual Thickness Used in Calculation 0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2202 \cdot 783 / (138 \cdot 1 - 0.6 \cdot 2202)$
= 12.6162 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2202 \cdot 22.1 / (118 \cdot 1 - 0.6 \cdot 2202)$
= 0.4168 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3490 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	89.9250 mm
Parallel to Vessel Wall	Rn+tn+t	44.9625 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.1860 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: Drain.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4168 mm
Wall Thickness per UG16(b),	tr16b = 5.7000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.8162 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.8162 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.7742 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.7000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[6.42, max(15.8, 5.7)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.42, 6.42)
= 6.4200 mm



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Nozzle Calcs.: Drain

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Available Nozzle Neck Thickness = $0.875 * 11.074 = 9.690$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 0.42, c = 3 mm, E* = 1

Thickness Ratio = $tr * E*/(tg - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.69, tr = 0.42, c = 3 mm, E* = 1

Thickness Ratio = $tr * E*/(tg - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.69, tr = 0.42, c = 3 mm, E* = 1

Thickness Ratio = $tr * E*/(tg - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10, c = 3.2 mm, E* = 1

Thickness Ratio = $tr * E*/(tg - c) = 0.85$, Temp. Reduction = 8 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.42, c = 3 mm, E* = 1

Thickness Ratio = $tr * E*/(tg - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2201.53/5100.00 = 0.432$

Weld Size Calculations, Description: Drain

Intermediate Calc. for nozzle/shell Welds	Tmin	8.0744 mm
Intermediate Calc. for pad/shell Welds	TminPad	10.0000 mm

Results Per UW-16.1:



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Nozzle Calcs.: Drain

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	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min.}$	$14.1400 = 0.7 * W_o$ mm
Pad Weld	$5.0000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5834 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.5834 mm

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Nozzle Calcs.: Shell outlet

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Input, Nozzle Desc: Shell outlet

From: 60

Pressure for Reinforcement Calculations	P	2200.163	KPa.
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	180	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1560.00	mm
Design Length of Section	L	6599.0000	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7794.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	697.2000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		120.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



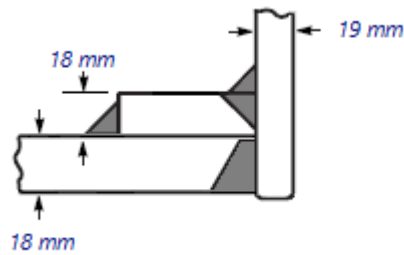
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Nozzle Calcs.: Shell outlet

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Reinforcement CALCULATION, Description: Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	18.000 in.
Actual Thickness Used in Calculation	0.750 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2200 \cdot 783 / (138 \cdot 1 - 0.6 \cdot 2200)$
 $= 12.6083 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2200 \cdot 229 / (118 \cdot 1 + 0.4 \cdot 2200)$
 $= 4.2311 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3814 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	850.2000 mm
Parallel to Vessel Wall, opening length	d	425.1000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



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Nozzle Calcs.: Shell outlet

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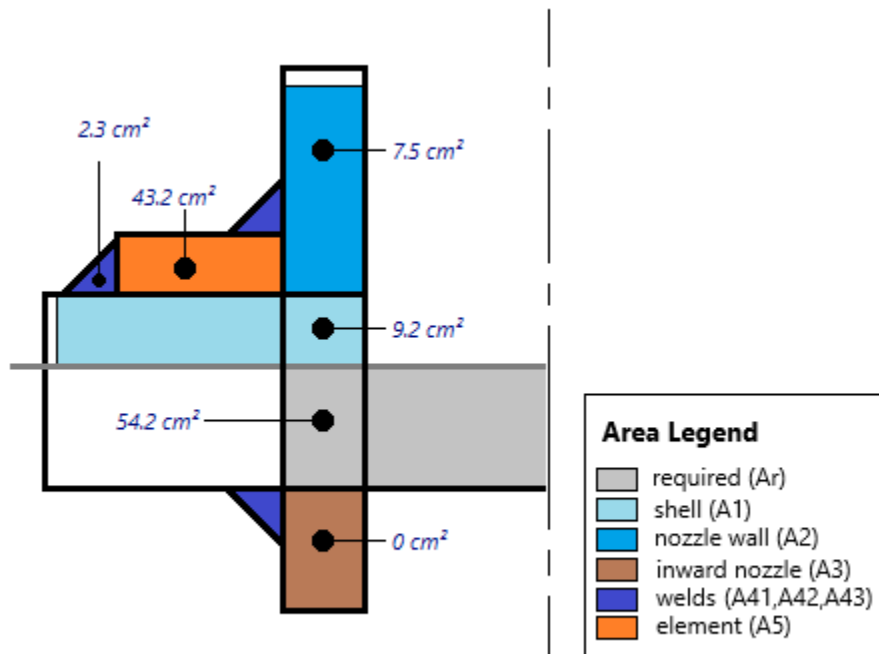


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	54.184	19.822	NA
Area in Shell	A1	9.215	23.440	NA
Area in Nozzle Wall	A2	7.478	9.282	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.295	2.295	NA
Area in Element	A5	43.200	43.200	NA
TOTAL AREA AVAILABLE	Atot	62.189	78.217	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 652.7327 18.0000 mm
Based on given Pad Diameter: 697.2000 14.6649 mm
Based on Shell or Nozzle Thickness: 652.7327 18.0000 mm

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)
= (425 • 12.6 • 1 + 2 • 16 • 12.6 • 1 • (1-0.86))
= 54.184 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)
= 425 (1 • 14.8 - 1 • 12.6) - 2 • 16
(1 • 14.8 - 1 • 12.6) • (1 - 0.86)



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Nozzle Calcs.: Shell outlet

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$$= 9.215 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot fr_2$$

$$= (2 \cdot 37) \cdot (16 - 4.23) \cdot 0.86$$

$$= 7.478 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$$

$$= 2.295 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (697 - 457) \cdot 18 \cdot 1$$

$$= 43.200 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 7.2311 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.7000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 15.8083 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 15.8083 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 3.7742 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.7000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(15.8, 5.7)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(7.23, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 19.050 = 16.669 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.0,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	268.9 N./mm ²	Passed
Occasional	:	14.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	9.7,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 16.7$, $t_r = 4.23$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.31$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $7 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16.7$, $t_r = 4.23$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.31$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-13 \text{ }^\circ\text{C}$



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Nozzle Calcs.: Shell outlet Nozl: 18 10:21am Sep 9, 2026

Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.7, tr = 4.23, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18, tr = 12.6, c = 3.2 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-19 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.7, tr = 4.23, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is:
Design Pressure/Ambient Rating = $2200.16/5100.00 = 0.431$

Weld Size Calculations, Description: Shell outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0500 mm
Intermediate Calc. for pad/shell Welds	TminPad	14.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$7.4000 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (54.2 - 9.22 + 2 \cdot 16 \cdot 0.86 \cdot (1 \cdot 14.8 - 12.6)) \cdot 138)$
 $= 64125.52 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_r2) \cdot S_v$
 $= (7.48 + 43.2 + 2.3 - 0 \cdot 0.86) \cdot 138$



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Nozzle Calcs.: Shell outlet

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= 74542.06 Kgf

Weld Load [W2]:

= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv

= (7.48 + 0 + 0.86 + (4.06)) • 138

= 17442.83 Kgf

Weld Load [W3]:

= (A2+A3+A4+A5+(2•tn•t•fr1))•S

= (7.48 + 0 + 2.3 + 43.2 + (4.06)) • 138

= 80258.33 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= (π/2) • Dlo • Wo • 0.49 • Snw

= (3.14/2.0) • 457 • 10 • 0.49 • 118

= 42342. Kgf

Shear, Pad Element Weld [Spew]:

= (π/2) • DP • WP • 0.49 • SEW

= (3.14/2.0) • 697 • 12 • 0.49 • 138

= 90614. Kgf

Shear, Nozzle Wall [Snw]:

= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn

= (3.14 • 221) • (19 - 3) • 0.7 • 118

= 93675. Kgf

Tension, Pad Groove Weld [Tpgw]:

= (π/2) • Dlo • Wgpn • 0.74 • Seg

= (3.14/2) • 457 • 18 • 0.74 • 138

= 134609. Kgf

Tension, Shell Groove Weld [Tngw]:

= (π/2) • Dlo • Wgnvi • 0.74 • Sng

= (3.14/2.0) • 457 • (14 - 0) • 0.74 • 138

= 104696. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SPEW + SNW) = (90614 + 93675) = 184290 Kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (42342 + 134609 + 104696 + 0) = 281646 Kgf

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (90614 + 104696 + 0) = 195310 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 184289 Kgf, must exceed W = 64125 Kgf or W1 = 74542 Kgf

Path 2-2 = 281645 Kgf, must exceed W = 64125 Kgf or W2 = 17442 Kgf

Path 3-3 = 195310 Kgf, must exceed W = 64125 Kgf or W3 = 80258 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2346 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 34.2507 mm



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Nozzle Calcs.: Shell outlet

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The Cut Length for this Nozzle is, Drop + Ho + H + T : 452.2507 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 54.68 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 425.10 mm

Input Echo, WRC297 Item 1, Description: Shell outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1560.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	457.200	mm
Nozzle Thickness	19.0500	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	697.200	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	2200.16	KPa.
Radial Load	P	-3058.00	Kgf
Circumferential Shear	Vc	2039.00	Kgf
Longitudinal Shear	Vl	2039.00	Kgf
Circumferential Moment	Mc	2283.00	Kg-m.
Longitudinal Moment	Ml	2283.00	Kg-m.
Torsional Moment	Mt	3262.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)} \\
 &= 457.2 + 2 \cdot 1.65 \cdot \sqrt{799.6(18.0 - 3.2)} \\
 &= 816.189 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1599.200	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	32.800	mm



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Nozzle Calcs.: Shell outlet

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Nozzle Thickness used (t) = 16.050 mm
 T / t = 2.044
 d / t = 28.486

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.996$$

Nr/P = 0.086
 Mr/P = 0.080
 M0/P = 0.047
 N0/P = 0.151
 MrD/Mc = 0.182
 NrDL/Mc = 0.112
 M0d/Mc = 0.124
 N0DL/Mc = 0.180
 MrD/Ml = 0.104
 NrDL/Ml = 0.068
 M0D/Ml = 0.063
 N0DL/Ml = 0.273

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	51	53	51	53
Outplane Stress Summary	33	54	93	45

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	25	25	25	25
Inplane Stress Summary	9	40	73	-10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	34	55	93	56
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-33	33	33	-33
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25
Outplane Stress Summary	-4	47	79	-3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-49	49	49	-49
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	53	51	53
Inplane Stress Summary	12	87	122	-2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	17	87	122	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36
Hoop Stress Summary	25	28	50	52



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-80	80	80	-80
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	-46	63	133	-78

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	7	7	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	73	65	134	131

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36
Hoop Stress Summary	29	32	46	48

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-144	144	144	-144
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	-109	127	197	-142

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary	4	4	7	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	139	127	197	191

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{790.6(18.0 - 3.2)}$$

$$= 814.163 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1581.200 mm
Nozzle Outside Diameter	(d) =	697.200 mm
Vessel Thickness used	(T) =	14.800 mm
Nozzle Thickness used	(t) =	136.050 mm
T / t	=	0.109
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.558$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.047
Mr/P	=	0.074
M0/P	=	0.022
N0/P	=	0.033
MrD/Mc	=	0.246
NrDL/Mc	=	0.098
M0d/Mc	=	0.074
N0DL/Mc	=	0.060
MrD/Ml	=	0.056
NrDL/Ml	=	0.050
M0D/Ml	=	0.017
N0DL/Ml	=	0.034

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	115	117	115	117
Outplane Stress Summary	117	114	157	94

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-48	48	48	-48
Inplane Pressure Stress	57	57	57	57

Inplane Stress Summary	68	45	180	-37
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	4	4	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	118	114	180	132
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-64	64	64	-64
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57

Outplane Stress Summary	6	100	153	-11
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-216	216	216	-216
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	115	117	115	117

Inplane Stress Summary	-48	265	413	-138
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
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Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	55	265	413	138

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	53	51	53	51	53	51	53
Circ. Pl (SUS)	-8	-8	16	16	-3	-3	7	7
Circ. Q (SUS)	-10	10	24	-24	-36	36	62	-62
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-1	-1	5	5	-4	-4	12	12
Long. Q (SUS)	-15	15	41	-41	-26	26	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	51.0	53.0	51.0	53.0	51.0	53.0	51.0	53.0
Pm+Pl (SUS)	43.0	45.0	67.0	69.0	48.0	50.0	58.0	60.0
Pm+Pl+Q (Total)	33.2	55.2	91.2	56.1	17.5	86.1	120.1	4.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed
Pm+Pl (SUS)	69.00	207.00	Passed
Pm+Pl+Q (TOTAL)	120.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	117	115	117	115	117	115	117
Circ. Pl (SUS)	0	0	8	8	-8	-8	20	20
Circ. Q (SUS)	3	-3	31	-31	-156	156	276	-276
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	-1	-1	13	13	-4	-4	12	12
Long. Q (SUS)	12	-12	108	-108	-47	47	81	-81



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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	115.0	117.0	115.0	117.0	115.0	117.0	115.0	117.0
Pm+P1 (SUS)	115.0	117.0	123.0	125.0	107.0	109.0	135.0	137.0
Pm+P1+Q (Total)	118.2	114.1	178.0	132.0	55.0	265.0	411.0	139.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	138.00	Passed
Pm+P1 (SUS)	137.01	207.00	Passed
Pm+P1+Q (TOTAL)	411.03	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	36	33	36	33	36	33	36
Circ. P1 (SUS)	-8	-8	16	16	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. P1 (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-55	55	105	-105	-119	119	169	-169
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	33.0	36.0	33.0	36.0	33.0	36.0	33.0	36.0
Pm+P1 (SUS)	25.1	28.0	49.0	52.0	29.0	32.0	45.1	48.0
Pm+P1+Q (Total)	73.3	64.3	131.3	131.4	140.4	127.3	195.3	191.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	118.00	Passed
Pm+P1 (SUS)	52.04	177.00	Passed
Pm+P1+Q (TOTAL)	195.33	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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ASME TS Calc: DESIGN 1

Case: 1 10:21a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell-1

Shell Maximum Design Pressure	Psd,max	2230.00 KPa.
Shell Maximum Operating Pressure	Psox,max	2230.00 KPa.
Shell Minimum Operating Pressure	Psox,min	101.13 KPa.
Shell Thickness	ts	18.0000 mm
Shell Internal Corrosion Allowance	cas	3.2000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1560.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	180.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1750.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1750.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	22.0000 mm
Channel Internal Corrosion Allowance	cac	3.2000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1560.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1780
Tube Wall Thickness	et	2.3200 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6706.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6466.00 mm
Design Temperature of the Tubes		180.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	155.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.3200 mm
Groove Weld Leg	ag	2.3200 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	726.462 mm



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ASME TS Calc: DESIGN 1

Case: 1 10:21a Sep 9,2026

Largest Center-to-Center Tube Distance	U1	41.2750 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	15161.1 cm ²
Perimeter of Tube Layout	Cp	4643.7 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	180.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	120.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1840.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	1632.00 mm
Step 1 Depth on the Tubeside	ht1	8.00 mm
Step 2 Diameter on the Tubeside	dt2	1557.00 mm
Step 2 Depth on the Tubeside	ht2	8.00 mm
Is the Tubesheet Flat on Shellside?		No
Calculated Tubesheet Diameter per Div 2, 4.18.4		1840.00 mm

Note: Tubesheet diameter is now: 1840.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	215.9 cm ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	647.700 mm
Tube End condition corresponding to Span (l)	k	0.80
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	-101.1300 KPa.
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	1840.000 mm
Flange Inside Diameter	B	1560.000 mm
Flange Face Outside Diameter	Fod	1635.000 mm
Flange Face Inside Diameter	Fid	1560.000 mm
Gasket Outside Diameter	Go	1624.000 mm
Gasket Inside Diameter	Gi	1592.000 mm
Small end Hub thk.	g0	22.0000 mm
Large end Hub thk.	g1	34.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option		Program Selects



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ASME TS Calc: DESIGN 1

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Bolting Information:

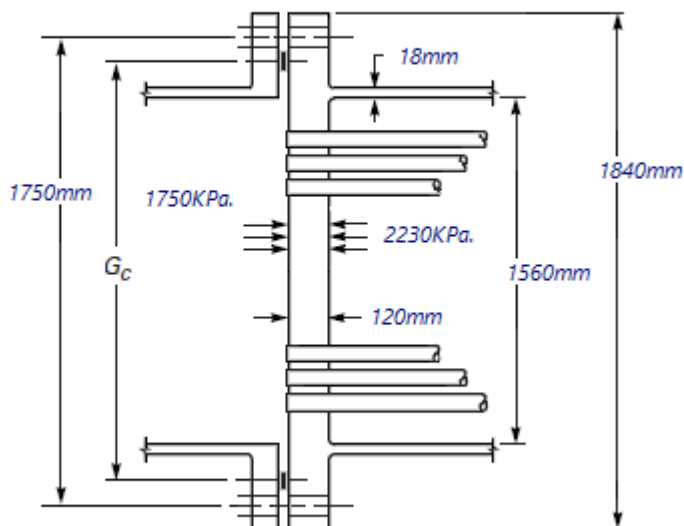
Diameter of Bolt Circle	C	1750.000 mm
Nominal Bolt Diameter	dB	28.5750 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	68
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 99.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19357E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
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Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19357E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19357E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	180.0 °C	0.19357E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= (P*(D/2-CAE)) / (S*E+0.4*P) per Appendix 1-1 (a)(1)

= (1851*(25.4/2-0))/(92.4*1+0.4*1851)

= 0.2524 + 0.0000 = 0.2524 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-1	at	180.00 °C
Elastic Modulus for Material			196084384.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.3200	25.40	6466.00	10.95	50.0000	0.0091770	92.27

EMAWP = (4*B)/(3*(D/T)) = (4 *92.3)/(3 *10.9) = 11237 KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6551	25.40	6466.00	38.77	50.0000	0.0007318	64.85

EMAWP = (4*B)/(3*(D/T)) = (4 *64.9)/(3 *38.8) = 2230 KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6551 mm
Allowable Internal Pressure at Corroded thickness	18208.61 KPa.
Required Internal Design Pressure	1851.13 KPa.
Allowable External Pressure at Corroded thickness	11236.96 KPa.
Required External Design Pressure	2230.00 KPa.
Required Thickness due to Shell Side pressure	0.6551 mm

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (Goc-Gic) / 2	16.000 mm
Basic Gasket Width,	b0 = N / 2	8.000 mm
Effective Gasket Width,	b = sqrt(b0) * 2.5	7.125 mm
Gasket Reaction Diameter,	G = Go-2.0*b	1609.750 mm



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Required Tubesheet Flanged Extension thk. per 4.18.5 (operating) [hropr]:

$$= \sqrt{(1.9 \cdot WM1 \cdot hg / (S \cdot G))}$$

$$= \sqrt{(1.9 \cdot 401777 \cdot 70.1 / (138 \cdot 1610))}$$

$$= 48.613 \text{ mm}$$

Required Tubesheet Flanged Extension thk. per 4.18.5 (seating) [hrseat]:

$$= \sqrt{(1.9 \cdot W \cdot hg / (Sa \cdot G))}$$

$$= \sqrt{(1.9 \cdot 560144 \cdot 70.1 / (138 \cdot 1610))}$$

$$= 57.400 \text{ mm}$$

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= ltx / h \text{ (modified for corrosion if present)}$$

$$= 0/114$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.32 (193573552/193573552) \cdot (92.4/138) \cdot (0); 25.4 - 2 \cdot 2.32)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 726 + 25.4$$

$$= 1478.324 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1478/2$$

$$= 739.1620 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 31.8 / \sqrt{(1 - 4 \cdot \min(216 \cdot 100 ; 4 \cdot 1478 \cdot 31.8) / (\pi \cdot 1478^2))}$$

$$= 31.9516 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p* - d*) / p*$$

$$= (32 - 25.4) / 32$$

$$= 0.2050$$

Ratio [ps]:

$$= as / ao$$

$$= 783/739$$

$$= 1.0596$$

Ratio [pc]:

$$= ac / ao$$

$$= 805/739$$

$$= 1.0889$$

Compute parameter [xt]:



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$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 1780 ((25.4 - 2 \cdot 2.32) / (2 \cdot 739))^2$$

$$= 0.6490$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 1780 (25.4 / (2 \cdot 739))^2$$

$$= 0.4745$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8 - 3); 0)$$

$$= 5.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts) Es / L$$

$$= \pi \cdot 14.8 (1566 + 14.8) 193573552 / 6472$$

$$= 224234.5469 \text{ Kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt) Et / L$$

$$= \pi \cdot 2.32 (25.4 - 2.32) 193573552 / 6472$$

$$= 513.0714 \text{ Kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 224235 / (1780 \cdot 513)$$

$$= 0.24553$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks / Kj)$$

$$= 1 / (1 + 224235 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1566 + 14.8) 14.8)$$

$$= 0.0119 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.012 \cdot 193573552 \cdot 14.8^3 / (6(1 - 0.3^2))$$

$$= 1365746176.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1566 + 14.8) \cdot 1365746176 / 114^3 \cdot (1 + 114 \cdot 0.012 + (114 \cdot 0.012)^2 / 2)$$

$$= 28618092.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1566^2 / (4 \cdot 193573552 \cdot 14.8) \cdot (1 - 1566 \cdot 0.3 / (2(1566 + 14.8)))$$

$$= 0.18228 \text{E+00 mm/N} \cdot \text{mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 3.590551$; $\mu^* = 0.205047$; $E^* = 31227670 \text{ KPa}$.

$E^*/E = 0.161322$; $\nu^* = 0.451291$

Note: As h/p (3.591) is > 2, data values for $h/p = 2$ were used.



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Tube Bundle Stiffness Factor [Xa]:

$$= \left(\frac{24(1 - \nu^2)Nt \cdot Et \cdot tt(dt - tt)ao^2}{E \cdot L \cdot H^3} \right)^{1/4}$$

$$= \left(\frac{24(1 - 0.45^2)1780 \cdot 193573552 \cdot 2.32}{(25.4 - 2.32)^{739^2} (31227670 \cdot 6472 \cdot 114^3)} \right)^{1/4}$$

$$= 5.0364$$

Values from Table 4.18.3

$$Zd = 0.011736 ; Zv = 0.038739 ; Zm = 0.280694$$

$$Za = 0.249680E+02 ; Zw = 0.038739$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 1840/1478$$

$$= 1.2447$$

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31227670 \cdot (28618092 + 0 + 193573552 \cdot \ln(1.24))$$

$$= 1.2473$$

Compute parameter [φ]:

$$= (1 + \nu) F$$

$$= (1 + 0.45) 1.25$$

$$= 1.8101$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.06 - 1 - 1.81 \cdot 0.039) / (1 + 1.81 \cdot 0.28)$$

$$= -0.006992$$

Compute Parameter [Qz1]:

$$= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$$

$$= (0.012 + -0.007 \cdot 0.039) / 2 \cdot 5.04^4$$

$$= 3.6885$$

Compute parameter [Qz2]:

$$= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$$

$$= (0.039 + -0.007 \cdot 0.28) / 2 \cdot 5.04^4$$

$$= 11.8310$$

Compute parameter [U]:

$$= (Zw + (ps - 1) Zm) Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.039 + (1.06 - 1) 0.28) 5.04^4 / (1 + 1.81 \cdot 0.28)$$

$$= 23.6620$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γb]:

$$= (Gc - C) / Do \text{ (config b)}$$

$$= (1610 - 1750) / 1478 = -0.095$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ωs]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.06 \cdot 1365746176 \cdot 0.012 \cdot 0.1822838 (1 + 114 \cdot 0.012)$$

$$= 7381.4111 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - \omega s$$



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$$= 739^2 (1.06^2 - 1) \cdot (1.06 - 1) / 4 - 7381$$

$$= -6382.8521 \text{ mm}^2$$

Parameter [wc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.09 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 114) -$$

$$0.000 (0/114 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 739^2 [(1.09^2 + 1) \cdot (1.09 - 1) / 4 - (1.06 - 1) / 2] - 0$$

$$= 10265.8027 \text{ mm}^2$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s -$$

$$[(\rho_s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= -101 \{ 0.47 + 2(1 - 0.47) 0.3 + [2/0.25 \cdot$$

$$(1566/1478)^2] 0.3 - [(1.06^2 - 1) / (1 \cdot 0.25)] -$$

$$[(1 - 1) / (2 \cdot 1 \cdot 0.25)] [(0^2 - (1566)^2) / 1478^2] \}$$

$$= -306.7880 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.65 + 2(1 - 0.65) 0.3 + 1/(1 \cdot 0.25)] 1750$$

$$= 8631.7197 \text{ KPa.}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 1780 \cdot 513 \cdot 0 / (3.14 \cdot 739^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0.095 \cdot 23.7 \cdot 560144 / (2 \cdot \pi \cdot 739^2)$$

$$= 3591.9236 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= -(23.7/739^2) \cdot (-6383 \cdot -101 - 10266 \cdot 1750)$$

$$= 750.0864 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P^* s - \omega_c \cdot P^* c)$$

$$= (23.7/739^2) \cdot (7381 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho_s - 1) Qz2)) \cdot$$

$$(P^* s - P^* t + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.25 / (1 + 1 \cdot 0.25 (3.69 + (1.06 -$$

$$1) 11.8)) \cdot (-307 - 8632 + 0 + 0 + 3592 + 750)$$

$$= -542.9266 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:



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Determine factor [Q2]:

$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega_s \cdot P * s - \omega_c \cdot P * c)) + W * \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-6383 \cdot -101 - 10266 \cdot 1750) - (7381 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 560144 \cdot -0.095 / (2 \cdot \pi)] / (1 + 1.81 \cdot 0.28)$$

$$= -6779.3369 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (P_e \cdot a_o^2)$$

$$= -0.007 + 2 \cdot -6779 / (-543 \cdot 739^2)$$

$$= 0.44124$$

Fm Value from Table 4.18.3 = 0.241776

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.24 / 0.21) \cdot (2 \cdot 739 / (114 - 5))^2 \cdot -543$$

$$= -176.6457 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.29 / 0.21) \cdot (2 \cdot 739 / (104 - 5))^2 \cdot -441$$

$$= -206.9945 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 104 + 3 + 3$$

$$= 110.3870 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.2 \cdot 114 / 739) \cdot 110$$

$$= 6.8106 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (15161 \cdot 1196) / (0.2 \cdot 18.3 \cdot 4644)$$

$$= 106.9562 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 18.3 + 3 + 3$$

$$= 24.2620 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(110; 24.3)$$

$$= 110.3870 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -2.16, F_{tmax} = 9.27$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((-101 \cdot 0.47 - 1750 \cdot 0.65) - (-543) \cdot -2.16) /$$



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$$\begin{aligned} & (0.65 - 0.47) \\ & = -13.5213 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} & = ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\ & = ((-101 \cdot 0.47 - 1750 \cdot 0.65) - (-543) \cdot 9.27) / \\ & \quad (0.65 - 0.47) \\ & = 22.0804 \text{ N./mm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 22.1 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92.3983 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} & = \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ & = \max((3.25 - 0.25(0.012 + 0.44 \cdot 0.039)5.04^4); 1.25) \\ & = 1.2500 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} & = (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4 \\ & = (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.32)^2}) / 4 \\ & = 8.2011 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} & = \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{yt}) } \\ & = \sqrt{ (2 \cdot \pi^2 \cdot 193573552 / 156) } \\ & = 156.7104 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} & = k \cdot L / r \\ & = 0.8 \cdot 648 / 8.2 \\ & = 63.1815 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} & = (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\ & = (156 / 1.25) \cdot (1 - 63.2 / (2 \cdot 157)) \\ & = 92.398 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)} \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} & = \sigma_{t,max} \cdot \text{Tube Area} \\ & = 22.1 \cdot 1.68 \\ & = 378.75 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} & = \pi (d_o - t) t \\ & = \pi (25.4 - 2.32) 2.32 \\ & = 1.682 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} & = \pi \cdot t (d_o - t) S \\ & = \pi \cdot 2.32 (25.4 - 2.32) 92.4 \\ & = 1584.927 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} & = 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)} \\ & = 0.85 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 92.4 \\ & = 1573.3376 \text{ Kg} \end{aligned}$$



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Weld Strength Factor [fw]:

= S / S_w where $S_w = \min(S \text{ and } S_t)$

= 92.4/92.4

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > F_t)

= $0.55 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 92.4$

= 1018.0419 Kgf

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$

= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.32 \cdot (25.4 - 2.32) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$

= 2.897 mm

Minimum Required Fillet Weld Leg [afr]: 1.4487 mm

Minimum Required Groove Weld Leg [agr]: 1.4487 mm

Tube-Tubesheet Jt allowable, 1585 is >= tube strength 1585 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

= $S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:

= $a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$

= $739^2 / ((1566 + 14.8) 14.8) \cdot [-543 + (1.06^2 - 1)$

$(-101 - 1750)] + 783^2 \cdot 1750 / ((1566 + 14.8) 14.8)$

= 27.8934 N./mm²

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 14.8 mm, the minimum Shell length adjacent to the tubesheet should be 274.066 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

= $a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$

= $739^2 / ((1566 + 14.8) 14.8) [-543 + (1.06^2 - 1)$

$(-101 - 1750)] + 783^2 \cdot 1750 / ((1566 + 14.8) 14.8)$

= 27.8934 N./mm²

Shell Bending Stress due to Joint Interaction [σ_s, b]:

= $6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E \cdot t_s)] +$

$6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$

= $6 \cdot 1365746176 / 14.8^2 \{ 0.012 [0.1822838 \cdot -101 + 783^2$

$\cdot 0 / (2.86E+09)] +$

$6(1 - 0.45^2) / (31227670) (739 / 114)^3 (1 + 114 \cdot$

$0.012 / 2) [-543 (0.039 + 0.28 \cdot -0.007) +$

$2 / 739^2 \cdot 0.28 \cdot -6779] \}$

= -239.2402 N./mm²

Shell Stress Summation vs. Allowable

:: $|\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$

:: $|27.9| + |-239| \leq 207 \text{ N./mm}^2$

:: 267, must be $\leq 207 \text{ N./mm}^2$ (See Elastic-Plastic iteration below)



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Stress/Force Summary for Loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	176.6 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	7.8 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	22.1 <=	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-13.5 <=	-92.4 N./mm ²	Pass
Maximum Force on any one Tube	378.7 <=	1584.9 Kg	Pass
Axial Membrane Stress in Shell	27.9 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	267.1 <=	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:

= min(Sys, SPS,s/2)

= min(227, 455/2) = 227 N./mm²

Compute Factor [facts]:

= min([1.4 - 0.4• abs(σs,b) / Sstars], 1)

= min([1.4 - 0.4• abs(-239)/227], 1) = 0.98

Compute Factor [factc]:

= 1 (for Configuration = B)

Compute Modified Modulus [Es]:

= Es • facts

= 193573552 • 0.98 = 189545024 KPa.

Tubesheet Bending Stress - final thickness [σf]:

= (1.5 • Fm / μ•) • (2 • ao/(h - h'g))² • Pe

= (1.5 • 0.29/0.21) • (2 • 739/(105 - 5))² • -441

= -206.9760 N./mm²

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

= h + Cats + Catc

= 105 + 3 + 3

= 110.5763 mm

For the Elastic-Plastic iteration, Shell Allowable = SPS,s

ShellAllow = 454.8170 N./mm²

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	177.2 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	7.8 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	22.1 <=	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-13.5 <=	-92.4 N./mm ²	Pass
Maximum Force on any one Tube	378.7 <=	1584.9 Kg	Pass
Axial Membrane Stress in Shell	27.9 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	267.1 <=	454.8 N./mm ²	Pass



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Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	16.000 mm
Basic Gasket Width,	$b0 = N / 2$	8.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	7.125 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	1609.750 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$= ltx / h$ (modified for corrosion if present)

$= 0/114$

$= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d*]$:

$= \max(dt - 2 * tt(EtT/E) * (StT/S) * (\rho); dt - 2 * tt)$

$= \max(25.4 - 2 * 2.32 (193573552/193573552) * (92.4/138) * (0); 25.4 - 2 * 2.32)$

$= 25.4000$ mm

Equivalent outer tube limit circle diameter $[Do]$:

$= 2 * ro + dt$

$= 2 * 726 + 25.4$

$= 1478.324$ mm

Determine the basic ligament efficiency for shear $[\mu]$:

$= (p - dt) / p$

$= (31.8 - 25.4) / 31.8$

$= 0.200$

Equivalent outer tube limit radius $[ao]$:

$= Do / 2$

$= 1478/2$

$= 739.1620$ mm

Effective tube pitch $[p*]$:

$= p / \sqrt{ (1 - 4 * \min(AL * CNVfactor; 4 * Do * p) / (\pi * Do^2)) }$

$= 31.8 / \sqrt{ (1 - 4 * \min(216 * 100 ; 4 * 1478 * 31.8) / (\pi * 1478^2)) }$

$= 31.9516$ mm

Effective ligament efficiency for bending $[\mu*]$:

$= (p* - d*) / p*$

$= (32 - 25.4) / 32$

$= 0.2050$

Ratio $[ps]$:

$= as / ao$

$= 783/739$

$= 1.0596$

Ratio $[pc]$:

$= ac / ao$

$= 805/739$

$= 1.0889$



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Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 1780 ((25.4 - 2 \cdot 2.32) / (2 \cdot 739))^2$$

$$= 0.6490$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 1780 (25.4 / (2 \cdot 739))^2$$

$$= 0.4745$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8 - 3); 0)$$

$$= 5.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts) Es / L$$

$$= \pi \cdot 14.8 (1566 + 14.8) 193573552 / 6472$$

$$= 224234.5469 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt) Et / L$$

$$= \pi \cdot 2.32 (25.4 - 2.32) 193573552 / 6472$$

$$= 513.0714 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 224235 / (1780 \cdot 513)$$

$$= 0.24553$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks / Kj)$$

$$= 1 / (1 + 224235 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1566 + 14.8) 14.8)$$

$$= 0.0119 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.012 \cdot 193573552 \cdot 14.8^3 / (6(1 - 0.3^2))$$

$$= 1365746176.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1566 + 14.8) \cdot 1365746176 / 114^3 \cdot (1 + 114 \cdot 0.012 + (114 \cdot 0.012)^2 / 2)$$

$$= 28618092.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1566^2 / (4 \cdot 193573552 \cdot 14.8) \cdot (1 - 1566 \cdot 0.3 / (2(1566 + 14.8)))$$

$$= 0.18228 \text{E+00 mm/N} \cdot \text{mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 3.590551$; $\mu^* = 0.205047$; $E^* = 31227670 \text{ KPa}$.

$E^*/E = 0.161322$; $\nu^* = 0.451291$



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Note: As h/p (3.591) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\frac{24(1 - \nu^2)Nt \cdot Et \cdot tt(dt - tt)ao^2}{E \cdot L \cdot H^3} \right)^{1/4}$$

$$= \left(\frac{24(1 - 0.45^2)1780 \cdot 193573552 \cdot 2.32}{(25.4 - 2.32)739^2(31227670 \cdot 6472 \cdot 114^3)} \right)^{1/4}$$

$$= 5.0364$$

Values from Table 4.18.3

$$Zd = 0.011736 ; Zv = 0.038739 ; Zm = 0.280694$$

$$Za = 0.249680E+02 ; Zw = 0.038739$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 1840/1478$$

$$= 1.2447$$

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31227670 \cdot (28618092 + 0 + 193573552 \cdot \ln(1.24))$$

$$= 1.2473$$

Compute parameter [φ]:

$$= (1 + \nu)F$$

$$= (1 + 0.45)1.25$$

$$= 1.8101$$

Compute parameter [Q1]:

$$= (\psi_s - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.06 - 1 - 1.81 \cdot 0.039) / (1 + 1.81 \cdot 0.28)$$

$$= -0.006992$$

Compute Parameter [Qz1]:

$$= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$$

$$= (0.012 + -0.007 \cdot 0.039) / 2 \cdot 5.04^4$$

$$= 3.6885$$

Compute parameter [Qz2]:

$$= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$$

$$= (0.039 + -0.007 \cdot 0.28) / 2 \cdot 5.04^4$$

$$= 11.8310$$

Compute parameter [U]:

$$= (Zw + (\psi_s - 1)Zm)Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.039 + (1.06 - 1)0.28)5.04^4 / (1 + 1.81 \cdot 0.28)$$

$$= 23.6620$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γb]:

$$= (Gc - C) / Do \text{ (config b)}$$

$$= (1610 - 1750) / 1478 = -0.095$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ωs]:

$$= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s)$$

$$= 1.06 \cdot 1365746176 \cdot 0.012 \cdot 0.1822838(1 + 114 \cdot 0.012)$$

$$= 7381.4111 \text{ mm}^2$$



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Calculate parameter [$\omega*s$]:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega s$$

$$= 739^2(1.06^2 - 1) \cdot (1.06 - 1) / 4 - 7381$$

$$= -6382.8521 \text{ mm}^2$$

Parameter [ωc]:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2))$$

$$= 1.09 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 114) -$$

$$0.000 (0/114 + 0 \cdot 0/2))$$

$$= 0.0000 \text{ mm}^2$$

Parameter [$\omega*c$]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega c$$

$$= 739^2 [(1.09^2 + 1) \cdot (1.09 - 1) / 4 - (1.06 - 1) / 2] - 0$$

$$= 10265.8027 \text{ mm}^2$$

Pressure [$P*S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P*C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= Ps \{ xs + 2(1 - xs) vt + [2 / Kst \cdot (Ds/Do)^2] vs -$$

$$[(ps^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(D)^2 - (Ds)^2 / Do^2] \}$$

$$= 2230 \{ 0.47 + 2(1 - 0.47) 0.3 + [2 / 0.25 \cdot$$

$$(1566/1478)^2] 0.3 - [(1.06^2 - 1) / (1 \cdot 0.25)] -$$

$$[(1 - 1) / (2 \cdot 1 \cdot 0.25)] [(0^2 - (1566)^2 / 1478^2)] \}$$

$$= 6764.9282 \text{ KPa.}$$

Pressure [$P't$]:

$$= [xt + 2(1 - xt) vt + 1 / (J \cdot Kst)] Pt$$

$$= [0.65 + 2(1 - 0.65) 0.3 + 1 / (1 \cdot 0.25)] 0$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 1780 \cdot 513 \cdot 0 / (3.14 \cdot 739^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0.095 \cdot 23.7 \cdot 0 / (2 \cdot \pi \cdot 739^2)$$

$$= 0.0000 \text{ KPa.}$$

Calculate the pressure [$Prim$]:

$$= - (U / ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= - (23.7 / 739^2) \cdot (-6383 \cdot 2230 - 10266 \cdot 0)$$

$$= 616.4412 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U / ao^2) \cdot (\omega s \cdot P*s - \omega c \cdot P*c)$$

$$= (23.7 / 739^2) \cdot (7381 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot$$

$$(P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.25 / (1 + 1 \cdot 0.25 (3.69 + (1.06 -$$

$$1) 11.8)) \cdot (6765 - 0 + 0 + 0 + 0 + 616)$$

$$= 871.8685 \text{ KPa.}$$



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ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega s \cdot P * s - \omega c \cdot P * c)) + W * \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-6383 \cdot 2230 - 10266 \cdot 0) - (7381 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 0 \cdot -0.095 / (2 \cdot \pi)] / (1 + 1.81 \cdot 0.28)$$

$$= -962.4717 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (P_e \cdot a_o^2)$$

$$= -0.007 + 2 \cdot -962 / (872 \cdot 739^2)$$

$$= -0.04662$$

Fm Value from Table 4.18.3 = 0.034898

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.035 / 0.21) \cdot (2 \cdot 739 / (114 - 5))^2 \cdot 872$$

$$= 40.9444 \text{ N./mm}^2$$

Allowable Tube sheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tube sheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.016 / 0.21) \cdot (2 \cdot 739 / (29.9 - 5))^2 \cdot 515$$

$$= 206.9962 \text{ N./mm}^2$$

Reqd Tube sheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 29.9 + 3 + 3$$

$$= 35.8840 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.2 \cdot 114 / 739) \cdot 110$$

$$= 6.8106 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tube sheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (15161 \cdot 282) / (0.2 \cdot 4.27 \cdot 4644)$$

$$= 107.8608 \text{ N./mm}^2$$

Reqd Tube sheet Thickness, for Shear Stress (including CA) [HReqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 4.27 + 3 + 3$$

$$= 10.2655 \text{ mm}$$

Reqd Tube sheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(35.9; 10.3)$$

$$= 35.8840 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.5, F_{tmax} = 3.19$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:



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$$\begin{aligned}
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs) \\
 &= ((2230 \cdot 0.47 - 0 \cdot 0.65) - (872) \cdot -0.5) / \\
 &\quad (0.65 - 0.47) \\
 &= 8.5754 \text{ N./mm}^2
 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned}
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs) \\
 &= ((2230 \cdot 0.47 - 0 \cdot 0.65) - (872) \cdot 3.19) / \\
 &\quad (0.65 - 0.47) \\
 &= -9.9009 \text{ N./mm}^2
 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 9.9 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= Sot = 92.3983 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned}
 &= \max((3.25 - 0.25(Zd + Q3 \cdot Zw)Xa^4); 1.25) \\
 &= \max((3.25 - 0.25(0.012 + -0.047 \cdot 0.039)5.04^4); 1.25) \\
 &= 1.6527 \text{ (Should be } \leq 2 \text{)}
 \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned}
 &= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4 \\
 &= (\sqrt{ 25.4^2 + (25.4 - 2 \cdot 2.32)^2 }) / 4 \\
 &= 8.2011 \text{ mm}
 \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned}
 &= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt) } \\
 &= \sqrt{ (2 \cdot \pi^2 \cdot 193573552 / 156) } \\
 &= 156.7104
 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned}
 &= k \cdot L / r \\
 &= 0.8 \cdot 648 / 8.2 \\
 &= 63.1815
 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned}
 &= (Sy_{t/Fs}) \cdot (1 - Ft / (2 \cdot Ct)) \\
 &= (156 / 1.65) \cdot (1 - 63.2 / (2 \cdot 157)) \\
 &= 75.169 \text{ N./mm}^2 \text{ (Never greater than Sot)}
 \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned}
 &= \sigma_{t,max} \cdot \text{Tube Area} \\
 &= 9.9 \cdot 1.68 \\
 &= 169.83 \text{ Kgf}
 \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned}
 &= \pi (do - t) t \\
 &= \pi (25.4 - 2.32) 2.32 \\
 &= 1.682 \text{ cm}^2
 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned}
 &= \pi \cdot t (do - t) S \\
 &= \pi \cdot 2.32 (25.4 - 2.32) 92.4 \\
 &= 1584.927 \text{ Kgf}
 \end{aligned}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft \text{)}$$



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ASME TS Calc: DESIGN 1

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$$= 0.85 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 92.4$$

$$= 1573.3376 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.4/92.4$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 92.4$$

$$= 1018.0419 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.32 \cdot (25.4 - 2.32) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.897 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.4487 mm

Minimum Required Groove Weld Leg [agr]: 1.4487 mm

Tube-Tubesheet Jt allowable, 1585 is $\geq$ tube strength 1585 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 739^2 / ((1566 + 14.8) 14.8) \cdot [872 + (1.06^2 - 1)(2230 - 0)] + 783^2 \cdot 0 / ((1566 + 14.8) 14.8)$$

$$= 26.7456 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 14.8 mm, the minimum Shell length adjacent to the tubesheet should be 274.066 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 739^2 / ((1566 + 14.8) 14.8) [872 + (1.06^2 - 1)(2230 - 0)] + 783^2 \cdot 0 / ((1566 + 14.8) 14.8)$$

$$= 26.7456 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 1365746176 / 14.8^2 \{ 0.012 [0.1822838 \cdot 2230 + 783^2 \cdot 0 / (2.86 \cdot 10^9)] + 6(1 - 0.45^2) / (31227670) (739 / 114)^3 (1 + 114 \cdot 0.012 / 2) [872 (0.039 + 0.28 \cdot -0.007) + 2 / 739^2 \cdot 0.28 \cdot -962] \}$$

$$= 239.2704 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$$



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:: | 26.7 | + | 239 | ≤ 207 N./mm²

:: 266, must be ≤ 207 N./mm² (See Elastic-Plastic iteration below)

Stress/Force Summary for Loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	40.9 ≤	207.0 N./mm ²	Pass
Tubesheet Shear Stress	12.5 ≤	110.4 N./mm ²	Pass
Maximum Tube Stress	9.9 ≤	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-9.9 ≤	-75.2 N./mm ²	Pass
Maximum Force on any one Tube	169.8 ≤	1584.9 Kgf	Pass
Axial Membrane Stress in Shell	26.7 ≤	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	266.0 ≤	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:

= min(Sys, SPS,s/2)

= min(227, 455/2) = 227 N./mm²

Compute Factor [facts]:

= min([1.4 - 0.4 • abs(σs,b) / Sstars], 1)

= min([1.4 - 0.4 • abs(239)/227], 1) = 0.98

Compute Factor [factc]:

= 1 (for Configuration = B)

Compute Modified Modulus [Es]:

= Es • facts

= 193573552 • 0.98 = 189534752 KPa.

Tubesheet Bending Stress - final thickness [σf]:

= (1.5 • Fm / μ*) • (2 • ao / (h - h'g))² • Pe

= (1.5 • 0.015/0.21) • (2 • 739 / (29.7 - 5))² • 512

= 206.9953 N./mm²

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

= h + Cats + Catc

= 29.7 + 3 + 3

= 35.7356 mm

For the Elastic-Plastic iteration, Shell Allowable = SPS,s

ShellAllow = 454.8170 N./mm²

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	41.0 ≤	207.0 N./mm ²	Pass
Tubesheet Shear Stress	12.5 ≤	110.4 N./mm ²	Pass
Maximum Tube Stress	9.9 ≤	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-9.9 ≤	-75.2 N./mm ²	Pass
Maximum Force on any one Tube	169.8 ≤	1584.9 Kgf	Pass
Axial Membrane Stress in Shell	26.7 ≤	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	266.0 ≤	454.8 N./mm ²	Pass



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ASME TS Calc: DESIGN 1

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Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	16.000 mm
Basic Gasket Width,	$b_0 = N / 2$	8.000 mm
Effective Gasket Width,	$b = \sqrt{b_0} * 2.5$	7.125 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	1609.750 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$= l_{tx} / h$ (modified for corrosion if present)

$= 0/114$

$= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:

$= \max(dt - 2 * tt(E_t T / E) * (S_t T / S) * (\rho); dt - 2 * tt)$

$= \max(25.4 - 2 * 2.32 (193573552 / 193573552) * (92.4 / 138) * (0); 25.4 - 2 * 2.32)$

$= 25.4000$ mm

Equivalent outer tube limit circle diameter $[D_o]$:

$= 2 * r_o + dt$

$= 2 * 726 + 25.4$

$= 1478.324$ mm

Determine the basic ligament efficiency for shear $[\mu]$:

$= (p - dt) / p$

$= (31.8 - 25.4) / 31.8$

$= 0.200$

Equivalent outer tube limit radius $[a_o]$:

$= D_o / 2$

$= 1478 / 2$

$= 739.1620$ mm

Effective tube pitch $[p^*]$:

$= p / \sqrt{ 1 - 4 * \min(AL * CNVfactor; 4 * D_o * p) / (\pi * D_o^2) }$

$= 31.8 / \sqrt{ (1 - 4 * \min(216 * 100 ; 4 * 1478 * 31.8) / (\pi * 1478^2)) }$

$= 31.9516$ mm

Effective ligament efficiency for bending $[\mu^*]$:

$= (p^* - d^*) / p^*$

$= (32 - 25.4) / 32$

$= 0.2050$

Ratio $[\rho_s]$:

$= a_s / a_o$

$= 783 / 739$

$= 1.0596$

Ratio $[\rho_c]$:

$= a_c / a_o$



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ASME TS Calc: DESIGN 1

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$$= 805/739$$

$$= 1.0889$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 1780 ((25.4 - 2 \cdot 2.32) / (2 \cdot 739))^2$$

$$= 0.6490$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 1780 (25.4 / (2 \cdot 739))^2$$

$$= 0.4745$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8 - 3); 0)$$

$$= 5.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 14.8 (1566 + 14.8)193573552/6472$$

$$= 224234.5469 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.32 (25.4 - 2.32)193573552/6472$$

$$= 513.0714 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 224235 / (1780 \cdot 513)$$

$$= 0.24553$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 224235/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - vs^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1566 + 14.8)14.8)$$

$$= 0.0119 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - vs^2))$$

$$= 0.012 \cdot 193573552 \cdot 14.8^3 / (6(1 - 0.3^2))$$

$$= 1365746176.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(1566 + 14.8) \cdot 1365746176/114^3 \cdot (1 + 114 \cdot 0.012 + (114 \cdot 0.012)^2/2)$$

$$= 28618092.0000 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot vs / (2(Ds + ts)))$$

$$= 1566^2 / (4 \cdot 193573552 \cdot 14.8) \cdot (1 - 1566 \cdot 0.3 / (2(1566 + 14.8)))$$

$$= 0.18228E+00 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:



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ASME TS Calc: DESIGN 1

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$h/p = 3.590551$; $\mu^* = 0.205047$; $E^* = 31227670$ KPa.
 $E^*/E = 0.161322$; $\nu^* = 0.451291$

Note: As h/p (3.591) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\left(24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.45^2)1780 \cdot 193573552 \cdot 2.32 \right. \right. \\ \left. \left. (25.4 - 2.32)739^2 \right) / (31227670 \cdot 6472 \cdot 114^3) \right)^{1/4}$$

$$= 5.0364$$

Values from Table 4.18.3

$Z_d = 0.011736$; $Z_v = 0.038739$; $Z_m = 0.280694$
 $Z_a = 0.249680E+02$; $Z_w = 0.038739$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 1840/1478$$

$$= 1.2447$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31227670 \cdot (28618092 + 0 + 193573552 \cdot \ln(1.24))$$

$$= 1.2473$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)^F$$

$$= (1 + 0.45)^{1.25}$$

$$= 1.8101$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.06 - 1 - 1.81 \cdot 0.039) / (1 + 1.81 \cdot 0.28)$$

$$= -0.006992$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4$$

$$= (0.012 + -0.007 \cdot 0.039) / 2 \cdot 5.04^4$$

$$= 3.6885$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4$$

$$= (0.039 + -0.007 \cdot 0.28) / 2 \cdot 5.04^4$$

$$= 11.8310$$

Compute parameter [U]:

$$= (Z_w + (ps - 1)Z_m)Xa^4 / (1 + \phi \cdot Z_m)$$

$$= (0.039 + (1.06 - 1)0.28)5.04^4 / (1 + 1.81 \cdot 0.28)$$

$$= 23.6620$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= (Gc - C) / Do \text{ (config b)}$$

$$= (1610 - 1750) / 1478 = -0.095$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.06 \cdot 1365746176 \cdot 0.012 \cdot 0.1822838 (1 + 114 \cdot 0.012)$$



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ASME TS Calc: DESIGN 1

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$$= 7381.4111 \text{ mm}^2$$

Calculate parameter $[\omega*s]$:

$$\begin{aligned} &= ao^2(\rho s^2 - 1) \cdot (\rho s - 1) / 4 - \omega s \\ &= 739^2(1.06^2 - 1) \cdot (1.06 - 1) / 4 - 7381 \\ &= -6382.8521 \text{ mm}^2 \end{aligned}$$

Parameter $[\omega c]$:

$$\begin{aligned} &= \rho c \cdot kc [\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)) \\ &= 1.09 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 114) - \\ &\quad 0.000 (0/114 + 0 \cdot 0/2)) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Parameter $[\omega*c]$:

$$\begin{aligned} &= ao^2[(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\rho s - 1) / 2] - \omega c \\ &= 739^2[(1.09^2 + 1) \cdot (1.09 - 1) / 4 - (1.06 - 1) / 2] - 0 \\ &= 10265.8027 \text{ mm}^2 \end{aligned}$$

Pressure $[P*S]$:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure $[P*C]$:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure $[P's]$:

$$\begin{aligned} &= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - \\ &\quad [(\rho s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(Dj^2 - (Ds)^2) / Do^2] \} \\ &= 2230 \{ 0.47 + 2(1 - 0.47) 0.3 + [2/0.25 \cdot \\ &\quad (1566/1478)^2] 0.3 - [(1.06^2 - 1) / (1 \cdot 0.25)] - \\ &\quad [(1 - 1) / (2 \cdot 1 \cdot 0.25)] [(0^2 - (1566)^2) / 1478^2] \} \\ &= 6764.9282 \text{ KPa.} \end{aligned}$$

Pressure $[P't]$:

$$\begin{aligned} &= [xt + 2(1 - xt) vt + 1 / (J \cdot Kst)] Pt \\ &= [0.65 + 2(1 - 0.65) 0.3 + 1 / (1 \cdot 0.25)] 1750 \\ &= 8631.7197 \text{ KPa.} \end{aligned}$$

Pressure $[P_y]$:

$$\begin{aligned} &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\ &= 1780 \cdot 513 \cdot 0 / (3.14 \cdot 739^2) \\ &= 0.0000 \text{ KPa.} \end{aligned}$$

Pressure $[P_w]$:

$$\begin{aligned} &= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2) \\ &= -0.095 \cdot 23.7 \cdot 560144 / (2 \cdot \pi \cdot 739^2) \\ &= 3591.9236 \text{ KPa.} \end{aligned}$$

Calculate the pressure $[Prim]$:

$$\begin{aligned} &= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt) \\ &= -(23.7/739^2) \cdot (-6383 \cdot 2230 - 10266 \cdot 1750) \\ &= 1394.4830 \text{ KPa.} \end{aligned}$$

Calculate the pressure $[P\omega]$:

$$\begin{aligned} &= (U/ao^2) \cdot (\omega s \cdot P*s - \omega c \cdot P*c) \\ &= (23.7/739^2) \cdot (7381 \cdot 0 - 0 \cdot 0) \\ &= 0.0000 \text{ KPa.} \end{aligned}$$

Determine the effective pressure $[Pe]$:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (\rho s - 1) Qz2)) \cdot \\ &\quad (P's - P't + P_y + P_w + P\omega + Prim) \\ &= 0.1000E+01 \cdot 0.25 / (1 + 1 \cdot 0.25 (3.69 + (1.06 - \\ &\quad 1) 11.8)) \cdot (6765 - 8632 + 0 + 0 + 3592 + 1394) \end{aligned}$$



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ASME TS Calc: DESIGN 1

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= 368.4809 KPa.

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega s \cdot P * s - \omega c \cdot P * c)) + W * \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-6383 \cdot 2230 - 10266 \cdot 1750) - (7381 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 560144 \cdot -0.095 / (2 \cdot \pi)] / (1 + 1.81 \cdot 0.28)$$

$$= -7785.4565 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (P_e \cdot a o^2)$$

$$= -0.007 + 2 \cdot -7785 / (368 \cdot 739^2)$$

$$= -0.76544$$

Fm Value from Table 4.18.3 = 0.382718

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.38 / 0.21) \cdot (2 \cdot 739 / (114 - 5))^2 \cdot 368$$

$$= 189.7763 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.35 / 0.21) \cdot (2 \cdot 739 / (109 - 5))^2 \cdot 403$$

$$= 206.9872 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 109 + 3 + 3$$

$$= 115.0590 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.2 \cdot 114 / 739) \cdot 110$$

$$= 6.8106 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (15161 \cdot 1589) / (0.2 \cdot 24.7 \cdot 4644)$$

$$= 104.9058 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 24.7 + 3 + 3$$

$$= 30.7335 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(115; 30.7)$$

$$= 115.0590 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -5.76, F_{tmax} = 3.14$$



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ASME TS Calc: DESIGN 1

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First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((2230 \cdot 0.47 - 1750 \cdot 0.65) - (368) \cdot -5.76) / (0.65 - 0.47)$$

$$= 11.7305 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((2230 \cdot 0.47 - 1750 \cdot 0.65) - (368) \cdot 3.14) / (0.65 - 0.47)$$

$$= -7.0844 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 11.7 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92.3983 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [F_s]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.012 + -0.77 \cdot 0.039)5.04^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [r_t]:

$$= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2}) / 4$$

$$= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.32)^2}) / 4$$

$$= 8.2011 \text{ mm}$$

Determine the factor [C_t]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{yt}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 193573552 / 156) }$$

$$= 156.7104$$

Determine the factor [F_t]:

$$= k \cdot L / r$$

$$= 0.8 \cdot 648 / 8.2$$

$$= 63.1815$$

Buckling Allowable Stress [S_{tb}]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (156 / 2) \cdot (1 - 63.2 / (2 \cdot 157))$$

$$= 62.116 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [W_t]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 11.7 \cdot 1.68$$

$$= 201.22 \text{ Kg}$$

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.32) 2.32$$

$$= 1.682 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.32 (25.4 - 2.32) 92.4$$

$$= 1584.927 \text{ Kg}$$



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Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 92.4$$

$$= 1573.3376 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.4/92.4$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 92.4$$

$$= 1018.0419 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\left((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d \right)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{\left((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.32 \cdot (25.4 - 2.32) 1 \cdot 1 \right)} - 0.75 \cdot 25.4 \right)$$

$$= 2.897 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.4487 mm

Minimum Required Groove Weld Leg [agr]: 1.4487 mm

Tube-Tubesheet Jt allowable, 1585 is $\geq$ tube strength 1585 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 739^2 / ((1566 + 14.8) 14.8) \cdot [368 + (1.06^2 - 1)(2230 - 1750)] + 783^2 \cdot 1750 / ((1566 + 14.8) 14.8)$$

$$= 55.8519 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 14.8 mm, the minimum Shell length adjacent to the tubesheet should be 274.066 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 739^2 / ((1566 + 14.8) 14.8) [368 + (1.06^2 - 1)(2230 - 1750)] + 783^2 \cdot 1750 / ((1566 + 14.8) 14.8)$$

$$= 55.8519 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E \cdot t_s)] + 6(1 - \nu^2) / (E \cdot t_s) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 1365746176 / 14.8^2 \{ 0.012 [0.1822838 \cdot 2230 + 783^2 \cdot 0 / (2.86E+09)] + 6(1 - 0.45^2) / (31227670) (739 / 114)^3 (1 + 114 \cdot 0.012 / 2) [368 (0.039 + 0.28 \cdot -0.007) + 2 / 739^2 \cdot 0.28 \cdot -7785] \}$$

$$= 10.8810 \text{ N./mm}^2$$



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Shell Stress Summation vs. Allowable

$||\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$
 $||55.9| + |10.9| \leq 207 \text{ N./mm}^2$
 $66.7, \text{ must be } \leq 207 \text{ N./mm}^2$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	189.8 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	5.3 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	11.7 <=	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-7.1 <=	-62.1 N./mm ²	Pass
Maximum Force on any one Tube	201.2 <=	1584.9 KgF	Pass
Axial Membrane Stress in Shell	55.9 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	66.7 <=	207.0 N./mm ²	Pass

Thickness Results for Loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	115.059	120.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.449	2.320	Ok
Tube-Tubesheet Groove Weld Leg :	1.449	2.320	Ok

Detailed Results for load Case D4 corr. (Psd,min + Ptd,min)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width, $N = (G_{oc} - G_{ic}) / 2$ 16.000 mm
 Basic Gasket Width, $b_0 = N / 2$ 8.000 mm
 Effective Gasket Width, $b = \text{sqrt}(b_0) * 2.5$ 7.125 mm
 Gasket Reaction Diameter, $G = G_{oc} - 2.0 * b$ 1609.750 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/114$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:
 $= \max(d_t - 2 * t_t (E_t T / E) * (S_t T / S) * (\rho); d_t - 2 * t_t)$
 $= \max(25.4 - 2 * 2.32 (193573552 / 193573552) * (92.4 / 138) * (0); 25.4 - 2 * 2.32)$
 $= 25.4000 \text{ mm}$

Equivalent outer tube limit circle diameter $[D_o]$:
 $= 2 * r_o + d_t$
 $= 2 * 726 + 25.4$
 $= 1478.324 \text{ mm}$

Determine the basic ligament efficiency for shear $[\mu]$:



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$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1478 / 2$$

$$= 739.1620 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 31.8 / \sqrt{(1 - 4 \cdot \min(216 \cdot 100 ; 4 \cdot 1478 \cdot 31.8) / (\pi \cdot 1478^2))}$$

$$= 31.9516 \text{ mm}$$

Effective ligament efficiency for bending [μ *]:

$$= (p* - d*) / p*$$

$$= (32 - 25.4) / 32$$

$$= 0.2050$$

Ratio [ps]:

$$= as / ao$$

$$= 783 / 739$$

$$= 1.0596$$

Ratio [pc]:

$$= ac / ao$$

$$= 805 / 739$$

$$= 1.0889$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 1780 ((25.4 - 2 \cdot 2.32) / (2 \cdot 739))^2$$

$$= 0.6490$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 1780 (25.4 / (2 \cdot 739))^2$$

$$= 0.4745$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8 - 3); 0)$$

$$= 5.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 14.8 (1566 + 14.8)193573552/6472$$

$$= 224234.5469 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.32 (25.4 - 2.32)193573552/6472$$

$$= 513.0714 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 224235 / (1780 \cdot 513)$$

$$= 0.24553$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kj)$$

$$= 1 / (1 + 224235/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$



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Shell coefficient [β_s]:

$$= \left(\left(12(1 - v_s^2) \right)^{1/4} \right) / \left((Ds + ts)ts \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \left((1566 + 14.8)14.8 \right)$$

$$= 0.0119 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - v_s^2))$$

$$= 0.012 \cdot 193573552 \cdot 14.8^3 / (6(1 - 0.3^2))$$

$$= 1365746176.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(1566 + 14.8) \cdot 1365746176/114^3 \cdot (1 + 114 \cdot 0.012 + (114 \cdot 0.012)^2/2)$$

$$= 28618092.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot E_s \cdot ts) \cdot (1 - Ds \cdot v_s / (2(Ds + ts)))$$

$$= 1566^2 / (4 \cdot 193573552 \cdot 14.8) \cdot (1 - 1566 \cdot 0.3 / (2(1566 + 14.8)))$$

$$= 0.18228\text{E}+00 \text{ mm/N} \cdot \text{mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 3.590551$; $\mu^* = 0.205047$; $E^* = 31227670 \text{ KPa}$.

$E^*/E = 0.161322$; $v^* = 0.451291$

Note: As h/p (3.591) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= \left(\left(24(1 - v^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.45^2)1780 \cdot 193573552 \cdot 2.32 \right. \right.$$

$$\left. \left(25.4 - 2.32 \right)739^2 \right) / (31227670 \cdot 6472 \cdot 114^3) \right)^{1/4}$$

$$= 5.0364$$

Values from Table 4.18.3

$$Z_d = 0.011736 ; Z_v = 0.038739 ; Z_m = 0.280694$$

$$Z_a = 0.249680\text{E}+02 ; Z_w = 0.038739$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 1840/1478$$

$$= 1.2447$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_C + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31227670 \cdot (28618092 + 0 + 193573552 \cdot \ln(1.24))$$

$$= 1.2473$$

Compute parameter [ϕ]:

$$= (1 + v^*)F$$

$$= (1 + 0.45)1.25$$

$$= 1.8101$$

Compute parameter [Q_1]:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.06 - 1 - 1.81 \cdot 0.039) / (1 + 1.81 \cdot 0.28)$$

$$= -0.006992$$

Compute Parameter [Q_{z1}]:



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$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.012 + -0.007 \cdot 0.039) / 2 \cdot 5.04^4$$

$$= 3.6885$$

Compute parameter [Qz2]:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.039 + -0.007 \cdot 0.28) / 2 \cdot 5.04^4$$

$$= 11.8310$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.039 + (1.06 - 1) 0.28) 5.04^4 / (1 + 1.81 \cdot 0.28)$$

$$= 23.6620$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (1610 - 1750) / 1478 = -0.095$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.06 \cdot 1365746176 \cdot 0.012 \cdot 0.1822838 (1 + 114 \cdot 0.012)$$

$$= 7381.4111 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega s$$

$$= 739^2 (1.06^2 - 1) \cdot (1.06 - 1) / 4 - 7381$$

$$= -6382.8521 \text{ mm}^2$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.09 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 114) - 0.000 (0/114 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega c$$

$$= 739^2 [(1.09^2 + 1) \cdot (1.09 - 1) / 4 - (1.06 - 1) / 2] - 0$$

$$= 10265.8027 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= -101 \{ 0.47 + 2(1 - 0.47) 0.3 + [2/0.25 \cdot (1566/1478)^2] 0.3 - [(1.06^2 - 1)/(1 \cdot 0.25)] - [(1 - 1)/(2 \cdot 1 \cdot 0.25)] [(0^2 - (1566)^2)/1478^2] \}$$

$$= -306.7880 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.65 + 2(1 - 0.65) 0.3 + 1/(1 \cdot 0.25)] - 0$$

$$= -0.0000 \text{ KPa.}$$



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Pressure [Py]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 1780 \cdot 513 \cdot 0 / (3.14 \cdot 739^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0.095 \cdot 23.7 \cdot 0 / (2 \cdot \pi \cdot 739^2)$$

$$= 0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(23.7/739^2) \cdot (-6383 \cdot -101 - 10266 \cdot -0)$$

$$= -27.9555 \text{ KPa.}$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega*s \cdot P*s - \omega*c \cdot P*c)$$

$$= (23.7/739^2) \cdot (7381 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + Py + Pw + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.25 / (1 + 1 \cdot 0.25(3.69 + (1.06 - 1)11.8)) \cdot (-307 - -0 + 0 + 0 + 0 + -28)$$

$$= -39.5390 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega*s \cdot P*s - \omega*c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-6383 \cdot -101 - 10266 \cdot -0) - (7381 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 0 \cdot -0.095 / (2 \cdot \pi)] / (1 + 1.81 \cdot 0.28)$$

$$= 43.6479 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.007 + 2 \cdot 43.6 / (-39.5 \cdot 739^2)$$

$$= -0.04662$$

Fm Value from Table 4.18.3 = 0.034898

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.035 / 0.21) \cdot (2 \cdot 739 / (114 - 5))^2 \cdot -39.5$$

$$= -1.8568 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.01 / 0.21) \cdot (2 \cdot 739 / (8.63 - 5))^2 \cdot -16.7$$

$$= -206.9855 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 8.63 + 3 + 3$$

$$= 14.6286 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:



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ASME TS Calc: DESIGN 1

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Shear Stress check [Tlimit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.2 \cdot 114/739) \cdot 110$$

$$= 6.8106 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 9.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 16.4325 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(14.6; 9.81)$$

$$= 14.6286 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -0.5, F_{tmax} = 3.19$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((-101 \cdot 0.47 - -0 \cdot 0.65) - (-39.5) \cdot -0.5) / (0.65 - 0.47)$$

$$= -0.3889 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((-101 \cdot 0.47 - -0 \cdot 0.65) - (-39.5) \cdot 3.19) / (0.65 - 0.47)$$

$$= 0.4490 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 0.45 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92.3983 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.012 + -0.047 \cdot 0.039)5.04^4); 1.25)$$

$$= 1.6527 \text{ (Should be } \leq 2)$$

Determine the factor [rt]:

$$= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2}) / 4$$

$$= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.32)^2}) / 4$$

$$= 8.2011 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 193573552 / 156)}$$

$$= 156.7104$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0.8 \cdot 648 / 8.2$$

$$= 63.1815$$

Buckling Allowable Stress [Stb]:

$$= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (156 / 1.65) \cdot (1 - 63.2 / (2 \cdot 157))$$



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= 75.169 N./mm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

= $\sigma_{t,max} \cdot \text{Tube Area}$

= 0.45 • 1.68

= 7.70 Kgf

Tube Metal Area [At]:

= $\pi (d_o - t) t$

= $\pi (25.4 - 2.32) 2.32$

= 1.682 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

= $\pi \cdot t (d_o - t) S$

= $\pi \cdot 2.32 (25.4 - 2.32) 92.4$

= 1584.927 Kgf

Groove Weld Strength [Fg]:

= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)

= $0.85 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 92.4$

= 1573.3376 Kgf

Weld Strength Factor [fw]:

= S / S_w where $S_w = \min(S \text{ and } S_t)$

= 92.4/92.4

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)

= $0.55 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 92.4$

= 1018.0419 Kgf

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$

= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.32 \cdot (25.4 - 2.32) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$

= 2.897 mm

Minimum Required Fillet Weld Leg [afr]: 1.4487 mm

Minimum Required Groove Weld Leg [agr]: 1.4487 mm

Tube-Tubesheet Jt allowable, 1585 is >= tube strength 1585 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

= $S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Shell [$\sigma_{s,m}$]:

= $a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$

= $739^2 / ((1566 + 14.8) 14.8) \cdot [-39.5 + (1.06^2 - 1)$

$(-101 - 0)] + 783^2 \cdot 0 / ((1566 + 14.8) 14.8)$

= -1.2129 N./mm²

Note: The Shell is under Axial Compression, checking the Compressive Stress



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For the Shell Attached to the Tubesheet, Factor A per UG-23:
 $= 0.125 / (S_{Ro}/t_s) = 0.125 / (798/14.8) = 0.0023$

Longitudinal Compressive Allowable Stress [Ss,b]:
 $= 99.7478 \text{ N./mm}^2$

Note: The Shell Axial Compressive stress is within limits.

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 14.8 mm, the minimum Shell length adjacent to the tubesheet should be 274.066 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 739^2 / ((1566 + 14.8) 14.8) [-39.5 + (1.06^2 - 1) (-101 - 0)] + 783^2 \cdot 0 / ((1566 + 14.8) 14.8)$
 $= -1.2129 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o/h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2/a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 1365746176 / 14.8^2 \{ 0.012 [0.1822838 \cdot -101 + 783^2 \cdot 0 / (2.86E+09)] + 6(1 - 0.45^2) / (31227670) (739/114)^3 (1 + 114 \cdot 0.012/2) [-39.5 (0.039 + 0.28 \cdot -0.007) + 2/739^2 \cdot 0.28 \cdot 43.6] \}$
 $= -10.8509 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$
 $:: |-1.21| + |-10.9| \leq 207 \text{ N./mm}^2$
 $:: 12.1, \text{ must be } \leq 207 \text{ N./mm}^2$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D4 corr. (Psd,min + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	1.9 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	0.6 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	0.4 <=	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-0.4 <=	-75.2 N./mm ²	Pass
Maximum Force on any one Tube	7.7 <=	1584.9 Kgf	Pass
Axial Membrane Stress in Shell	1.2 <=	138.0 N./mm ²	Pass
Axial Compressive Shell Stress	-1.2 <=	-99.7 N./mm ²	Pass
Shell Stress (jt. inter.)	12.1 <=	207.0 N./mm ²	Pass

Thickness Results for Loadcase D4 corr. (Psd,min + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	14.629	120.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.449	2.320	Ok
Tube-Tubesheet Groove Weld Leg :	1.449	2.320	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

--Reqd. Thk. + CA ---- Tubesheet Stresses Case Pass/



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Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	103.471	57.400	158	207	7	110	Ps+Pt-Th	D1 Ok
D2uc	39.831	...	37	207	12	110	Ps+Pt-Th	D2 Ok
D3uc	111.925	...	180	207	5	110	Ps+Pt-Th	D3 Ok
D4uc	12.555	...	2	207	1	110	Ps+Pt-Th	D4 Ok
O1uc	59.950	...	138	431	7	110	Ps+Pt+Th	O1 Ok
O2uc	60.688	...	141	431	18	110	Ps+Pt+Th	O2 Ok
O3uc	66.252	...	157	431	5	110	Ps+Pt+Th	O3 Ok
O4uc	54.122	...	110	431	6	110	Ps+Pt+Th	O4 Ok
D1c	110.576	57.400	177	207	8	110	Ps+Pt-Th-c	D1 Ok
D2c	35.884	...	41	207	12	110	Ps+Pt-Th-c	D2 Ok
D3c	115.059	...	190	207	5	110	Ps+Pt-Th-c	D3 Ok
D4c	14.629	...	2	207	1	110	Ps+Pt-Th-c	D4 Ok
O1c	70.072	...	161	431	7	110	Ps+Pt+Th-c	O1 Ok
O2c	67.180	...	151	431	19	110	Ps+Pt+Th-c	O2 Ok
O3c	74.021	...	173	431	5	110	Ps+Pt+Th-c	O3 Ok
O4c	63.128	...	126	431	6	110	Ps+Pt+Th-c	O4 Ok
Max:	115.0590	57.400	mm	0.917		0.168	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	23	137	...	...		...	...	...	...			Ok
D2uc	22	137	...	...		...	...	...	...			Ok
D3uc	46	137	...	...		...	...	...	...			Ok
D4uc	1	137	-1	-103		...	...	...	...			Ok
O1uc	24	454	...	...		...	...	...	...			Ok
O2uc	30	454	...	...		...	...	...	...			Ok
O3uc	46	454	...	...		...	...	...	...			Ok
O4uc	8	454	...	...		...	...	...	...			Ok
D1c	28	137	...	...		...	...	...	...			Ok
D2c	27	137	...	...		...	...	...	...			Ok
D3c	56	137	...	...		...	...	...	...			Ok
D4c	1	137	-1	-99		...	...	...	...			Ok
O1c	29	454	...	...		...	...	...	...			Ok
O2c	37	454	...	...		...	...	...	...			Ok
O3c	56	454	...	...		...	...	...	...			Ok



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04c	10	454	...	...	...	...	...	...	...	Ok
-----	----	-----	-----	-----	-----	-----	-----	-----	-----	----

Max RATIO	0.405	0.012	...	...
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Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass											
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	19	92	-13	-92	328	1584	243	454	...	...	Ok
D2uc	9	92	-9	-69	151	1584	220	454	...	...	Ok
D3uc	10	92	-7	-62	170	1584	59	206	...	...	Ok
D4uc	...	92	...	-69	7	1584	10	206	...	...	Ok
O1uc	19	184	-13	-92	321	3169	235	454	...	...	Ok
O2uc	7	184	-6	-62	129	3169	130	454	...	...	Ok
O3uc	10	184	-7	-62	170	3169	59	454	...	...	Ok
O4uc	5	184	-5	-62	79	3169	106	454	...	...	Ok
D1c	22	92	-14	-92	379	1584	267	454	...	...	Ok
D2c	10	92	-10	-75	170	1584	266	454	...	...	Ok
D3c	12	92	-7	-62	201	1584	67	206	...	...	Ok
D4c	...	92	...	-75	8	1584	12	206	...	...	Ok
O1c	22	184	-13	-92	371	3169	257	454	...	...	Ok
O2c	8	184	-6	-62	133	3169	169	454	...	...	Ok
O3c	12	184	-7	-62	201	3169	67	454	...	...	Ok
O4c	5	184	-5	-62	93	3169	117	454	...	...	Ok

Max RATIO	0.239	0.146	0.239	0.587	...
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Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	115.059	120.000	Ok
Tubesheet thickness flanged extension :	57.400	99.000	Ok
Tube thickness :	0.655	2.320	Ok
Tube-tubesheet fillet weld leg :	1.449	2.320	Ok
Tube-tubesheet groove weld leg :	1.449	2.320	Ok

Min Shell length of thk, (18.000) adj. to tubesheet: 301.628 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.102 mm
 Due to Pressure + Thermal Shell Compresses by : -0.102 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	2472.46	1.000	11273.84	1.000
Tubesheet Shear Stress	15343.34	1.000	18632.53	1.000
Tube Tensile Stress	8044.06	1.000	20811.10	1.000
Tube Compressive Stress	8729.10	0.575	16602.28	1.000
Tube-Tubesheet Joint load	8044.06	1.000	20811.10	1.000
Shell Stress (Axial, Junction)	2472.46	1.000	3812.70	1.000
Tube Pressure Stress	18208.59	1.000	11236.95	1.000
Tubesheet Extension Stress	2439.78	...	No Calc	No Calc
Minimum MAWP	2439.78		3812.70	

Tubesheet MAPnc used to Compute Hydrotest Pressure:



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ASME TS Calc: DESIGN 1

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Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	2769.74	0.999	12488.92	1.000
Tubesheet Shear Stress	23290.88	1.000	20250.86	1.000
Tube Tensile Stress	9252.01	1.000	23365.33	1.000
Tube Compressive Stress	13668.27	1.000	19960.77	1.000
Tube-Tubesheet Joint load	9252.01	1.000	23365.32	1.000
Shell Stress (Axial, Junction)	2769.74	0.999	5241.77	0.996
Tube Pressure Stress	18208.59	1.000	11671.23	1.000
Tubesheet Extension Stress	2439.78	...	No Calc	No Calc
Minimum MAPnc	2439.78		5241.77	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0350	0.0351	...	...	...	...
2	-0.1020	-0.1021	...	...	...	...
3	-0.0717	-0.0717	...	...	...	...
4	0.0046	0.0046	...	...	...	...
5	0.0304	0.0304	...	...	...	...
6	-0.1020	-0.1021	...	...	...	...
7	-0.0717	-0.0717	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

= max(tubesheet thk/4, min(tubesheet thk, shell thickness))

= max(120/4, min(120, 18))

= 30.000 mm

Thickness Ratio = 0.72, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Channel Side MDMT calculation:

Governing thickness for the channel side:

= tubesheet thickness/4

= 120/4

= 30.000 mm

Thickness Ratio = 0.72, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(1750/2440, 2230/5242)

= 0.717

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Problems/Failures Summary:

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: